

Chapter 2: Plane Electromagnetic Waves

Opening Words

- Solutions to Eqs. (6.25) & (6.26) represent waves
- Those waves with just 1D spatial dependence: *plane waves*.
- Uniform plane waves: particular solution of Maxwell's equations with $\mathbf{E}$ assuming same direction, magnitude, & phase in infinite planes perpendicular to propagation direction; and same thing for $\mathbf{H}$.
- True plane waves nonexistent in reality $\therefore$ source infinite in extent needed to create them ²

Part 1: Plane Waves in Lossless Media

One-Dimensionality

Source-free wave equation (6.25) for free-space becomes homogeneous vector Helmholtz equation: $\nabla^2 \mathbf{E} + k_0^2 \mathbf{E} = 0$ (1.1)

as of (7.18) in Chapter 1

$\downarrow$
free-space wavenumber

Free-space wavenumber: $k_0 = \omega \sqrt{\mu_0 \epsilon_0} = \omega/c$ (1.2)

(rad/m)

In Cartesian coordinates, (1.1) represents 3 *scalar* Helmholtz's equations, one each in components E_x , E_y , & E_z .

For $w = x, y$ or z , scalar equation:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k_0^2 \right) E_w = 0 \quad (1.3)$$

Consider uniform plane wave with non-zero E_x only, with constant magnitude & phase over planar surfaces parallel to xy plane perpendicular to propagation direction z , i.e.,

$$\partial^2 E_x / \partial x^2 \quad \& \quad \partial^2 E_x / \partial y^2 = 0$$

Eq. (1.3) simplifies to: $\frac{\partial^2 E_x(z)}{\partial z^2} + k_0^2 E_x(z) = 0$ (1.4)

Instantaneous (Real Time) Function

One particular solution of (1.4):

$$E_x(z) = E_x^+(z) + E_x^-(z) = E_0^+ e^{-jk_0 z} + E_0^- e^{+jk_0 z} \quad (1.5)$$

where E_0^+ & E_0^- arbitrary (generally complex) constants – must be determined by boundary conditions

Now, examine 1st phasor term on RHS of (1.4) in real time

Use $\cos \omega t$ as reference & assuming E_0^+ real constant (zero ref phase at $z = 0$):

$$E_x^+(z, t) = \text{Re} \left[\underbrace{E_0^+ e^{-jk_0 z}}_{E_0^+ e^{j(\omega t - k_0 z)}} e^{j\omega t} \right] = E_0^+ \cos(\omega t - k_0 z) \quad (1.6)$$

Traveling Wave

- Plot (1.6) for $E_x^+(z, t)$ vs. z for several t (below)
- At $t = 0$: $E_x^+(z, 0) = E_0^+ \cos k_0 z$: cosine curve with amplitude E_0^+ .
- Successive times: curve travels in $+z$ direction: *traveling wave*.

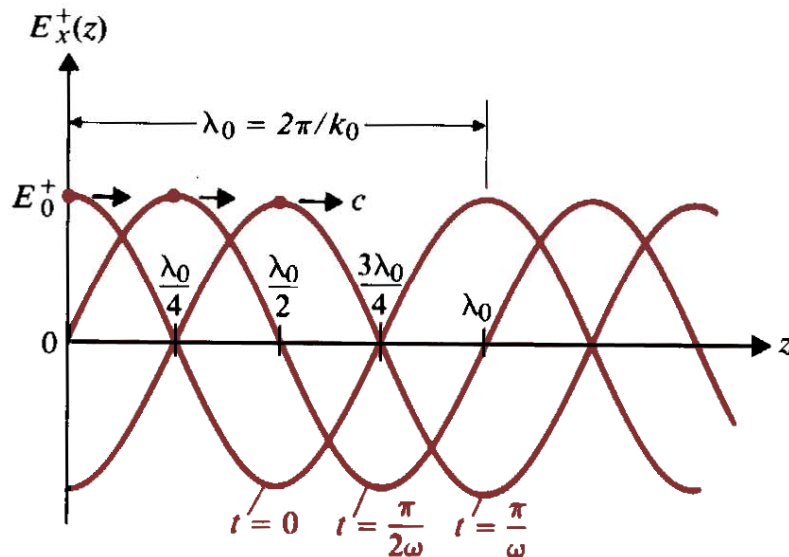


FIGURE 8-1

Wave traveling in positive z direction

$E_x^+(z, t) = E_0^+ \cos(\omega t - k_0 z)$, for several values of t .

Phase Velocity, Wavelength & Wavenumber

- If fix attention on certain point (of particular phase) on wave:
 - Set $\cos(\omega t - k_0 z) = \text{constant}$, i.e. $\omega t - k_0 z = \text{constant}$
 - ◆ $z = (\omega t - \text{constant})/k_0$.
 - ◆ $dz/dt = \omega/k_0 = \omega/[\omega\sqrt{(\mu_0 \epsilon_0)}] = 1/\sqrt{(\mu_0 \epsilon_0)} = u_p = c$. (1.7)
 - This $u_p = c = \text{phase velocity}$
- This (1.7) assures propagation velocity of equiphase front (phase velocity) in free-space equals speed of light, $c = 299792458 \text{ m/s}$
- Since wavelength $\lambda_0 = c/f$ & $k_0 = 2\pi f/c$ from (1.2),
 $\therefore k_0 = 2\pi/\lambda_0 \quad (\text{rad/m})$ (1.8)

wavenumber: measures amount of phase progression per unit distance

Comments

- Obvious, without replotting (1.6), that 2nd phasor term on RHS of (1.5), $E_0^- \exp(+jk_0 z)$, represents cosinusoidal wave traveling in $-z$ direction with same velocity c .
- If concerned only with wave traveling in $+z$ direction, then $E_0^- = 0$.
- However, if discontinuities in medium, reflected waves in opposite direction must also be considered (as seen later)

Associated H-Field

From (7.14) in Chapter 1: $\nabla \times \vec{E} = -j\omega\mu\vec{H}$ **(7.14)repeated**

$$\therefore \nabla \times \vec{E} = \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} \times \begin{pmatrix} E_x^+ \\ E_y^+ \\ E_z^+ \end{pmatrix} = \begin{pmatrix} \cancel{\partial E_z^+ / \partial y} - \cancel{\partial E_y^+ / \partial z} \\ \partial E_x^+ / \partial z - \cancel{\partial E_z^+ / \partial x} \\ \cancel{\partial E_y^+ / \partial x} - \cancel{\partial E_x^+ / \partial y} \end{pmatrix} \stackrel[\text{Chapter 1}]{\text{from (7.14)}} = -j\omega\mu \begin{pmatrix} H_x^+ \\ H_y^+ \\ H_z^+ \end{pmatrix}$$

in which all slashes (forward & backward) denote cancellation to zero;

- forward slashes (/) due to vanishing partial derivatives with respect to x & y ($\therefore$ independence of them),
- whereas backward slashes (\) $\therefore$ only $E_x \neq 0$.

$$\therefore H_x^+ \equiv H_z^+ \equiv 0; \quad H_y^+ = -\frac{1}{j\omega\mu_0} \frac{\partial E_x^+}{\partial z} \quad \textbf{(1.9)}$$

Intrinsic Wave Impedance

Since $E_x^+(z) = E_0^+ e^{-jk_0 z}$ from (1.5)

$$\frac{\partial E_x^+(z)}{\partial z} = \frac{\partial}{\partial z} \left(E_0^+ e^{-jk_0 z} \right) = -jk_0 \overbrace{E_0^+ e^{-jk_0 z}}^{E_x^+(z)}$$

(1.9) yields

$$H_y^+(z) = -\frac{1}{j\omega\mu_0} \left[\overbrace{-jk_0 E_x^+(z)}^{\partial E_x^+ / \partial z} \right] = \frac{\overbrace{\omega\sqrt{\mu_0\epsilon_0}}^{k_0}}{\omega\mu_0} E_x^+(z) = \sqrt{\frac{\epsilon_0}{\mu_0}} E_x^+(z) \stackrel{(1.10)}{=} \frac{\overbrace{E_0^+ e^{-jk_0 z}}^{E_x^+(z)}}{\underbrace{\sqrt{\mu_0/\epsilon_0}}_{\substack{\text{define as} \\ \eta_0}}} \quad (\text{A/m})$$

$$\boxed{\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \cong 120\pi \cong 377 \quad (\Omega)} \quad (1.11)$$

intrinsic wave impedance of free space

Instantaneous H -Field

- Instantaneous (real-time) expression for H :

$$\vec{H}(z,t) = \vec{a}_y H_y^+(z,t) = \vec{a}_y \operatorname{Re} \left[\underbrace{\overbrace{H_y^+(z) e^{j\omega t}}^{\frac{E_0^+}{\eta_0} \exp[j(\omega t - k_0 z)]}}_{\substack{\text{from } \Downarrow (1.10) \\ \underbrace{E_x^+(z) / \eta_0}_{E_0^+ e^{-jk_0 z}}}} \right] = \vec{a}_y \frac{E_0^+}{\eta_0} \cos(\omega t - k_0 z) \quad (\text{A/m}) \quad (1.12)$$

Comments & Observations

- Since η_0 real, $H_y^+(z)$ in phase with $E_x^+(z)$
- For uniform plane wave, ratio of magnitudes of E & H = intrinsic wave impedance of medium
- E & H perpendicular to each other, both perpendicular to propagation direction
- Showcased E_x & H_y just arbitrary; could have been any other pair of orthogonal directions not necessarily parallel to xy plane, both perpendicular to propagation direction not necessarily along z .

Example: Uniform Plane Wave in Lossless Medium (#1)

Uniform plane wave, $\mathbf{E} = \mathbf{a}_x E_x$, propagates in lossless simple medium ($\epsilon_r = 4$, $\mu_r = 1$, $\sigma = 0$) in +z direction. Assume E_x sinusoidal with freq 100 MHz, max value 10^{-4} V/m at time $t = 0$ & $z = 1/8$ m.

- a) Write instantaneous $\mathbf{E}$ for any t & z .
- b) Write instantaneous $\mathbf{H}$.
- c) Determine locations where E_x a positive max when $t = 10^{-8}$ sec

Solution

Preliminary:

$$\begin{aligned} \text{First, } k &= \omega \sqrt{\mu\epsilon} = 2\pi f \sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0} = 2\pi(100 \times 10^6) \sqrt{\mu_0 \times 4\epsilon_0} = \\ &= 4 \times 10^8 \pi \sqrt{\mu_0 \epsilon_0} = 4 \times 10^8 \pi / c = 4 \times 10^8 \pi / (3 \times 10^8) = 4\pi/3 \text{ rad/m} \end{aligned}$$

Example: Uniform Plane Wave in Lossless Medium (#2)

Solution (a)

Using $\cos \omega t$ as reference, instantaneous E :

$$E(z, t) = a_x E_x(z, t) = a_x \underbrace{10^{-4}}_{\substack{\text{given max} \\ \text{(amplitude)} \\ \text{of cosinusoid}}} \cos(\underbrace{2\pi 10^8 t - kz + \psi}_{\omega})$$

Since $E_x = 10^{-4}$ (max) when argument of cosine function is 0,

i.e., $2\pi 10^8 t - kz + \psi = 0$ when $t = 0$ & $z = 1/8$

$$\Rightarrow \psi = (1/8)k = \boxed{4\pi/3} \times (1/8) = \pi/6 \text{ rad} \quad \xrightarrow{k \text{ found prev slide}}$$

$$\therefore E(z, t) = a_x 10^{-4} \cos[2\pi 10^8 t - (4\pi/3)z + \pi/6]$$

$$= a_x \underbrace{10^{-4} \cos[2\pi 10^8 t - (4\pi/3)(z - 1/8)]}_{E_x(z, t)} \quad (\text{V/m})$$

This expression shows shift of $1/8$ m in $+z$ direction, and could have been written down directly from statement of problem

Example: Uniform Plane Wave in Lossless Medium (#3)

Solution (b)

Phasor $\mathbf{H} = \mathbf{a}_y H_y = \mathbf{a}_y E_x / \eta$, since learned earlier that $\mathbf{H} \perp \mathbf{E}$

where $\eta = \sqrt{\mu/\epsilon} = \eta_0/\sqrt{\epsilon_r} = 120\pi/\sqrt{4} = 60\pi \text{ } (\Omega)$
from solution (a) previous slide
 $E_x(z,t)$

$$\vec{H}(z,t) = \vec{a}_y \underbrace{\frac{1}{60\pi}}_{\eta} 10^{-4} \cos \left[\overbrace{2\pi 10^8 t - \frac{4\pi}{3} \left(z - \frac{1}{8} \right)}^{E_x(z,t)} \right] \quad (\text{A/m})$$

Solution (c)

At $t = 10^{-8}$, equate argument of cosine function to $\pm 2n\pi$ in order to make E_x positive maximum:

$\therefore$ the 2 in $2n\pi$,
 otherwise if abs
 max, then just $n\pi$.

$$2\pi 10^8 (10^{-8}) - \frac{4\pi}{3} \left(z_m - \frac{1}{8} \right) = \pm 2n\pi$$

from which: $z_m = \frac{13}{8} \pm \frac{3}{2}n \text{ (m)}, \quad n = 0, 1, 2, \dots; \quad z_m > 0$

Example: Uniform Plane Wave in Lossless Medium (#4 Further Insights)

- Wavelength in given medium, $\lambda = 2\pi/k = 3/2$
- $\therefore$ positive max of E_x occurs at: $4\pi/3$ from earlier

$$z_m = \frac{13}{8} \pm n \overset{1.5}{\lambda} \quad (\text{m}), \quad n = 0, 1, 2, \dots; \quad z_m > 0$$

- E & H fields shown below as functions of z for reference time $t = 0$.

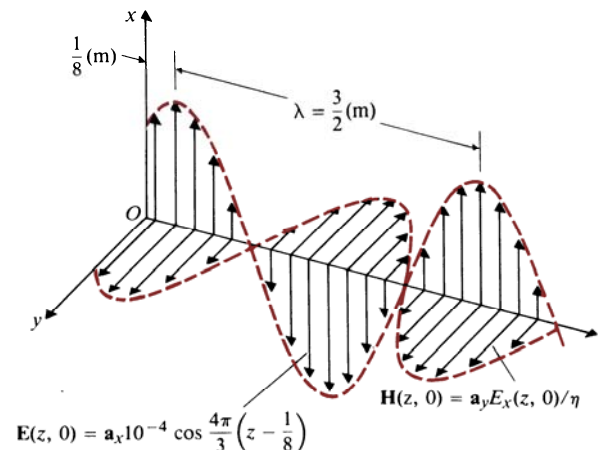


FIGURE 8-2
E and H fields of a uniform plane wave at $t = 0$ (Example 8-1).

Part 1.1

Transverse Electromagnetic (TEM) Waves

Opening Remarks

- Previously, uniform plane wave characterized by $\mathbf{E} = \mathbf{a}_x E_x$ propagating in +z direction with associated $\mathbf{H} = \mathbf{a}_y H_y$.
- $\therefore \mathbf{E} \perp \mathbf{H} \perp$ (*transverse* to) propagation direction
- $\therefore$ a particular case of *transverse electromagnetic (TEM)* wave
- Phasor fields functions of only distance z along single coordinate axis (propagation direction)
- Now consider uniform plane wave propagation along arbitrary direction not necessarily coincident with any coordinate axis

Expression of $\vec{E}$ for General Plane Wave

Phasor $\vec{E}$ for uniform plane wave propagating in +z direction:

$$\vec{E}(z) = \underbrace{\vec{E}_0}_{\substack{\text{const} \\ \text{vector}}} e^{-jkz} \quad (1.13)$$

More general form of (1.13):

$$\vec{E}(x, y, z) = \underbrace{\vec{E}_0}_{\substack{\text{const} \\ \text{vector}}} e^{-jk_x x - jk_y y - jk_z z} \quad (1.14)$$

Easily proved by direct substitution that this (1.14) satisfies homogeneous Helmholtz's equation (7.18): $(\nabla^2 + k^2)\vec{E} = 0$, provided

$$k^2 = k_x^2 + k_y^2 + k_z^2 = \omega^2 \mu \epsilon. \quad (1.15)$$

Wavenumber Vector & Phase-Front

- Define

- *wavenumber vector*: $\mathbf{k} = \mathbf{a}_x k_x + \mathbf{a}_y k_y + \mathbf{a}_z k_z = k \mathbf{a}_n$. (1.16)

- radius vector from origin: $\mathbf{R} = \mathbf{a}_x x + \mathbf{a}_y y + \mathbf{a}_z z$. (1.17)

- Eq. (1.14) written compactly as:

$$\vec{E}(\vec{R}) = \overbrace{\vec{E}_0}^{\text{const vector}} e^{-j\vec{k} \cdot \vec{R}} = \vec{E}_0 e^{-j\overbrace{k\vec{a}_n}^{\vec{k}} \cdot \vec{R}} \quad (\text{V/m}) \quad (1.18)$$

- Wavenumber components:

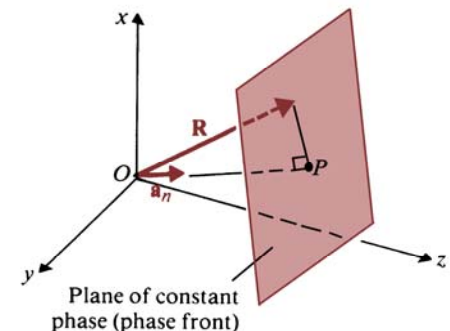
- $k_x = \mathbf{k} \cdot \mathbf{a}_x = k \mathbf{a}_n \cdot \mathbf{a}_x$,

- $k_y = \mathbf{k} \cdot \mathbf{a}_y = k \mathbf{a}_n \cdot \mathbf{a}_y$,

- $k_z = \mathbf{k} \cdot \mathbf{a}_z = k \mathbf{a}_n \cdot \mathbf{a}_z$,

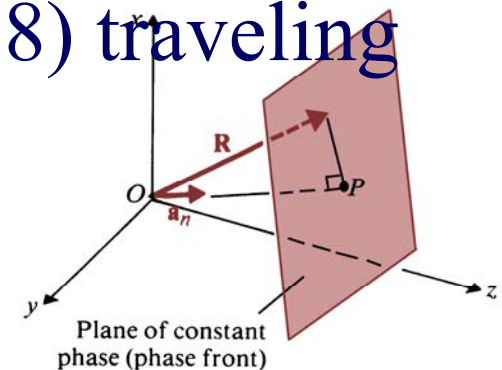
$\mathbf{a}_n$ = unit vector in direction of propagation, $\perp$ planar surface of constant phase (wavefront)

direction cosines of $\mathbf{a}_n$.



Further Notes

- $\underbrace{a_n \cdot R = \text{length } \overline{OP}}_{\text{equation of plane normal to } a_n \rightarrow \text{propagation direction}}$ (a constant).
- $z = \text{constant} \rightarrow$ plane of constant phase & uniform amplitude (*phase-front* or *wave-front*) for plane wave in (1.13) propagating along z direction
- $a_n \cdot R = \text{constant} \rightarrow$ plane of constant phase & uniform amplitude for wave in (1.18) traveling along a_n direction



Use of Vector Identity

- From identity: $\nabla \cdot (\psi \mathbf{A}) = \psi \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla \psi$, let $\mathbf{E}(\mathbf{R}) = \mathbf{E}_0 \exp(-j\mathbf{k} \cdot \mathbf{R})$ of (1.18) be $\psi \mathbf{A}$, i.e.
 - $\psi = \exp(-j\mathbf{k} \cdot \mathbf{R})$
 - $\mathbf{A} = \mathbf{E}_0$ $0 \because \mathbf{E}_0 \text{ const}$
- $\therefore \nabla \cdot \mathbf{E}(\mathbf{R}) = \exp(-j\mathbf{k} \cdot \mathbf{R}) \nabla \cdot \mathbf{E}_0 + \mathbf{E}_0 \cdot \nabla \exp(-j\mathbf{k} \cdot \mathbf{R})$,
(1.19)
- Since in charge-free region, i.e. $\rho = 0$, $\therefore$ Gauss' Law: $\nabla \cdot \mathbf{E} = \rho/\epsilon = 0 \rightarrow$ LHS of (1.19) above = 0
- $\therefore \mathbf{E}_0 \cdot \nabla \exp(-j\mathbf{k} \cdot \mathbf{R}) = 0$ (1.20)

E-Field Perpendicular to Propagation Direction

$$\begin{aligned}\nabla(e^{-j\overbrace{\vec{k}}^{\vec{k}}\vec{a}_n\cdot\vec{R}}) &= \left(\vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z} \right) e^{-j(k_x x + k_y y + k_z z)} \\ &= -j \underbrace{(\vec{a}_x k_x + \vec{a}_y k_y + \vec{a}_z k_z)}_{k\vec{a}_n = \vec{k}} e^{-j(k_x x + k_y y + k_z z)} = -jk\vec{a}_n e^{-j\overbrace{\vec{k}}^{\vec{k}}\vec{a}_n\cdot\vec{R}}\end{aligned}$$

$\therefore$ (1.20) written as:

$$-jk(\overbrace{\vec{E}_0}^{\text{const vector}} \cdot \vec{a}_n) e^{-j\overbrace{\vec{k}}^{\vec{k}}\vec{a}_n\cdot\vec{R}} = 0$$

which requires $\mathbf{a}_n \cdot \mathbf{E}_0 = 0$ **(1.21)**

$\Rightarrow$ plane-wave solution in (1.18) implies $\mathbf{E}_0$ transverse ($\perp$) to propagation direction $\mathbf{a}_n$.

Associated H -Field

From (7.14) in Chapter 1: $\nabla \times \vec{E} = -j\omega\mu\vec{H}$ **(7.14)repeated**

$$\begin{aligned}
 \vec{H}(\vec{R}) &= -\frac{1}{j\omega\mu} \nabla \times \vec{E}(\vec{R}) = -\frac{1}{j\omega\mu} \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} \times \begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix} e^{-j(k_x x + k_y y + k_z z)} \\
 &= -\frac{1}{j\omega\mu} \begin{pmatrix} -jk_y E_{0z} + jk_z E_{0y} \\ -jk_z E_{0x} + jk_x E_{0z} \\ -jk_x E_{0y} + jk_y E_{0x} \end{pmatrix} e^{-j(k_x x + k_y y + k_z z)} = \frac{1}{\omega\mu} \underbrace{\begin{pmatrix} k_y E_{0z} - k_z E_{0y} \\ k_z E_{0x} - k_x E_{0z} \\ k_x E_{0y} - k_y E_{0x} \end{pmatrix}}_{\begin{pmatrix} k_x \\ k_y \\ k_z \end{pmatrix} \times \begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix}} e^{-j(k_x x + k_y y + k_z z)} \\
 &= \frac{1}{\omega\mu} \underbrace{\begin{pmatrix} k_x \\ k_y \\ k_z \end{pmatrix}}_{\vec{k}} \times \underbrace{\begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix}}_{\vec{E}(\vec{R})} e^{-j(k_x x + k_y y + k_z z)} = \frac{1}{\omega\mu} \underbrace{\vec{k}}_{\text{intrinsic impedance of medium}} \vec{a}_n \times \vec{E}(\vec{R}) = \frac{\overbrace{\omega\sqrt{\mu\epsilon}}^k}{\underbrace{\omega\mu}_{\frac{1}{\sqrt{\mu/\epsilon}} = \frac{1}{\eta}}}} \vec{a}_n \times \vec{E}(\vec{R}) = \frac{1}{\eta} \vec{a}_n \times \vec{E}(\vec{R})
 \end{aligned}$$

$\therefore \vec{H}(\vec{R}) = \frac{1}{\eta} \vec{a}_n \times \vec{E}(\vec{R}) \quad (\text{A/m}) \quad (1.22)$

$\eta = \omega\mu/k = \sqrt{\mu/\epsilon} \quad (\Omega) \quad (1.23)$

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Uniform Plane Wave is TEM

Substitute (1.18) into (1.22):

$$\vec{H}(\vec{R}) = \frac{1}{\eta} \vec{a}_n \times \overbrace{\vec{E}_0}^{\vec{E}(\vec{R})} e^{-j \overbrace{k \vec{a}_n \cdot \vec{R}}^{\vec{k} \cdot \vec{R}}} \quad (1.24)$$

$\therefore$ uniform plane wave propagating in arbitrary direction, $\mathbf{a}_n$, is TEM wave with $\mathbf{E} \perp \mathbf{H}$, both normal to $\mathbf{a}_n$.

Example: Express E of TEM Wave

If $E(\mathbf{R})$ of TEM wave given, as in (1.18), $H(\mathbf{R})$ found by (1.22). Obtain relation expressing $E(\mathbf{R})$ in terms of $H(\mathbf{R})$.

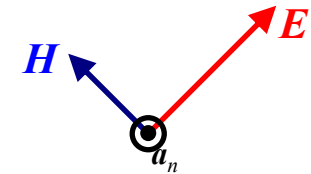
Solution

Assume $H(\mathbf{R})$ of the form: $\vec{H}(\vec{R}) = \vec{H}_0 e^{-j\vec{k}\vec{a}_n \cdot \vec{R}}$ (1.25)

From (7.15) in Chapter 1: $\nabla \times \vec{H} = j\omega\epsilon\vec{E}$ (7.15)repeated

$$\vec{E}(\vec{R}) = \frac{1}{j\omega\epsilon} \nabla \times \vec{H}(\vec{R}) = \frac{1}{j\omega\epsilon} \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} \times \begin{pmatrix} H_{0x} \\ H_{0y} \\ H_{0z} \end{pmatrix} e^{-j(k_x x + k_y y + k_z z)}$$

$$= \frac{1}{j\omega\epsilon} \begin{pmatrix} -jk_y H_{0z} + jk_z H_{0y} \\ -jk_z H_{0x} + jk_x H_{0z} \\ -jk_x H_{0y} + jk_y H_{0x} \end{pmatrix} e^{-j(k_x x + k_y y + k_z z)} = -\frac{1}{\omega\epsilon} \underbrace{\begin{pmatrix} k_y H_{0z} - k_z H_{0y} \\ k_z H_{0x} - k_x H_{0z} \\ k_x H_{0y} - k_y H_{0x} \end{pmatrix}}_{\begin{pmatrix} k_x \\ k_y \\ k_z \end{pmatrix} \times \begin{pmatrix} H_{0x} \\ H_{0y} \\ H_{0z} \end{pmatrix}} e^{-j(k_x x + k_y y + k_z z)}$$



could have also been directly obtained by intuitive deduction from above triple-vector diagram relating E, H & propag. direction

$$= -\frac{1}{\omega\epsilon} \underbrace{\begin{pmatrix} k_x \\ k_y \\ k_z \end{pmatrix}}_{\vec{k}} \times \underbrace{\begin{pmatrix} H_{0x} \\ H_{0y} \\ H_{0z} \end{pmatrix}}_{\vec{H}(\vec{R})} e^{-j(k_x x + k_y y + k_z z)} = -\frac{1}{\omega\epsilon} \vec{k} \times \vec{H}(\vec{R}) = -\frac{\omega\sqrt{\mu\epsilon}}{\omega\epsilon} \vec{a}_n \times \vec{H}(\vec{R}) = -\eta \vec{a}_n \times \vec{H}(\vec{R}) \quad (1.26)$$

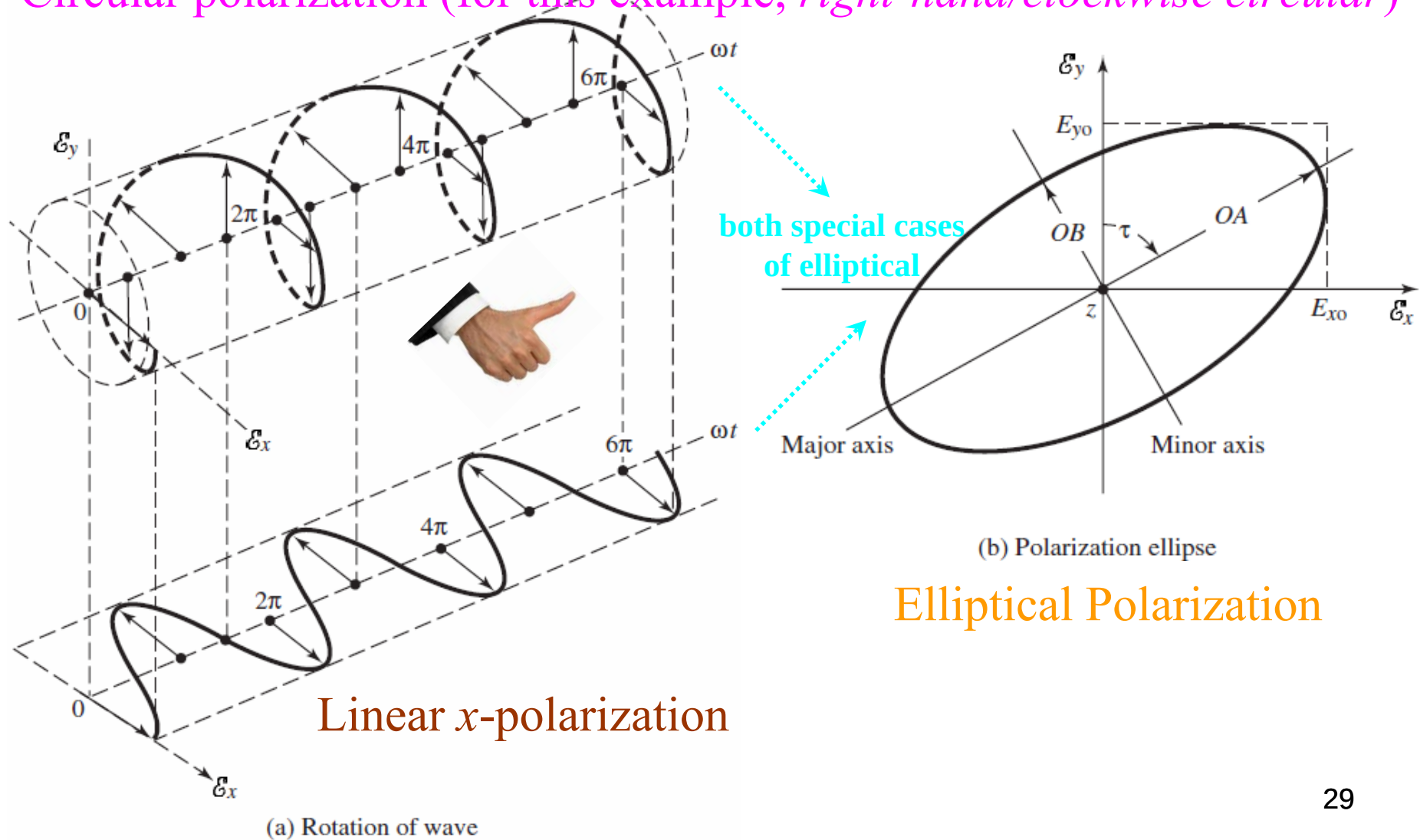
Part 2: Polarization

Definitions

- Polarization of *antenna* in given direction defined as: *polarization of wave* transmitted by antenna (when direction not stated, assumed that of max gain)
- Polarization of *radiated wave* defined as:
 - figure traced as function of time by extremity of time-varying E -field vector at fixed location in space, and the *sense* (*clockwise or anticlockwise*) in which it is traced, as observed along propagation direction
 - Curve traced by end-point of arrow (vector) representing instantaneous E -field

Three Main Types of Polarization

Circular polarization (for this example, *right-hand/clockwise circular*)



Part 2.1: Linear Polarization

Co & Cross Polar Components

Opening Remarks

- E & H fields have both x & y comps – EM wave said to be polarized
 - can simultaneously modulate different signals on E_x & E_y at same frequency
 - reuse frequencies in comms system: double capacity
 - Can use any pair of orthogonal polarizations, not only x & y → transmit & receive antennas must be designed for good isolation between both polarizations, minimal mutual interference
- Co (desired) & cross (undesired) polar components
 - the one excited is co-polar
 - the other is cross-polar

Co- & Cross-Polar Field Components

- Co-polar (desired) field component E_{co} parallel with co-polar unit vector $\hat{co}$
- Cross-polar (undesired) field component E_{xp} parallel with cross-polar unit vector $\hat{xp}$, orthogonal to $\hat{co}$
- Both $\hat{co}$ & $\hat{xp}$ orthogonal (*transverse*) to propag direction, for present case, z-direction (but applies to all other propag directions)

Total E -field stated as: $E = (E_{co}\hat{co} + E_{xp}\hat{xp})e^{-jkz}$ (2.11)

Co & cross polar unit vectors generally as orthonormal complex vectors

$$|\hat{co}|^2 = \hat{co} \cdot \hat{co}^* = 1 \quad \hat{co} \cdot \hat{xp}^* = 0 \quad (2.12)$$

$$|\hat{xp}|^2 = \hat{xp} \cdot \hat{xp}^* = 1 \quad \hat{xp} \cdot \hat{co}^* = 0 \quad (2.13)$$

Co & cross polar components of *transverse* E -field (E_t)

$$E_{co} = E_t \cdot \hat{co}^*; \quad E_{xp} = E_t \cdot \hat{xp}^* \quad (2.14)$$

Linear Polarization

When desired E -field y -directed: linearly y -polarized:

$$\hat{co} = \hat{y}; \quad \hat{xp} = \hat{x} \quad (2.15)$$

$$E_{co} = \mathbf{E}_t \cdot \hat{y}^* = \mathbf{E}_t \cdot \hat{y} = E_y \quad (2.16)$$

$$E_{xp} = \mathbf{E}_t \cdot \hat{x}^* = \mathbf{E}_t \cdot \hat{x} = E_x \quad (2.17)$$

(We could also have chosen $\hat{xp} = -\hat{x}$ or more generally $\hat{xp} = e^{j\phi}\hat{x}$ with ϕ on arbitrary phase.)

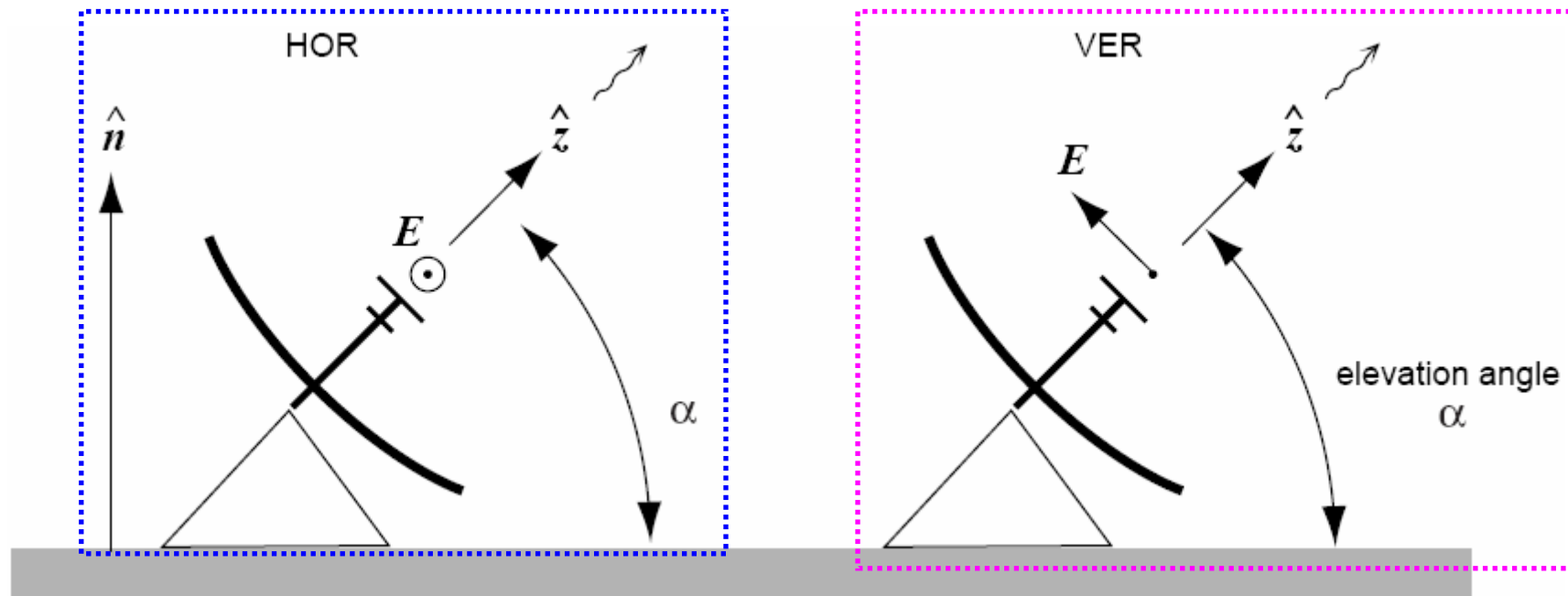
For co-polar linear x -polarization, $\hat{co} = \hat{x}; \quad \hat{xp} = -\hat{y} \quad (2.18)$

(We could also here have chosen $\hat{xp} = \hat{y}$ or $\hat{xp} = e^{j\phi}\hat{y}$ in the same way as for the co-polar y -polarization.)

Vertical & Horizontal Polarizations

Horizontal (or perpendicular):

$\mathbf{E} \perp \hat{\mathbf{n}} \perp \hat{\mathbf{z}}$ where $\hat{\mathbf{n}}$ is the normal to the ground and $\hat{\mathbf{z}}$ is the direction of propagation



Vertical (or parallel):

E -field not necessarily normal to ground, i.e. normally do not require $\mathbf{E} \perp \hat{\mathbf{n}}$ but rather that $\mathbf{E}$ lies in plane defined by $\hat{\mathbf{z}}$ & $\hat{\mathbf{n}}$.

Arbitrary Polarization

- Generally, arbitrary linear polarization defined by co & cross polar unit vectors of the form:

$$\hat{co} = \cos \xi \hat{x} + \sin \xi \hat{y} \quad (2.19)$$

$$\hat{xp} = \sin \xi \hat{x} - \cos \xi \hat{y} \quad (2.20)$$

- When $\xi = \pi/2 \rightarrow y$ -polarization
- When $\xi = 0 \rightarrow x$ -polarization

when ξ anywhere in between $\rightarrow$ tilted polarization (neither purely y nor purely x polarized), with co-polar field having both x & y components (some blend/mixture of both), and likewise for cross-polar field, being perpendicular to co-polar field

Part 2.2: Circular Polarization

General Features

- Plane wave whose E -field rotates angle 2π in xy plane as time passes through period T .
more generally, transverse plane perpendicular to propagation direction (here, z direction of propagation)
- Rotation may either be observed
 - at given position in space as function of increasing time
 - at given time as function of space: propag z -direction
- Both these observation methods give different rotation directions, right or left hand (fingers)
- 1st observation method (function of time) chosen for *definition by convention* (see diagram later)

Right-Hand Circular (RHC) and its Co-Polar Unit Vector

Define right-hand circular (RHC) polarization for waves propagating in z-direction by unit-vector: $\hat{c}_o = (\hat{x} - j\hat{y}) / \sqrt{2}$ (2.21)

which means y-comp has phase factor $-j = e^{-j\pi/2}$ compared to x-comp

In other words, y-comp of field delayed by $t = T/4$ (quarter period) compared to x-comp

Notice that $|c_o| = \sqrt{\left(1/\sqrt{2}\right)^2 + \left|(-j/\sqrt{2})\right|^2} = 1$ as required

RHC-Polarized Plane Wave

Study RHC-polarized plane-wave:

$$\mathbf{E} = \hat{\mathbf{c}} \mathbf{o} e^{-jkz} = (\hat{\mathbf{x}} - j\hat{\mathbf{y}})e^{-jkz} / \sqrt{2} \quad (2.22)$$

Time-varying field:

$$\vec{E}(z, t) = \text{Re}\{\mathbf{E}e^{j\omega t}\} = E_x(z, t)\hat{\mathbf{x}} + E_y(z, t)\hat{\mathbf{y}} \quad (2.23)$$

with

$$E_x(z, t) = \text{Re}\{\mathbf{E} \cdot \hat{\mathbf{x}} e^{j\omega t}\} = \cos(\omega t - kz) / \sqrt{2} \quad (2.24a)$$

$$E_y(z, t) = \text{Re}\{\mathbf{E} \cdot \hat{\mathbf{y}} e^{j\omega t}\} = \cos\left(\omega t - \frac{\pi}{2} - kz\right) / \sqrt{2} =$$

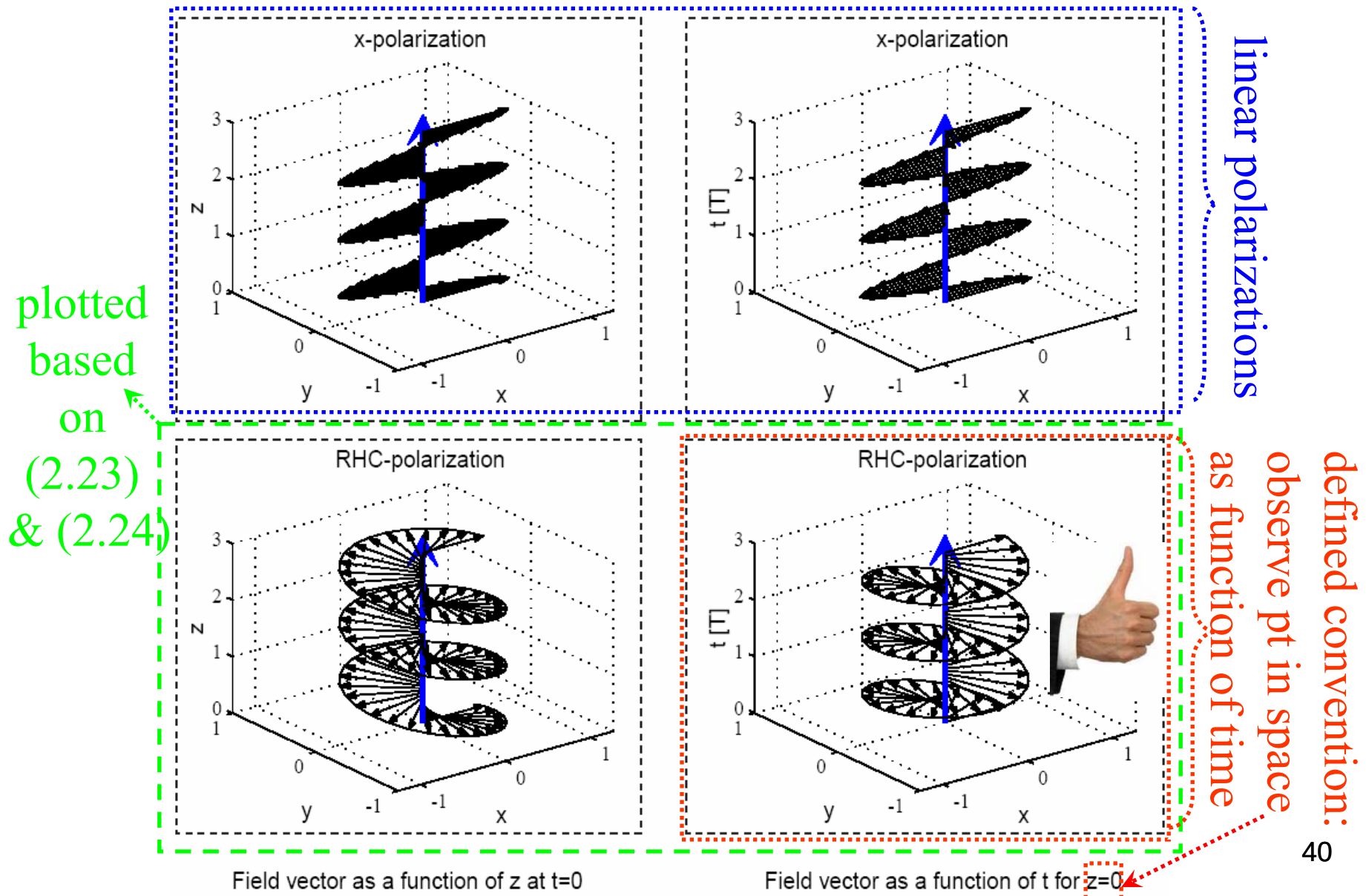
since $\omega = 2\pi f = 2\pi/T, \therefore \omega T/4 = \pi/2$

$$= \cos\left(\omega\left(t - \frac{T}{4}\right) - kz\right) / \sqrt{2} = \cos\left(\omega t - k\left(z + \frac{\lambda}{4}\right)\right) / \sqrt{2} \quad (2.24b)$$

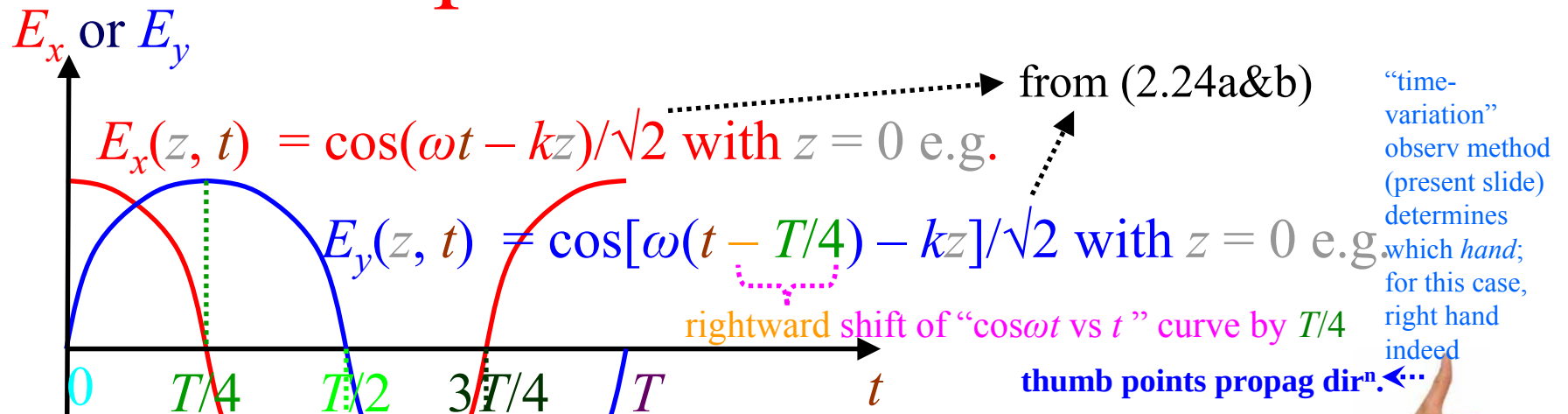
↘ $2\pi/\lambda$

This $E(z, t)$ of above (2.23) plotted in following diagram as function of i) z for $t = 0$, and ii) t for $z = 0$

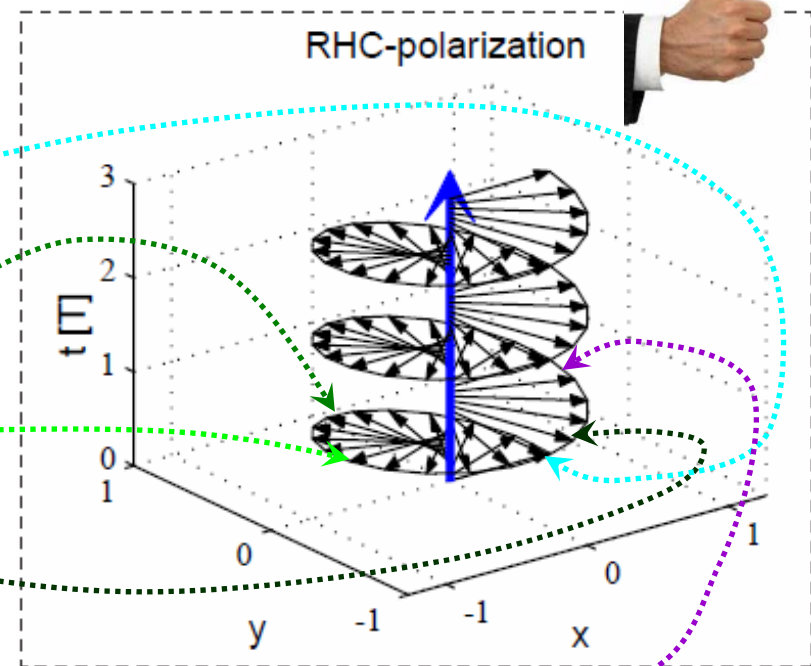
Illustration – RHC



Detailed Explanation: Time Variation

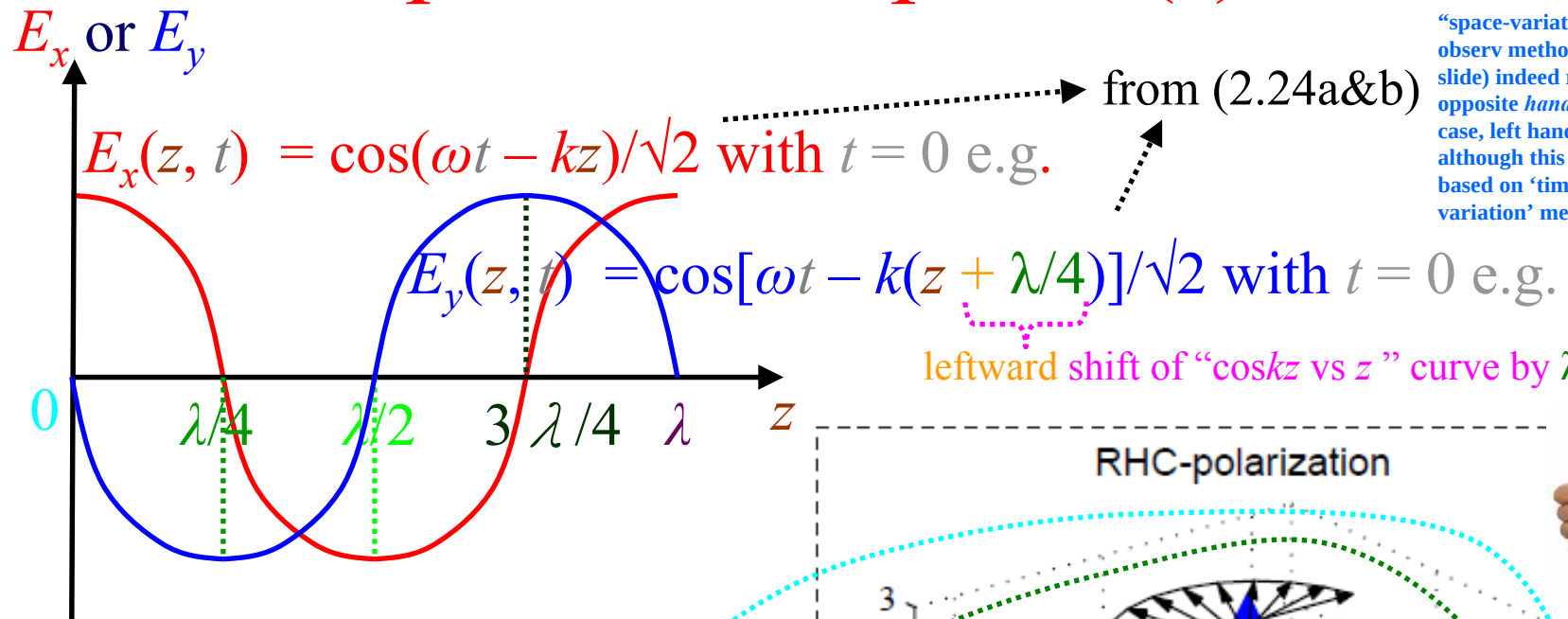


- At $t = 0$, $E_x = \text{pos max}$, $E_y = 0$
- At $t = T/4$, $E_x = 0$, $E_y = \text{pos max}$
- At $t = T/2$, $E_x = \text{neg max}$, $E_y = 0$
- At $t = 3T/4$, $E_x = 0$, $E_y = \text{neg max}$
- At $t = T$, $E_x = \text{pos max}$, $E_y = 0$

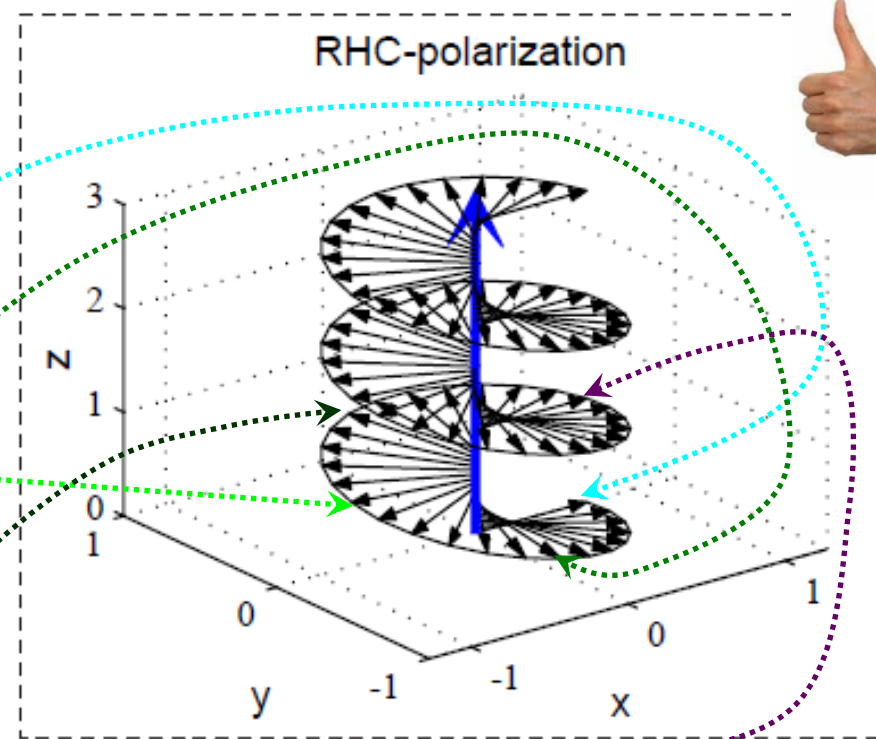


Field vector as a function of t for $z=0$

Detailed Explanation: Spatial (z) Variation



- At $z = 0$, $E_x = \text{pos max}$, $E_y = 0$
- At $z = \lambda/4$, $E_x = 0$, $E_y = \text{neg max}$
- At $z = \lambda/2$, $E_x = \text{neg max}$, $E_y = 0$
- At $z = 3\lambda/4$, $E_x = 0$, $E_y = \text{pos max}$
- At $z = \lambda$, $E_x = \text{pos max}$, $E_y = 0$



Cross-Polar Unit Vector (of RHC)

Cross-polar unit vector corresponding to $\hat{c}o$ in (2.21)

$$\hat{x}p = (\hat{x} + j\hat{y})/\sqrt{2} \quad (2.25)$$

(We could also have chosen $\hat{x}p = (\hat{y} - j\hat{x})/\sqrt{2}$ or (2.25) multiplied with any complex constant of unit amplitude).

When propagating E -field of form (2.11): $E = (E_{co}\hat{c}o + E_{xp}\hat{x}p)e^{-jkz}$

complex amplitudes of co & cross polar fields are derived in the following two slides ...

Co-Polar Field (for RHCP)

$$\begin{aligned}
 E_{co} &= \underbrace{\vec{E}}_{\substack{\text{of} \\ (2.11)}} \cdot \underbrace{\hat{c}\hat{o}}_{\substack{\text{of} \\ (2.21)}}^* = \left[\left(E_{co} \frac{\hat{x} - jy}{\sqrt{2}} + E_{xp} \frac{\hat{x} + jy}{\sqrt{2}} \right) e^{-jkz} \right] \cdot \left(\frac{\hat{x} - jy}{\sqrt{2}} \right)^* = \\
 &= \frac{e^{-jkz}}{\sqrt{2}\sqrt{2}} \underbrace{\begin{pmatrix} E_{co} + E_{xp} \\ -j(E_{co} - E_{xp}) \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ j \\ 0 \end{pmatrix}}_{2E_{co}} = \boxed{E_{co} e^{-jkz}} = \\
 &= \frac{E_{co} \cancel{(2/\sqrt{2})} e^{-jkz}}{\cancel{\sqrt{2}}} = \frac{\overbrace{\left[(E_{co} + E_{xp}) e^{-jkz} / \sqrt{2} \right] + \left[(E_{co} - E_{xp}) e^{-jkz} / \sqrt{2} \right]}^{2E_{co} e^{-jkz} / \sqrt{2}}}{\sqrt{2}} = \\
 &= \frac{\overbrace{\left[(E_{co} + E_{xp}) e^{-jkz} / \sqrt{2} \right]}^{E_x} + j \overbrace{\left[-j(E_{co} - E_{xp}) e^{-jkz} / \sqrt{2} \right]}^{E_y}}{\sqrt{2}} = \boxed{\frac{E_x + jE_y}{\sqrt{2}}} \quad (2.26)
 \end{aligned}$$

seems pretty 'trivial' or 'obvious', but the rigorous math details to prove it are not that 'trivial'

and this final form is certainly not very obvious at all

Cross-Polar Field (for RHCP)

$$\begin{aligned}
 E_{xp} &= \underbrace{\vec{E}}_{\substack{\text{of} \\ (2.11)}} \cdot \underbrace{x\hat{p}}_{\substack{\text{of} \\ (2.25)}}^* = \left[\left(E_{co} \frac{x - jy}{\sqrt{2}} + E_{xp} \frac{x + jy}{\sqrt{2}} \right) e^{-jkz} \right] \cdot \left(\frac{x + jy}{\sqrt{2}} \right)^* = \\
 &= \frac{e^{-jkz}}{\sqrt{2}\sqrt{2}} \underbrace{\begin{pmatrix} E_{co} + E_{xp} \\ -j(E_{co} - E_{xp}) \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -j \\ 0 \end{pmatrix}}_{2E_{xp}} = \boxed{E_{xp} e^{-jkz}} = \\
 &= \frac{E_{xp} \cancel{(2/\sqrt{2})} e^{-jkz}}{\cancel{\sqrt{2}}} = \frac{\overbrace{\left[(E_{co} + E_{xp}) e^{-jkz} / \sqrt{2} \right] - \left[(E_{co} - E_{xp}) e^{-jkz} / \sqrt{2} \right]}^{2E_{xp} e^{-jkz} / \sqrt{2}}}{\sqrt{2}} = \\
 &= \frac{\overbrace{\left[(E_{co} + E_{xp}) e^{-jkz} / \sqrt{2} \right]}^{E_x} - j \overbrace{\left[-j(E_{co} - E_{xp}) e^{-jkz} / \sqrt{2} \right]}^{E_y}}{\sqrt{2}} = \boxed{\frac{E_x - jE_y}{\sqrt{2}}} \quad (2.27)
 \end{aligned}$$

seems pretty 'trivial' or 'obvious', but the rigorous math details to prove it are not that 'trivial'

and this final form is certainly not very obvious at all

Complex values in terms of amplitudes & phases

$$E_x = |E_x|e^{j\phi_x}; \quad E_y = |E_y|e^{j\phi_y} \quad (2.28)$$

Must Dot with Conjugate of Unit Vectors

Noting that

$$\underbrace{\left(\frac{x - jy}{\sqrt{2}}\right)}_{\hat{c}o} \cdot \underbrace{\left(\frac{x - jy}{\sqrt{2}}\right)^*}_{\hat{c}o^*} = \left(\frac{x - jy}{\sqrt{2}}\right) \cdot \left(\frac{x + jy}{\sqrt{2}}\right) = \frac{1}{2} \underbrace{\begin{pmatrix} 1 \\ -j \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ j \\ 0 \end{pmatrix}}_{1 + \underbrace{(-j)j}_{=2}} = 1$$

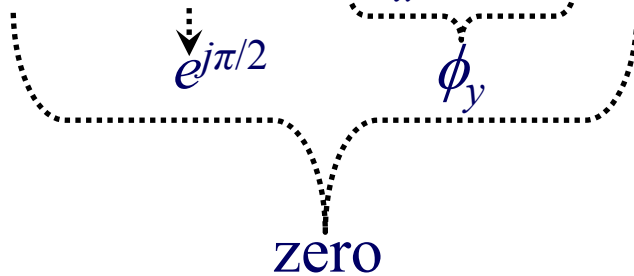
and

$$\underbrace{\left(\frac{x + jy}{\sqrt{2}}\right)}_{\hat{x}p} \cdot \underbrace{\left(\frac{x + jy}{\sqrt{2}}\right)^*}_{\hat{x}p^*} = \left(\frac{x + jy}{\sqrt{2}}\right) \cdot \left(\frac{x - jy}{\sqrt{2}}\right) = \frac{1}{2} \underbrace{\begin{pmatrix} 1 \\ j \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -j \\ 0 \end{pmatrix}}_{1 + \underbrace{j(-j)}_{=2}} = 1$$

RHCP by Two Orthogonal Equal Linear Polarizations

If $|E_x| = |E_y|$ and $\phi_y - \phi_x = -\pi/2$, $E_{xp} = 0$, since from (2.27) $E_{xp} = \frac{E_x - jE_y}{\sqrt{2}}$

$$E_{xp} = |E_x| (e^{j\phi_x} - j \exp[j(\phi_x - \pi/2)]) / \sqrt{2} = 0$$



RHC polarized wave (without any cross-polar component: $E_{xp} = 0$)

can be excited by two orthogonal but otherwise equal linearly-polarized antennas

in terms of both direction & phase; for the latter, $\phi_y = \phi_x - \pi/2$

- Assume two antennas x & y polarized
- 90° delay on y -polarized antenna relative to x -polarized one obtained by connecting x polarized antenna to in-phase (0°) & y -polarized one to quadrature (−90°) outputs of 3dB hybrid power divider

Left-Hand Circular (LHC) Polarization

Correspondingly, for LHC polarization, co & cross polar unit vectors:

$$\hat{co} = (\hat{x} + j\hat{y}) / \sqrt{2} \text{ and} \quad (2.29)$$

$$\hat{xp} = (\hat{x} - j\hat{y}) / \sqrt{2} \quad (2.30)$$

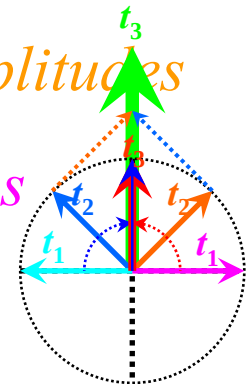
Elliptical Polarization

$$\hat{co} = [\hat{x} + Ae^{j\Delta\phi}\hat{y}] / \sqrt{1 + A^2} \quad (2.31)$$

$$\hat{xp} = [-Ae^{-j\Delta\phi}\hat{x} + \hat{y}] / \sqrt{1 + A^2} \quad (2.32)$$

where A & $\Delta\phi$ are real constants

- However, elliptical polarization of **no practical interest** as desired polarization, although **in practice**, **linearly & circularly** polarized fields often **elliptically polarized**, as **cross-pol** never identically zero
- Obtained by **adding** either
 - two *orthogonal circular* polarizations with **different amplitudes**
 - two *orthogonal linear* polarizations with **different phases**
- **Adding** two orthog circ pol with **equal amp** gives **linear pol**
- **Adding** 2 orthog lin pol with **90° phase diff & equal amp** gives **circ pol**



Part 3: Plane Waves in Lossy Media

Complex Wavenumber k_c .

- In source-free lossy medium, homogeneous vector Helmholtz's equation to be solved:
$$\nabla^2 \mathbf{E} + k_c^2 \mathbf{E} = 0 \quad (3.1)$$
- where $k_c = \omega \sqrt{\mu \epsilon_c}$ = complex wavenumber, as of (7.31) in Chapter 1
- Derivations & discussions of plane-waves in lossless medium in Part 1 of present chapter modified to apply to wave propagation in lossy medium by simply replacing k with k_c .

Complex Propagation Constant

- Define propagation constant: $\gamma = jk_c = j\omega\sqrt{\mu\epsilon_c}$ (3.2)
- Since γ complex, write using (7.27) of Chapter 1

$$\gamma \underset{\text{define as}}{=} \alpha + j\beta \underset{\substack{\text{from (3.2) above}}}{=} j\omega \underbrace{\sqrt{\mu\epsilon_c}}_{k_c} \overset{\substack{\text{use (7.27) Chapter 1}}}{=} j\omega \underbrace{\sqrt{\mu\epsilon \left(1 - j\frac{\sigma}{\omega\epsilon}\right)}}_{\sqrt{\mu\epsilon - j\mu\sigma/\omega}} = j\omega\sqrt{\mu\epsilon} \sqrt{1 + \frac{\sigma}{j\omega\epsilon}} \quad (3.3)$$

- From (7.31) of Chapter 1, repeated: $k_c = \omega\sqrt{\mu\epsilon_c} = \omega\sqrt{\mu(\epsilon' - j\epsilon'')}$,

$$\gamma = \alpha + j\beta \underset{\substack{\text{from (3.2) above}}}{=} j \underbrace{k_c}_{\omega\sqrt{\mu\epsilon_c}} \overset{\substack{\text{use (7.31) Chapter 1}}}{=} j\omega \underbrace{\sqrt{\mu(\epsilon' - j\epsilon'')}}_{\sqrt{\mu\epsilon' \left(1 - j\frac{\epsilon''}{\epsilon'}\right)}} = j\omega\sqrt{\mu\epsilon'} \sqrt{1 - j\frac{\epsilon''}{\epsilon'}} \quad (3.4)$$

Lossless media: $\sigma = 0$ ($\epsilon'' = 0$, $\epsilon = \epsilon'$), $\alpha = 0$, $\beta = k = \omega\sqrt{\mu\epsilon}$

Attenuation & Phase Constants

- Helmholtz's Eq. (3.1) becomes: $\nabla^2 \mathbf{E} - \gamma^2 \mathbf{E} = 0$ (3.5)
- Solution of this (3.5) in form of plane-wave propagating in +z direction: $\mathbf{E} = \mathbf{a}_x E_x = \mathbf{a}_x E_0 e^{-\gamma z}$, (3.6) assuming wave linearly-polarized in x-direction
- Propagation factor: $e^{-\gamma z}$, as product of 2 factors: $E_x = E_0 e^{-\alpha z} e^{-j\beta z}$,
 - $e^{-\alpha z}$: decreases as z increases → *attenuation factor*; with $\alpha = \textbf{attenuation constant}$ (Neper per meter, i.e. Np/m)
 - $e^{-j\beta z}$: phase factor; with $\beta = \textbf{phase constant}$ (rad/m) → expresses amount of phase shift that occurs as wave travels one meter

Part 3.1: Low-Loss Dielectrics

Propag. Constant (low-loss dielectrics)

- Low-loss dielectrics: good but imperfect insulators with non-zero equivalent conductivity
- $\varepsilon'' \ll \varepsilon'$ [see (3.4)] or $\sigma/\omega\varepsilon \ll 1$ [see (3.3)]
- γ of (3.4) approximated by binomial expansion:

$$\begin{aligned}
 \gamma = \alpha + j\beta &= j\omega\sqrt{\mu\varepsilon'} \underbrace{\left\{1 + \left[-\left(j\varepsilon''/\varepsilon'\right)\right]\right\}^{1/2}}_{\text{binom} \updownarrow \text{expan}} \\
 &= j\omega\sqrt{\mu\varepsilon'} \left[1 + \underbrace{\left(\frac{\frac{1}{2}}{1}\right)\left(-j\frac{\varepsilon''}{\varepsilon'}\right)}_{-j\frac{\varepsilon''}{2\varepsilon'}} + \underbrace{\left(\frac{\frac{1}{2}}{2}\right)\left(-j\frac{\varepsilon''}{\varepsilon'}\right)^2}_{\frac{1}{2}\left(\frac{-1}{2}\right)\frac{1}{2!}\left(\frac{\varepsilon''}{\varepsilon'}\right)^2} + \dots \right] \\
 &\cong j\omega\sqrt{\mu\varepsilon'} \left[1 - j\frac{\varepsilon''}{2\varepsilon'} + \frac{1}{8}\left(\frac{\varepsilon''}{\varepsilon'}\right)^2 \right] = \underbrace{\omega\sqrt{\mu\varepsilon'} \frac{\varepsilon''}{2\varepsilon'}}_{\substack{\text{(3.5a)} \\ \frac{\omega\varepsilon''}{2} \sqrt{\frac{\mu}{\varepsilon'}} \\ \text{approx} \\ \alpha}} + j\omega\sqrt{\mu\varepsilon'} \underbrace{\left[1 + \frac{1}{8}\left(\frac{\varepsilon''}{\varepsilon'}\right)^2 \right]}_{\substack{\text{approx} \\ \beta} \text{(3.5b)}} \quad \text{(3.5)}
 \end{aligned}$$

Attenuation & Phase Constants of Low-Loss Dielectrics

Eqs. (3.5a) & (3.5b) give approximations for attenuation & phase constants:

$$\alpha \cong \frac{\omega \varepsilon''}{2} \sqrt{\frac{\mu}{\varepsilon'}} \quad (\text{Np/m}) \quad \text{(3.5a) repeat}$$

- positive (> 0)
- $\approx$ directly prop. to freq $f = \omega/(2\pi)$

$$\beta \cong \omega \sqrt{\mu \varepsilon'} \left[1 + \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'} \right)^2 \right] \quad (\text{rad/m}) \quad \text{(3.5b) repeat}$$

Deviates only slightly from $\omega \sqrt{\mu \varepsilon}$ for perfect lossless dielectric

Intrinsic Impedance (low-loss dielectric)

$$\begin{aligned}
 \eta_c &= \sqrt{\frac{\mu}{\epsilon_c}} \stackrel{\epsilon_c = \epsilon' - j\epsilon''}{=} \sqrt{\frac{\mu}{\epsilon' \left(1 - j\frac{\epsilon''}{\epsilon'}\right)}} \\
 &= \sqrt{\frac{\mu}{\epsilon'}} \underbrace{\left\{1 + \left[-\left(j\epsilon''/\epsilon'\right)\right]\right\}^{-1/2}}_{\substack{\text{binom} \updownarrow \text{expan}}} \cong \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j\frac{\epsilon''}{2\epsilon'}\right) \quad (\Omega) \quad (3.6) \\
 &\quad \left[1 + \underbrace{\left(\frac{-\frac{1}{2}}{1}\right)\left(-j\frac{\epsilon''}{\epsilon'}\right)}_{j\frac{\epsilon''}{2\epsilon'}} + \underbrace{\left(\frac{\frac{3}{8}}{2}\right)\left(-j\frac{\epsilon''}{\epsilon'}\right)^2}_{\frac{-\frac{1}{2}\left(-\frac{3}{2}\right)}{2!} - \left(\frac{\epsilon''}{\epsilon'}\right)^2} + \dots \right]
 \end{aligned}$$

$\therefore$ intrinsic impedance for uniform plane wave $= E_x/H_y$,

$\therefore$ electric & magnetic fields in lossy dielectric NOT in time phase, as they are in lossless medium

Phase Velocity (low-loss dielectric)

As was in (1.7): phase velocity $u_p = \omega/\beta$, $\therefore$ using (3.5b):

$$\begin{aligned}
 u_p = \frac{\omega}{\beta} &\cong \frac{\omega}{\underbrace{\omega\sqrt{\mu\epsilon'} \left[1 + \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right]}_{\substack{\beta \\ \text{from (3.5b)}}}} \\
 &= \frac{1}{\sqrt{\mu\epsilon'}} \underbrace{\left[1 + \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right]^{-1}}_{\substack{\updownarrow \\ 1 - \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 + \left[\frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right]^2 - \left[\frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right]^3 + \dots}} \cong \frac{1}{\sqrt{\mu\epsilon'}} \left[1 - \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right] \quad (\text{m/s}) \quad (3.7)
 \end{aligned}$$

using well-known binomial expansion:
 $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$

Part 3.2: Good Conductors

Propag. Constant (good conductors)

- Good conductor: $\sigma/\omega\epsilon \gg 1$
- Using (3.3) and in it, neglect 1 as compared to $\sigma/(j\omega\epsilon)$:

$$\overset{\alpha+j\beta}{\underbrace{\gamma}_{(3.3) \text{ from}}} = j\omega\sqrt{\mu\epsilon} \sqrt{1 + \frac{\sigma}{j\omega\epsilon}} \underset{\frac{\sigma}{\omega\epsilon} \gg 1}{\cong} j\omega\sqrt{\mu\epsilon} \sqrt{\frac{\sigma}{j\omega\epsilon}} = j\omega \underbrace{\sqrt{\mu\sigma/(j\omega)}}_{\text{see (3.2)}_{\epsilon_c}} \sqrt{\mu(-j\sigma/\omega)} \quad (3.8a)$$

$$= \underbrace{\sqrt{j}}_{\substack{\sqrt{e^{j\pi/2}} = e^{j\pi/4} \\ = \cos\frac{\pi}{4} + j\sin\frac{\pi}{4} \\ = \frac{1+j}{\sqrt{2}}}} \sqrt{\omega\mu\sigma} = \frac{1+j}{\sqrt{2}} \sqrt{\underbrace{\omega}_{2\pi f} \mu\sigma} = \underbrace{(1+j)\sqrt{\pi f \mu\sigma}}_{\alpha+j\beta} \quad (3.8b) \quad (3.8)$$

- $\alpha \approx \beta \approx \sqrt[4]{(\pi f \mu \sigma)}$ for good conductors
- both $\uparrow$ as $\sqrt[4]{f}$ & $\sqrt[4]{\sigma}$.

Intrinsic Impedance (good conductor)

From (3.8a) $\epsilon_c = \frac{\sigma}{j\omega}$ for good conductors **(3.8a) repeat**

$$\begin{aligned} \therefore \eta_c &= \sqrt{\frac{\mu}{\epsilon_c}} \stackrel{\epsilon_c = \frac{\sigma}{j\omega}}{=} \sqrt{\frac{j\omega\mu}{\sigma}} = \underbrace{\sqrt{j}}_{\substack{\sqrt{e^{j\pi/2}} = e^{j\pi/4} \\ = \cos\frac{\pi}{4} + j\sin\frac{\pi}{4} \\ \underbrace{\cos\frac{\pi}{4}}_{1/\sqrt{2}} \quad \underbrace{j\sin\frac{\pi}{4}}_{1/\sqrt{2}} \\ = \frac{1+j}{\sqrt{2}}}} \sqrt{\frac{2\pi f \mu}{\sigma}} = (1+j) \underbrace{\sqrt{\frac{\pi f \mu}{\sigma}}}_{\substack{\text{from (3.8b)} \\ \alpha=\beta \\ \frac{\sqrt{\pi f \mu \sigma}}{\sqrt{\sigma^2}} = \frac{\alpha}{\sigma}}}} \stackrel{\substack{\text{from (3.8b)} \\ \alpha=\beta \\ = \sqrt{\pi f \mu \sigma}}}{=} (1+j) \frac{\alpha}{\sigma} \quad (\Omega) \end{aligned} \quad \textbf{(3.9)}$$

with phase angle of 45° (due to $1 + j$ term) $\Rightarrow$ magnetic field lags electric field by 45° $\therefore \eta = E/H$ (phase of H must be less than that of E by 45°)

Phase Velocity (good conductor)

As was in (1.7): phase velocity $u_p = \omega/\beta$, $\therefore$ using (3.8b) for β :

$$u_p = \frac{\omega}{\beta} \cong \frac{\overbrace{2\pi f}^{\omega}}{\underbrace{\sqrt{\pi f \mu \sigma}}_{\substack{\text{approx} \\ \beta \cong \alpha}}} \stackrel[\substack{\text{numer} \\ \& \text{denom} \\ \text{by} \\ \sqrt{2}}]{\text{multiply}} \frac{\overbrace{\sqrt{2} 2\pi f}^{\cancel{\sqrt{\omega}} \cancel{\sqrt{\omega}}}}{\underbrace{\sqrt{2\pi f \mu \sigma}}_{\cancel{\sqrt{\omega}} \sqrt{\mu \sigma}}} = \sqrt{\frac{2\omega}{\mu \sigma}} \quad (\text{m/s}) \quad (3.10)$$

which is proportional to $\sqrt[3]{f}$ and $1/\sqrt{\sigma}$.

$\therefore$ higher the conductivity σ , slower the phase velocity

Wavelength (good conductor)

Wavelength of plane wave in good conductor:

$$\lambda = \frac{2\pi}{\beta} \overset{\substack{\text{use} \\ (3.8b)}}{\underset{\substack{\text{for} \\ \text{approx} \\ \beta}}{\cong}} \frac{2\pi}{\underbrace{\sqrt{\pi f \mu \sigma}}_{\substack{\text{approx} \\ \beta \cong \alpha}}} = 2\sqrt{\frac{\pi}{f \mu \sigma}} \quad (\text{m})$$

$$\text{OR : } \lambda = \frac{u_p}{f} \overset{\substack{\text{use} \\ (3.10)}}{\underset{\substack{\text{for} \\ u_p}}{=}} \frac{1}{f} \sqrt{\frac{\overbrace{2(2\pi f)}^{\omega}}{\mu \sigma}} = 2\sqrt{\frac{\pi}{f \mu \sigma}} \quad (\text{m}) \quad (3.11)$$

Example of Copper: plane wave propagation in it

- Parameters:
 - Conductivity: $\sigma = 5.8 \times 10^7$ S/m
 - Permeability: $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m
- At $f = 3$ MHz $= 3 \times 10^6$ Hz
 - phase velocity $u_p = 720$ m/s [use (3.10)]
 - wavelength $\lambda = 0.24$ mm [use (3.11)]
- Remarks (as compared to plane propagation in free-space at same 3 MHz):
 - much slower (being 299792458 m/s in vacuum)
 - much shorter wavelength (being 100 m in vacuum)

Field Function: $E(z) = A_o e^{-\alpha z}$

- Consider wave amplitude, say A_i , at any initial location along z , say z_i , i.e., $E(z = z_i) = A_o e^{-\alpha z_i} = A_i$, being smaller than original amplitude A_o at origin $z = 0$ by factor of $e^{-\alpha z_i}$.
- Upon traveling distance (along z) of $\delta = 1/\alpha$, to arrive at $z = z_i + (1/\alpha)$: amplitude attenuates (decays) to $A_{i+\delta} = A_o e^{-\alpha[z_i + (1/\alpha)]} = \overbrace{A_o e^{-\alpha z_i}}^{A_i} e^{-1} = A_i e^{-1}$, being 0.368 of that initial A_i (at initial point z_i)
- For copper:
 - at 3 MHz, $\alpha \cong \sqrt{\pi f \mu \sigma} = 2.62 \times 10^4 \Rightarrow \delta = 0.038 \text{ mm}$
 - at 10 GHz, $\delta = 0.66 \text{ } \mu\text{m} \rightarrow \text{very small!}$

Skin Depth (Depth of Penetration)

- $\therefore$ EM wave at high frequency (large f) attenuates very rapidly as it propagates in good conductor (high σ)
- Distance δ through which amplitude of traveling plane wave decreases by factor of $e^{-1} \approx 0.368$ defined as: **skin depth** or **depth of penetration**:
$$\delta = \frac{1}{\alpha} \stackrel{\text{from (3.8b)}}{=} \frac{1}{\sqrt{\pi f \mu \sigma}} \quad (\text{m}) \quad (3.12)$$
- $\therefore \alpha \approx \beta$ for good conductors [from (3.8b)], can also write:
$$\delta \stackrel{\text{good}}{\underset{\text{conductors}}{\cong}} \frac{1}{\beta} = \frac{\lambda}{2\pi} \quad (3.13)$$

Skin Depths of Various Metals

Skin Depths, δ in (mm), of Various Materials

Material	σ (S/m)	$f = 60$ (Hz)	1 (MHz)	1 (GHz)
Silver	6.17×10^7	8.27 (mm)	0.064 (mm)	0.0020 (mm)
Copper	5.80×10^7	8.53	0.066	0.0021
Gold	4.10×10^7	10.14	0.079	0.0025
Aluminum	3.54×10^7	10.92	0.084	0.0027
Iron ($\mu_r \cong 10^3$)	1.00×10^7	0.65	0.005	0.00016

At microwave frequencies, skin depth of a good conductor so small that fields & currents can be considered as, for all practical purposes, confined in a very thin layer (i.e., in the *skin*) of the conductor surface

Example: Seawater (#1)

Electric field of linearly-polarized uniform plane wave propagating in +z direction in seawater is $\mathbf{E} = \mathbf{a}_x 100 \cos(10^7 \pi t)$ V/m at $z = 0$.

Parameters of seawater: $\epsilon_r = 72$, $\mu_r = 1$, $\sigma = 4$ (S/m).

- Find attenuation constant, phase constant, intrinsic impedance, phase velocity, wavelength, and skin depth.
- Find distance at which amplitude of $\mathbf{E}$ is 1% of its value at $z = 0$.
- Write expressions for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$ at $z = 0.8$ (m) as functions of t .

Solution

Preliminaries: $\omega = 10^7 \pi$ (rad/s), $f = \frac{\omega}{2\pi} = 5 \times 10^6$ (Hz)

$$\frac{\sigma}{\omega \epsilon} = \frac{\sigma}{\omega \underbrace{\epsilon_0 \epsilon_r}_{\epsilon}} = \frac{\overset{\sigma}{4}}{\underbrace{10^7 \pi}_{\omega} \underbrace{\left(\frac{10^{-9}}{36\pi} \right)}_{\epsilon_0} \underbrace{(72)}_{\epsilon_r}} = 200 \gg 1$$

$\therefore$ can use formulas for good conductors

Example: Seawater (#2)

(a) i) Attenuation constant α : use (3.8b):

$$\alpha \stackrel{\text{use}}{\underset{(3.8b)}{\cong}} \sqrt{\pi f \mu \sigma} = \sqrt{\pi (5 \times 10^6) (4\pi \times 10^{-7}) (4)} = 8.89 \text{ (Np/m)} \quad \textbf{(Ans a-i)}$$

ii) Phase constant β : from (3.8b) as well:

$$\beta \stackrel{\text{from}}{\underset{(3.8b)}{\cong}} \sqrt{\pi f \mu \sigma} \stackrel{\text{from}}{\underset{\text{above}}{=}} 8.89 \text{ (rad/m)} \quad \textbf{(Ans a-ii)}$$

iii) Intrinsic impedance η_c : use (3.9):

$$\eta_c = (1+j) \sqrt{\frac{\pi f \mu}{\sigma}} = (1+j) \sqrt{\frac{\pi (5 \times 10^6) (4\pi \times 10^{-7})}{4}} = \pi \frac{1+j}{\sqrt{2}} = \pi \underbrace{\frac{1}{\sqrt{2}} + j \frac{1}{\sqrt{2}}}_{\cos \frac{\pi}{4} + j \sin \frac{\pi}{4}} e^{j\pi/4} \quad (\Omega)$$

$\underbrace{\pi \sqrt{5 \times 10^{-1}} = \pi \sqrt{0.5} = \frac{\pi}{\sqrt{2}} = \pi \cos \frac{\pi}{4} = \pi \sin \frac{\pi}{4}}_{\downarrow}$

$$\stackrel{\text{alter}}{\text{native}} = (1+j) \frac{\alpha}{\sigma} \stackrel{\text{use}}{\underset{\text{answer above}}{=}} (1+j) \frac{\overbrace{8.89}^{\alpha}}{\underbrace{4}_{\sigma}} = 2.22(1+j) \quad (\Omega) \quad \textbf{(Ans a-iii)}$$

Example: Seawater (#3)

(a) iv) Phase velocity u_p : use (3.10):

$$u_p = \frac{\omega}{\beta} \stackrel{\text{use}}{\underset{\substack{\text{answer} \\ \text{for} \\ \beta}}{8.89}} = \frac{10^7 \pi}{\underbrace{8.89}_{\beta}} = 3.53 \times 10^6 \text{ (m/s)} \quad \textbf{(Ans a-iv)}$$

v) Wavelength λ : use (3.11):

$$\lambda = \frac{2\pi}{\beta} \stackrel{\text{use}}{\underset{\substack{\text{answer} \\ \text{for} \\ \beta}}{8.89}} = \frac{2\pi}{\underbrace{8.89}_{\beta}} = 0.707 \text{ (m)} \quad \textbf{(Ans a-v)}$$

vi) Skin depth δ : use (3.12):

$$\delta = \frac{1}{\alpha} \stackrel{\text{use}}{\underset{\substack{\text{answer} \\ \text{for} \\ \alpha}}{8.89}} = \frac{1}{8.89} = 0.112 \text{ (m)} \quad \textbf{(Ans a-vi)}$$

Example: Seawater (#4)

(b) Distance z_1 at which wave amplitude decreases to 1% of its value at $z = 0$:

$$e^{-\alpha z_1} = 0.01 \Rightarrow e^{\alpha z_1} = 100 \xRightarrow[\substack{\text{natural} \\ \log \\ \text{both} \\ \text{sides}}]{\text{take}} \alpha z_1 = \ln 100 \Rightarrow z_1 = \frac{\overbrace{\ln 100}^{4.605}}{\underbrace{\alpha}_{8.89}} = 0.518 \text{ (m)} \quad \textbf{(Ans b)}$$

(c) Expressions for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$ at $z = 0.8$ (m) as functions of t :

(i) Given: $\mathbf{E} = \mathbf{a}_x 100 \cos(10^7 \pi t)$ V/m at $z = 0$

In phasor notation: $\vec{E}(z) = \vec{a}_x 100 e^{-\alpha z} e^{-j\beta z}$ **(Ans ci)**

Instantaneous form:

$$\vec{E}(z, t) = \text{Re} \left[\underbrace{\vec{E}(z)}_{\vec{a}_x 100 e^{-\alpha z} e^{-j\beta z}} e^{j\omega t} \right] = \vec{a}_x 100 e^{-\alpha z} \underbrace{\text{Re} \left[e^{j(\omega t - \beta z)} \right]}_{\cos(\omega t - \beta z)} = \vec{a}_x 100 e^{-\alpha z} \cos(\omega t - \beta z) \quad \textbf{(Ans ci-)}$$

At $z = 0.8$ (m),

$$\vec{E}(z = 0.8, t) = \vec{a}_x 100 e^{-\overbrace{\alpha}^{8.89} \underbrace{z}_{0.8}} \cos \left[\underbrace{\omega}_{10^7 \pi} t - \underbrace{\beta}_{8.89} \underbrace{(0.8)}_z \right] = \vec{a}_x 0.082 \cos(10^7 \pi t - 7.11) \text{ (V/m)} \quad \textbf{(Final Ans ci=)} \quad 72$$

Example: Seawater (#5)

(cii) To find corresponding H-field:

Facts to note:

- uniform plane wave is TEM with $\mathbf{E} \perp \mathbf{H}$, and both $\perp$ propagation direction $\Rightarrow \mathbf{H} = \mathbf{a}_y H_y$ for present x-polarized E-field propagating along +z direction
- formula required: ratio of E to H fields (in *phasor* form, **not** instantaneous) gives wave impedance η :

$$\text{both phasor form} \quad \boxed{H_y(z)} = \frac{\boxed{E_x(z)}}{\eta_c}$$

Example: Seawater (#6)

$\therefore$ instantaneous $H_y(z, t) = \text{Re}[H_y(z)\exp(j\omega t)]$

$$H_y(z, t) = \text{Re}\left[\underbrace{E_x(z)/\eta_c}_{\substack{\text{use phasor in} \\ \text{(Ans ci)}}} e^{j\omega t}\right] = \text{Re}\left[\frac{\underbrace{E_x(z)}_{\substack{\text{use} \\ \text{(Ans a-iii)}}}}{\underbrace{\eta_c}_{\substack{\text{use} \\ \text{(Ans a-iii)}}}} e^{j\omega t}\right] = \text{Re}\left[\frac{100e^{-\underbrace{\alpha}_{8.89}z} e^{-j\underbrace{\beta}_{8.89}z}}{\underbrace{\pi e^{j\pi/4}}_{\eta_c}} e^{j\omega t}\right] = \frac{100}{\pi} e^{-\underbrace{\alpha}_{8.89}z} \text{Re}\left[e^{j(\underbrace{\omega}_{10^7}t - \underbrace{\beta}_{8.89}z - \frac{\pi}{4})}\right]$$

$$\therefore H_y(z, t) = \frac{100}{\pi} e^{-\underbrace{\alpha}_{8.89}z} \cos(\underbrace{\omega}_{10^7}t - \underbrace{\beta}_{8.89}z - \frac{\pi}{4})$$

Then at the required $z = 0.8$,

$$H_y(z = 0.8, t) = \underbrace{\frac{100}{\pi}}_{0.026} e^{-\underbrace{\alpha}_{8.89} \underbrace{z}_{0.8}} \cos(\underbrace{\omega}_{10^7}t - \underbrace{\beta}_{8.89} \underbrace{z}_{0.8} - \frac{\pi}{4}) = 0.026 \cos(10^7 \pi t - 1.61) \quad (\text{A/m}) \quad \textbf{(Ans cii)}$$

$\underbrace{-7.9 \text{ radians}}_{\substack{\text{also equals to} \\ -1.61 \text{ radians} \\ (\because \text{cyclic in } 2\pi)}}$

Part 4: Group & Phase Velocity

Linearity of Phase Velocity with Freq for Plane Waves in Lossless Media

- Defined earlier from (1.7) this chapter: phase velocity, u_p , of single-frequency plane-wave as velocity of propagation of equiphase wavefront:

$$u_p = \omega/\beta \quad (\text{m/s}) \quad (4.1)$$

- For plane waves in lossless medium, $\beta = \omega\sqrt{\mu\epsilon}$ [see (1.2) this chapter] $\rightarrow$ linear function of ω .
- $\therefore$ phase velocity $u_p = 1/\sqrt{\mu\epsilon} \rightarrow$ constant, indep of freq

Dispersive Media

- But some cases, e.g. wave propagation in lossy dielectric, along transmission line, or waveguide (covered later), phase constant not linear function of $\omega \rightarrow$ waves of different frequencies propagate with different velocities
- All info-bearing signals comprise band of frequencies $\rightarrow$ waves of component frequencies travel with different phase velocities $\rightarrow$ distortion in signal wave shape $\rightarrow$ signal “disperses”
- Phenomenon of signal distortion caused by dependence of phase velocity on freq = *dispersion*

Group Velocity

- Any info-bearing signal: normally has small spread of frequencies (sidebands) around a high carrier freq
- Comprises “group” of frequencies, forms wave packet
- *Group velocity*: velocity of propagation of wave-packet envelope (group of frequencies)

Simplest Case: 2-Freq Wave Packet

- Simplest case of wave packet: two traveling waves, equal amplitude, slightly different angular frequencies: $\omega_0 + \Delta\omega$ & $\omega_0 - \Delta\omega$ ($\Delta\omega \ll \omega_0$)
- Phase constants (functions of freq) also slightly different
- Let phase constants corresponding to those 2 frequencies be: $\beta_0 + \Delta\beta$ & $\beta_0 - \Delta\beta$.

Instantaneous Field Variation: Illustration

- Instantaneous (real-time) E -fields for both freq:

$$E(z, t) = E_0 \cos[(\omega_0 + \Delta\omega)t - (\beta_0 + \Delta\beta)z] + \dots$$

$$\dots + E_0 \cos[(\omega_0 - \Delta\omega)t - (\beta_0 - \Delta\beta)z]$$

standard

trigo
identities

$$= 2E_0 \cos[(\Delta\omega)t - (\Delta\beta)z] \cos(\omega_0 t - \beta_0 z) \quad (4.2)$$

- Since $\Delta\omega \ll \omega_0$, this (4.2) represents rapidly-oscillating wave (solid) with angular freq ω_0 & amplitude that varies slowly (dashed) with angular freq $\Delta\omega$.

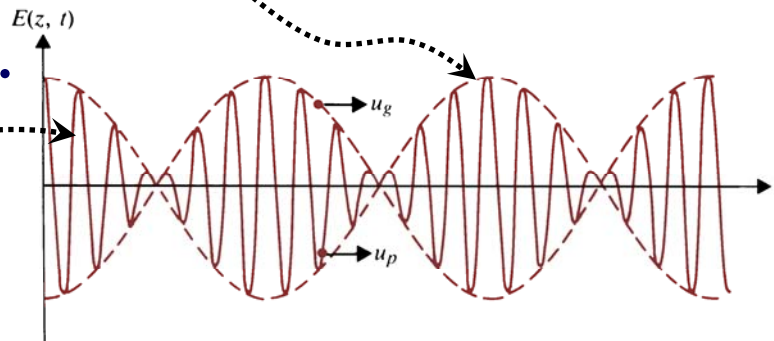


FIGURE 8-6

Sum of two time-harmonic traveling waves of equal amplitude and slightly different frequencies at a given t .

Phase & Group Velocities

- Wave inside envelope propagates with phase velocity found by setting $\boxed{\omega_0 t - \beta_0 z}$ arg of 2nd cos term in (4.2) = constant,

$$\beta_0 z = \omega_0 t - \text{constant} \Rightarrow \frac{dz}{dt} = \frac{\omega_0}{\beta_0} \quad (4.3)$$

- Velocity of envelope (group velocity, u_g) found by setting **argument of 1st cosine factor** in (4.2) to constant: $t \Delta\omega - z \Delta\beta = \text{constant}$

$$(\Delta\beta)z = (\Delta\omega)t - \text{constant} \Rightarrow \frac{dz}{dt} = \frac{\Delta\omega}{\Delta\beta} = \frac{1}{\Delta\beta/\Delta\omega} \quad (4.4)$$

- In limit $\Delta\omega \rightarrow 0$, obtain formula for group velocity in dispersive medium:

$$\boxed{u_g = \frac{1}{d\beta/d\omega}} \quad (\text{m/s}) \quad (4.5) \quad 81$$

Description of Characteristics

- Group velocity: velocity of a point on envelope of wave packet
- Velocity of narrow-band signal
- If ω plotted vs. β ,
 - ✓ slope of straight line drawn from origin to any point on graph = phase velocity: ω/β .
 - ✓ local slope of tangent to graph at point = group velocity: $d\omega/d\beta$.

Part 5: Flow of EM Power & Poynting Vector

Opening Words

- EM waves carry with them EM power
- Energy transported through space to distant receiving points by EM waves

Use of a Vector Identity

Maxwell's curl equations from (3.10) & (3.11) of Chapter 1:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (5.1)$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad (5.2)$$

$$\text{Vector identity: } \nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H}) \quad (5.3)$$

Substitute (5.1) & (5.2) into (5.3):

$$\begin{aligned} \nabla \cdot (\vec{E} \times \vec{H}) &= \vec{H} \cdot \overbrace{\left(-\frac{\partial \vec{B}}{\partial t} \right)}^{\nabla \times \vec{E}} - \vec{E} \cdot \overbrace{\left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)}^{\nabla \times \vec{H}} \\ \Rightarrow \nabla \cdot (\vec{E} \times \vec{H}) &= -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} - \vec{E} \cdot \vec{J} \end{aligned} \quad (5.4)$$

Components of (5.4)

For simple medium, ϵ , μ , & σ independent of time $\rightarrow$ write components of (5.4):

$$\vec{H} \cdot \frac{\partial \overset{\mu \vec{H}}{\vec{B}}}{\partial t} = \mu \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} = \frac{1}{2} \mu (2) \begin{pmatrix} H_x \\ H_y \\ H_z \end{pmatrix} \cdot \begin{pmatrix} \partial H_x / \partial t \\ \partial H_y / \partial t \\ \partial H_z / \partial t \end{pmatrix} = \frac{1}{2} \mu \left(2H_x \frac{\partial H_x}{\partial t} + 2H_y \frac{\partial H_y}{\partial t} + 2H_z \frac{\partial H_z}{\partial t} \right)$$

$$\begin{aligned} &\text{by chain rule of} \\ &= \frac{\mu}{2} \frac{\partial}{\partial t} (H_x^2 + H_y^2 + H_z^2) = \frac{1}{2} \frac{\partial (\mu \vec{H} \cdot \vec{H})}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\mu}{2} H^2 \right) \end{aligned} \quad (5.5)$$

$$\vec{E} \cdot \frac{\partial \overset{\epsilon \vec{E}}{\vec{D}}}{\partial t} = \epsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} = \frac{1}{2} \epsilon (2) \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} \cdot \begin{pmatrix} \partial E_x / \partial t \\ \partial E_y / \partial t \\ \partial E_z / \partial t \end{pmatrix} = \frac{1}{2} \epsilon \left(2E_x \frac{\partial E_x}{\partial t} + 2E_y \frac{\partial E_y}{\partial t} + 2E_z \frac{\partial E_z}{\partial t} \right)$$

$$\begin{aligned} &\text{by chain rule of} \\ &= \frac{\epsilon}{2} \frac{\partial}{\partial t} (E_x^2 + E_y^2 + E_z^2) = \frac{1}{2} \frac{\partial (\epsilon \vec{E} \cdot \vec{E})}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\epsilon}{2} E^2 \right) \end{aligned} \quad (5.6)$$

$$\vec{E} \cdot \vec{J} = \vec{E} \cdot (\sigma \vec{E}) = \sigma E^2. \quad (5.7)$$

Power Conservation Statement

$\therefore$ Eq. (5.4) written as:

$$\nabla \cdot (\vec{E} \times \vec{H}) = -\overbrace{\frac{\partial}{\partial t} \left(\frac{\mu}{2} H^2 \right)}^{\vec{H} \cdot \frac{\partial \vec{B}}{\partial t}} - \overbrace{\frac{\partial}{\partial t} \left(\frac{\epsilon}{2} E^2 \right)}^{\vec{E} \cdot \frac{\partial \vec{D}}{\partial t}} - \underbrace{\sigma E^2}_{(5.8)} = -\frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) - \sigma E^2$$

which is point-function relationship

Integral form of this (5.8) obtained by integrating both sides over volume of concern:

$$\underbrace{\int_V \nabla \cdot (\vec{E} \times \vec{H}) dv}_{\substack{\text{divergence theorem} \\ \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s}}} = -\overbrace{\frac{\partial}{\partial t} \int_V \frac{1}{2} \epsilon E^2 dv}^{\substack{\text{learned from EM1:} \\ \text{energy stored in} \\ \text{electric field}}} - \overbrace{\frac{\partial}{\partial t} \int_V \frac{1}{2} \mu H^2 dv}^{\substack{\text{learned from EM1:} \\ \text{energy stored in} \\ \text{magnetic field}}} - \underbrace{\int_V \sigma E^2 dv}_{\substack{\text{ohmic power dissipated} \\ \text{in volume due to flow} \\ \text{of conduction current} \\ \text{density } \sigma E \text{ in} \\ \text{presence of E-field}}} \quad (5.9)$$

time rate of change of energy stored in electric field time rate of change of energy stored in magnetic field

Interpretation of (5.9); Poynting Vector

- RHS of this (5.9): rate of decrease of electric & magnetic energies stored, subtracted by ohmic power dissipated as heat in volume V .
- For consistency with energy conservation law, this RHS must equal power (rate of energy) leaving volume V through its bounding surface S .
- $\therefore$ surface integrand: $\mathbf{E} \times \mathbf{H}$ on LHS of (5.9) = vector representing power flow per unit area
- $\therefore$ define: $\mathbf{P} = \mathbf{E} \times \mathbf{H}$. (W/m²) (5.10)

Poynting vector: power density vector
associated with EM field

Poynting Theorem

- Assertion that surface integral of Poynting vector $\mathbf{P}$ over closed surface [LHS of (5.9)] equals power leaving enclosed volume = *Poynting's Theorem*.
- This statement *not* limited to plane waves
- $\mathbf{P}$ perpendicular to both $\mathbf{E}$ & $\mathbf{H}$.
- If region lossless, $\sigma = 0$

Power Balance Eq. – Another Form (Reverse Flow Concept)

Another form of (5.9):

$$\underbrace{-\oint_S \vec{P} \cdot d\vec{S}}_{\text{total power flowing into closed surface}} = \underbrace{\frac{\partial}{\partial t} \int_V (\underbrace{\frac{1}{2}\epsilon E^2}_{w_e} + \underbrace{\frac{1}{2}\mu H^2}_{w_m}) dv}_{\text{time rate of change of energy stored in both electric \& magnetic fields}} + \underbrace{\int_V \overbrace{p_\sigma}^{\sigma E^2} dv}_{\text{ohmic power dissipated in volume due to flow of conduction current density } \sigma E \text{ in presence of E-field}} \quad (5.11)$$

$$w_e = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \epsilon \vec{E} \cdot \vec{E}^* = \text{electric energy density} \quad (5.12)$$

$$w_m = \frac{1}{2} \mu H^2 = \frac{1}{2} \mu \vec{H} \cdot \vec{H}^* = \text{magnetic energy density} \quad (5.13)$$

$$p_\sigma = \sigma E^2 \underset{J=\sigma E}{=} J^2 / \sigma = \sigma \underbrace{\vec{E} \cdot \vec{E}^*}_{E^2} = \vec{J} \cdot \vec{J}^* / \sigma = \text{ohmic power density} \quad (5.14)$$

Above (5.11) states: total power flowing into closed surface at any instant = sum of rates of increase of stored electric & magnetic energies & ohmic power dissipated within enclosed volume

Example: Verify Poynting Theorem (#1)

Find Poynting vector on surface of long, straight conducting wire (radius b , conductivity σ) that carries direct current I . Verify Poynting's theorem.

Solution

Since dc, current in wire uniformly distributed over cross-sectional area
Assume wire axis coincides with z -axis.

Drawing beside shows segment of length l of long wire.

Write:

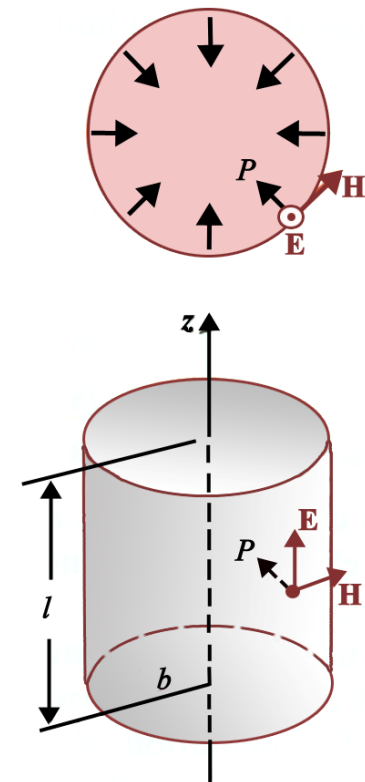
$$\mathbf{J} = \mathbf{a}_z I / (\pi b^2)$$

$$\mathbf{E} = \mathbf{J} / \sigma = \mathbf{a}_z I / (\sigma \pi b^2)$$

On wire surface:

$$\mathbf{H} = \mathbf{a}_\phi I / (2\pi b)$$

from Ampere's law learned in EM1



Example: Verify Poynting Theorem (#2)

Poynting vector at wire surface:

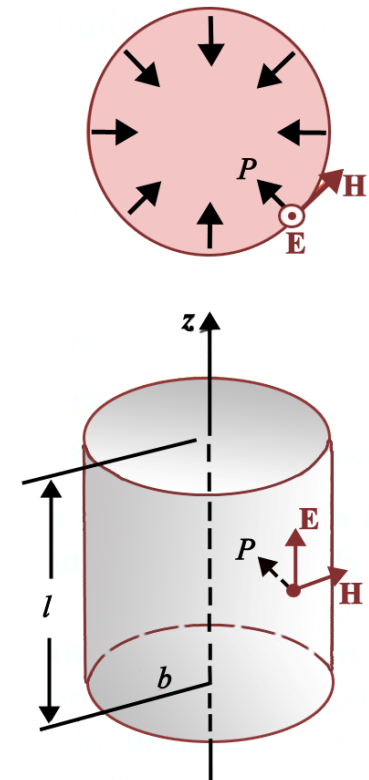
$$\vec{P} = \underbrace{\vec{E}}_{\vec{a}_z \frac{I}{\sigma \pi b^2}} \times \underbrace{\vec{H}}_{\vec{a}_\phi \frac{I}{2\pi b}} = \underbrace{(\vec{a}_z \times \vec{a}_\phi)}_{-\vec{a}_r} \frac{I^2}{2\sigma \pi^2 b^3} = -\vec{a}_r \frac{I^2}{2\sigma \pi^2 b^3}$$

which is everywhere radially directed into wire surface

To verify Poynting's theorem, integrate $\vec{P}$ over wall of wire segment: see drawing beside

$$-\oint_S \underbrace{\vec{P}}_{\substack{\text{from above} \\ -\vec{a}_r \frac{I^2}{2\sigma \pi^2 b^3}}} \cdot \underbrace{d\vec{S}}_{\vec{a}_r ds} = \left(\frac{I^2}{2\sigma \pi^2 b^3} \right) 2\pi b \ell = I^2 \underbrace{\left(\frac{\ell}{\sigma \pi b^2} \right)}_{\substack{\text{"}R=\ell/\sigma S\text{"} \\ \text{resistance of} \\ \text{straight wire,} \\ \text{learned in EM1}}} = I^2 R$$

$\therefore$ affirms that: negative surface integral of Poynting vector = $I^2 R$ ohmic power loss in conducting wire
 $\therefore$ Poynting theorem verified!



Instantaneous Power

$$P(t) = \iint_S \underbrace{\left[\vec{E}(x, y, z, t) \times \vec{H}(x, y, z, t) \right]}_{\text{instantaneous Poynting vector, } \vec{P}^{den}(x, y, z, t)} \cdot \hat{n} dS \quad (2.5)$$

instantaneous Poynting vector, $\vec{P}^{den}(x, y, z, t)$ in [W/m²]

Fields are periodic with period T
 $\Rightarrow$ power averaged over a period
 T must be constant & indep of t .

$\therefore$ power averaged over period $T =$

$$= P_{av} = \frac{1}{T} \int_0^T P(t) dt = \frac{1}{T} \int_0^T \iint_S \underbrace{\left[\vec{E}(x, y, z, t) \times \vec{H}(x, y, z, t) \right]}_{\text{instantaneous Poynting vector, } \vec{P}^{den}(x, y, z, t)} \cdot \hat{n} dS dt$$

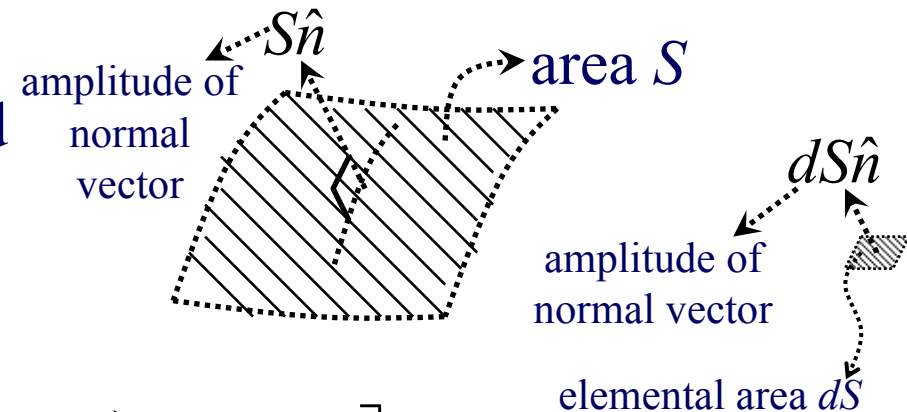
$$= \iint_S \frac{1}{T} \int_0^T \left\{ \underbrace{\left[\vec{E}(x, y, z, t) \times \vec{H}(x, y, z, t) \right]}_{\text{instantaneous Poynting vector, } \vec{P}^{den}(x, y, z, t)} dt \right\} \cdot \hat{n} dS = \iint_S \left(\vec{W}_{av} \cdot \hat{n} \right) dS \quad (2.6)$$

units [W] i.e. a power quantity in watts

see subsequent slides for derivation

$$\vec{W}_{av} = \frac{1}{2} \text{Re} \left[\vec{E}(x, y, z) \times \vec{H}^*(x, y, z) \right] \quad (2.7)$$

power density vector = time-average Poynting vector in [W/m²]

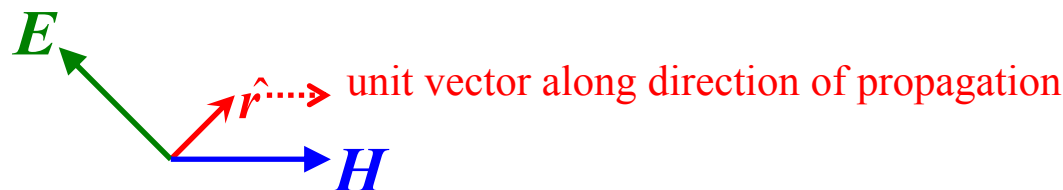


Working (1)

From (2.2) & (2.3), for plane-wave propagating along direction $\boldsymbol{\beta}_r = \mathbf{a}_x \beta_x + \mathbf{a}_y \beta_y + \mathbf{a}_z \beta_z$, with $\beta_r = |\boldsymbol{\beta}_r| = \sqrt{\beta_x^2 + \beta_y^2 + \beta_z^2}$

$$\vec{E}(x, y, z, t) = \text{Re} \left[\begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix} e^{-jk_r r} e^{j\omega t} \right] \begin{matrix} \text{include losses, i.e.} \\ k_r = \beta_r - j\alpha_r \\ = \\ \text{where} \\ \beta_r = |\vec{\beta}_r| = \sqrt{\beta_x^2 + \beta_y^2 + \beta_z^2} \\ \text{with} \\ \vec{\beta}_r = \hat{x}\beta_x + \hat{y}\beta_y + \hat{z}\beta_z \end{matrix} e^{-\alpha_r r} \begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix} \cos(\omega t - \beta_r r) \quad (2.4\text{ai})$$

$$\vec{H}(x, y, z, t) = \text{Re} \left[\begin{pmatrix} H_{0x} \\ H_{0y} \\ H_{0z} \end{pmatrix} e^{-jk_r r} e^{j\omega t} \right] \begin{matrix} \text{include losses, i.e.} \\ k_r = \beta_r - j\alpha_r \\ = \\ \text{where} \\ \beta_r = |\vec{\beta}_r| = \sqrt{\beta_x^2 + \beta_y^2 + \beta_z^2} \\ \text{with} \\ \vec{\beta}_r = \hat{x}\beta_x + \hat{y}\beta_y + \hat{z}\beta_z \end{matrix} e^{-\alpha_r r} \begin{pmatrix} H_{0x} \\ H_{0y} \\ H_{0z} \end{pmatrix} \cos(\omega t - \beta_r r) \quad (2.4\text{bi})$$

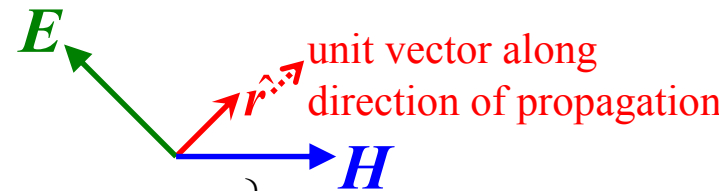


$$\frac{1}{\eta} \left\{ \frac{\vec{\beta}_r}{|\vec{\beta}_r|} \times \begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix} \right\}$$

where $\eta = \sqrt{\mu/\epsilon}$
= wave impedance

Working (2)

Instantaneous Poynting vector (or instantaneous power density vector)



$$\vec{P}^{den}(x, y, z, t) = \overbrace{\vec{E}(x, y, z, t)}^{\text{of (2.4ai)}} \times \overbrace{\vec{H}(x, y, z, t)}^{\text{of (2.4bi)}}$$

$$= \left\{ e^{-\alpha_r r} \begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix} \cos(\omega t - \beta_r r) \right\} \times \left\{ e^{-\alpha_r r} \begin{pmatrix} H_{0x} \\ H_{0y} \\ H_{0z} \end{pmatrix} \cos(\omega t - \beta_r r) \right\}$$

$$= \frac{e^{-2\alpha_r r}}{\eta} \underbrace{\cos^2(\omega t - \beta_r r)}_{\frac{1 + \cos[2(\omega t - \beta_r r)]}{2}} \underbrace{\begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix} \times \left\{ \frac{\vec{\beta}_r}{|\vec{\beta}_r|} \times \begin{pmatrix} E_{0x} \\ E_{0y} \\ E_{0z} \end{pmatrix} \right\}}_{\hat{r} |\vec{E}(x, y, z)|^2} = \hat{r} \frac{|\vec{E}(x, y, z)|^2}{2\eta} e^{-2\alpha_r r} \{1 + \cos[2(\omega t - \beta_r r)]\}$$

(★) [W/m²]

where $\eta = \sqrt{\mu/\epsilon}$
= wave impedance

Working (3)

Average value of power more important than its instantaneous value

Time-average Poynting vector $\mathbf{P}_{av}$:

$$\begin{aligned}
 \vec{P}_{av}^{den}(x, y, z) &= \frac{1}{T} \int_0^T \overbrace{\vec{P}^{den}(x, y, z, t)}^{\text{of previous slide}} dt = \hat{r} \frac{|\vec{E}(x, y, z)|^2 e^{-2\alpha_r r}}{2\eta T} \int_0^T \{1 + \cos[2(\omega t - \beta_r r)]\} dt \\
 &= \hat{r} \frac{|\vec{E}(x, y, z)|^2 e^{-2\alpha_r r}}{2\eta T} \left\{ t + \frac{\sin[2(\omega t - \beta_r r)]}{2\omega} \right\}_{t=0}^{t=T} \\
 &= \hat{r} \frac{|\vec{E}(x, y, z)|^2 e^{-2\alpha_r r}}{2\eta T} \left\{ \cancel{T} + \frac{\overbrace{\sin[2(\omega T - \beta_r r)]}^{\text{these two quantities } \dots \rightarrow} - \overbrace{\sin[2(0 - \beta_r r)]}^{\leftarrow \dots \text{ are EQUAL!!!}}}{2\omega} \right\} = \hat{r} \frac{|\vec{E}(x, y, z)|^2 e^{-2\alpha_r r}}{2\eta} \quad \text{(\textcolor{green}{*})}
 \end{aligned}$$

$f(t = t_0) = t(t = t_0 \pm T)$
 THEREFORE ZERO!!!

where $T = 2\pi/\omega = \text{time-period of wave}$

Working (4)

Now, consider two general complex vectors $\vec{A}$ & $\vec{B}$. We know that:

$$\text{Re}(\vec{A}) = \frac{1}{2}(\vec{A} + \vec{A}^*) \text{ and } \text{Re}(\vec{B}) = \frac{1}{2}(\vec{B} + \vec{B}^*)$$

$$\therefore \text{Re}(\vec{A}) \times \text{Re}(\vec{B}) = \frac{1}{4} \left[(\vec{A} \times \vec{B}^* + \vec{A}^* \times \vec{B}) + (\vec{A} \times \vec{B} + \vec{A}^* \times \vec{B}^*) \right]$$

$$= \frac{1}{2} \underbrace{\text{Re}(\vec{A} \times \vec{B}^* + \vec{A} \times \vec{B})}_{\substack{\frac{1}{2} \left\{ [\vec{A} \times \vec{B}^* + \vec{A} \times \vec{B}] + [\vec{A} \times \vec{B}^* + \vec{A} \times \vec{B}]^* \right\} \\ \text{according to first line above}}} \quad (\Theta)$$

Working (5)

From (★) involving (2.4ai) & (2.4bi), and using (⊕) of previous slide

$$\vec{P}^{den}(x, y, z, t) = \text{Re} \left[\overbrace{\underbrace{\vec{E}(x, y, z)}_{\text{electric-field vector}} e^{j\omega t}}^{\text{time-harmonic}} \right] \times \text{Re} \left[\overbrace{\underbrace{\vec{H}(x, y, z)}_{\text{magnetic-field vector}} e^{j\omega t}}^{\text{time-harmonic}} \right]$$

$$\stackrel{\substack{\text{use} \\ (\ominus) \\ \text{prev} \\ \text{slide}}}{=} \frac{1}{2} \text{Re} \left\{ \underbrace{\left(\vec{E}(x, y, z) e^{j\omega t} \times \left[\vec{H}(x, y, z) e^{j\omega t} \right]^* \right)}_{\vec{E}(x, y, z) \times \vec{H}^*(x, y, z)} + \underbrace{\left[\vec{E}(x, y, z) e^{j\omega t} \times \vec{H}(x, y, z) e^{j\omega t} \right]}_{\vec{E}(x, y, z) \times \vec{H}(x, y, z) e^{j2\omega t}} \right\}$$

Working (6)

Then as was (*), average power density vector $\mathbf{P}_{av}(x, y, z)$ obtained by integrating $\mathbf{P}(x, y, z, t)$ of previous slide over period T .

$$\begin{aligned}
 \vec{P}_{av}^{den}(x, y, z) &= \frac{1}{T} \int_0^T \overbrace{\vec{P}^{den}(x, y, z, t)}^{\text{of previous slide}} dt \\
 &\quad \downarrow \\
 &\quad \frac{1}{2} \operatorname{Re} \left\{ \left[\vec{E}(x, y, z) \times \vec{H}^*(x, y, z) \right] + \dots \right. \\
 &\quad \left. \dots + \left[\vec{E}(x, y, z) \times \vec{H}(x, y, z) e^{j2\omega t} \right] \right\} \\
 &= \frac{1}{2T} \operatorname{Re} \left\{ \left[\vec{E}(x, y, z) \times \vec{H}^*(x, y, z) \right] \int_0^T dt \right\} + \underbrace{\frac{1}{2T} \operatorname{Re} \left\{ \left[\vec{E}(x, y, z) \times \vec{H}(x, y, z) \right] \int_0^T e^{j2\omega t} dt \right\}}_{\text{VANISHES}} \\
 &\quad \left(\underbrace{\int_0^T e^{j2\omega t} dt}_{\text{ZERO}} = \frac{e^{j2\omega T} - 1}{j2\omega} = 0 \text{ since } \omega T = 2\pi \right) \\
 &= \frac{1}{2} \operatorname{Re} \left[\vec{E}(x, y, z) \times \vec{H}^*(x, y, z) \right] \stackrel{\text{define}}{=} \vec{W}_{av}(x, y, z) \quad \text{thus deriving (2.7)}
 \end{aligned}$$

Part 6: Plane-Wave Incidence on Plane Interface Boundary

Opening Words

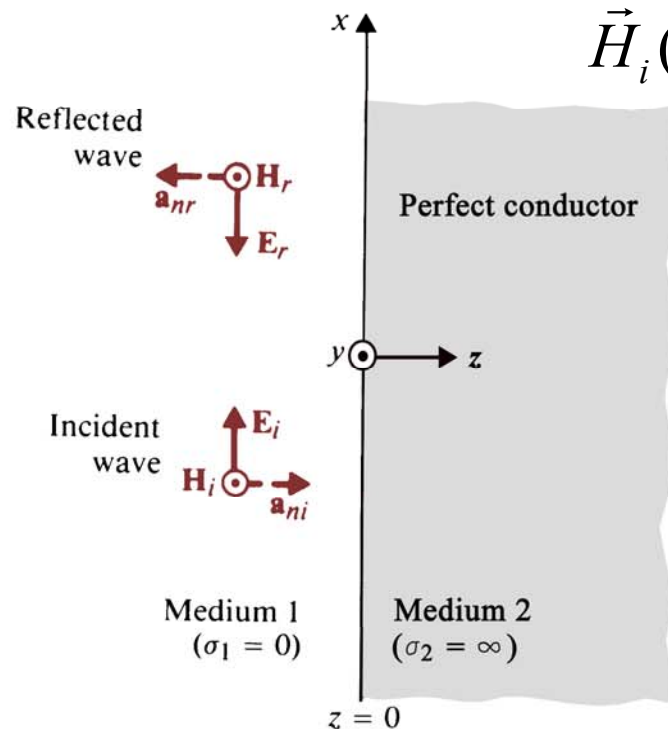
- So far, discussed propagation of uniform plane waves in unbounded homogeneous medium
- In reality, waves often propagate in bounded regions where several media with different material parameters present
- When EM wave traveling in one medium impinges on another with different intrinsic impedance → reflection experienced
- Two types of plane-wave incidences studied:
 - on plane conducting boundary
 - on interface between two dielectric media

Part 6.1: Normal Incidence on Plane Conducting Boundary

Incident Fields in Medium#1

$$\vec{E}_i(z) = \vec{a}_x \overbrace{E_{i0}}^{\text{field amplitude}} e^{-j \overbrace{\beta_1}^{\text{phase const of medium \#1}} z} \quad (6.1.1)$$

$$\vec{H}_i(z) = \vec{a}_y \left(\frac{\overbrace{E_{i0}}^{\text{field amplitude}}}{\underbrace{\eta_1}_{\text{intrinsic impedance of medium \#1}}} \right) e^{-j \overbrace{\beta_1}^{\text{phase const of medium \#1}} z} \quad (6.1.2)$$



Poynting vector of incid wave:
 $\mathbf{P}_i(z) = \mathbf{E}_i(z) \times \mathbf{H}_i(z)$ in $\mathbf{a}_z$ direction.

FIGURE 8-9
 Plane wave incident normally on a plane conducting boundary.

Null Transmitted Fields in Medium#2

- Inside medium#2 (perfect electric conductor), both E - & H -fields vanish (learned in EM1) $\rightarrow$
 $E_2 = H_2 = 0$
- $\therefore$ no waves transmitted across boundary into $z > 0$ region

Reflected E-Field Back into Medium #1

- Incident wave reflected $\rightarrow$ reflected wave ($\vec{E}_r$, $\vec{H}_r$)
- Reflected E -field:

$$\vec{E}_r(z) = \vec{a}_x \overbrace{E_{r0}}^{\text{field amplit}} e^{+j \overbrace{\beta_1}^{\text{phase const of medium \#1}} z} \quad (6.1.3)$$

plus sign $\rightarrow$ reflected waves travels in $-z$ direction

Total E-Field in Medium #1

- Total E-field in medium#1 = sum of E_i & E_r :

$$\vec{E}_1(z) = \vec{E}_i(z) + \vec{E}_r(z) = \vec{a}_x \left(E_{i0} e^{-j\beta_1 z} + E_{r0} e^{+j\beta_1 z} \right) \quad (6.1.4)$$

- Continuity of tangential component of E -field at boundary $z = 0$:
null field in
conductor medium#1

$$\begin{aligned} \text{➤ } E_1(0) &= E_i(0) + E_r(0) = a_x(E_{i0} + E_{r0}) = \overbrace{E_2(0)}^{\text{null field in conductor medium\#1}} = 0 \\ \text{➤ } \therefore E_{r0} &= -E_{i0}. \end{aligned} \quad (6.1.5)$$

- Sub this (6.1.5) into (6.1.4):

$$\vec{E}_1(z) = \vec{a}_x E_{i0} \underbrace{\left(e^{-j\beta_1 z} - e^{+j\beta_1 z} \right)}_{-2j \sin \beta_1 z} = -\vec{a}_x j 2 E_{i0} \sin \beta_1 z \quad (6.1.6)$$

Reflected & Total H-Field in Medium #1

Reflected magnetic field $\vec{H}_r$ related to reflected electric field, $\vec{E}_r$, via (1.22) earlier this chapter:

$$\begin{aligned}\vec{H}_r(z) &= \frac{1}{\eta_1} \underbrace{\vec{a}_{nr}}_{-\vec{a}_z} \times \vec{E}_r(z) = \frac{1}{\eta_1} (-\vec{a}_z) \times \underbrace{\vec{E}_r(z)}_{\vec{a}_x E_{r0} e^{+j\beta_1 z}} \\ &\quad \because \text{reflected wave in } -z \text{ direction} \\ &= -\vec{a}_y \frac{E_{r0}}{\eta_1} e^{+j\beta_1 z} \underset{\substack{\because E_{r0} = -E_{i0} \\ \text{from (6.1.5)}}}{=} \vec{a}_y \frac{E_{i0}}{\eta_1} e^{+j\beta_1 z} \quad (6.1.7)\end{aligned}$$

Total $\vec{H}$ -field in medium #1:

$$\vec{H}_1(z) = \underbrace{\vec{H}_i(z)}_{\substack{\vec{a}_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} \\ \text{from (6.1.2)}}} + \underbrace{\vec{H}_r(z)}_{\substack{\frac{E_{i0}}{\eta_1} e^{+j\beta_1 z} \\ \text{from (6.1.7)}}} = \vec{a}_y 2 \frac{E_{i0}}{\eta_1} \cos \beta_1 z \quad (6.1.8)$$

Instantaneous Real-Time Fields

To examine space-time behavior of total field in medium#1, write instantaneous E & H fields

$$\vec{E}_1(z, t) = \text{Re} \left[\underbrace{\vec{E}_1(z)}_{\substack{-\vec{a}_x j 2 E_{i0} \sin \beta_1 z \\ \text{of (6.1.6)}}} \underbrace{e^{j\omega t}}_{\cos \omega t + j \sin \omega t} \right] = \vec{a}_x 2 E_{i0} \sin \beta_1 z \sin \omega t \quad (6.1.9)$$

$$\vec{H}_1(z, t) = \text{Re} \left[\underbrace{\vec{H}_1(z)}_{\substack{\vec{a}_y 2 \frac{E_{i0}}{\eta_1} \cos \beta_1 z \\ \text{of (6.1.8)}}} \underbrace{e^{j\omega t}}_{\cos \omega t + j \sin \omega t} \right] = \vec{a}_y 2 \frac{E_{i0}}{\eta_1} \cos \beta_1 z \cos \omega t \quad (6.1.10)$$

Zeros & Maxima of Wave Function

Both $\vec{E}_1(z, t)$ & $\vec{H}_1(z, t)$ possess zeros & maxima at fixed distances from conducting boundary for all t , as follow:

$$\begin{array}{l}
 \left. \begin{array}{l}
 \text{Zeros of } \overbrace{\vec{E}_1(z, t)}^{\text{involves sines}} \\
 \text{Maxima of } \underbrace{\vec{H}_1(z, t)}_{\text{involves cossines}}
 \end{array} \right\} \text{ occur at } \beta_1 z = -n\pi, \text{ or } z = -n\frac{\lambda}{2}, \quad n = 0, 1, 2, \dots
 \end{array}$$

$$\begin{array}{l}
 \left. \begin{array}{l}
 \text{Maxima of } \overbrace{\vec{E}_1(z, t)}^{\text{involves sines}} \\
 \text{Zeros of } \underbrace{\vec{H}_1(z, t)}_{\text{involves cossines}}
 \end{array} \right\} \text{ occur at } \beta_1 z = -(2n+1)\frac{\pi}{2}, \text{ or } z = -(2n+1)\frac{\lambda}{4}, \quad n = 0, 1, 2, \dots
 \end{array}$$

Standing Wave

- Total wave in medium#1 not traveling type, rather, it's a *standing wave*, resulting from superposition of two waves traveling in opposite directions
- For given time instance t , both E_1 & H_1 vary sinusoidally with distance measured from boundary plane
- Next slide for illustration ...

Standing Waves - Illustration

Standing waves of $E_1 = \mathbf{a}_x E_1$ & $H_1 = \mathbf{a}_y H_1$ shown below for several values of ωt .

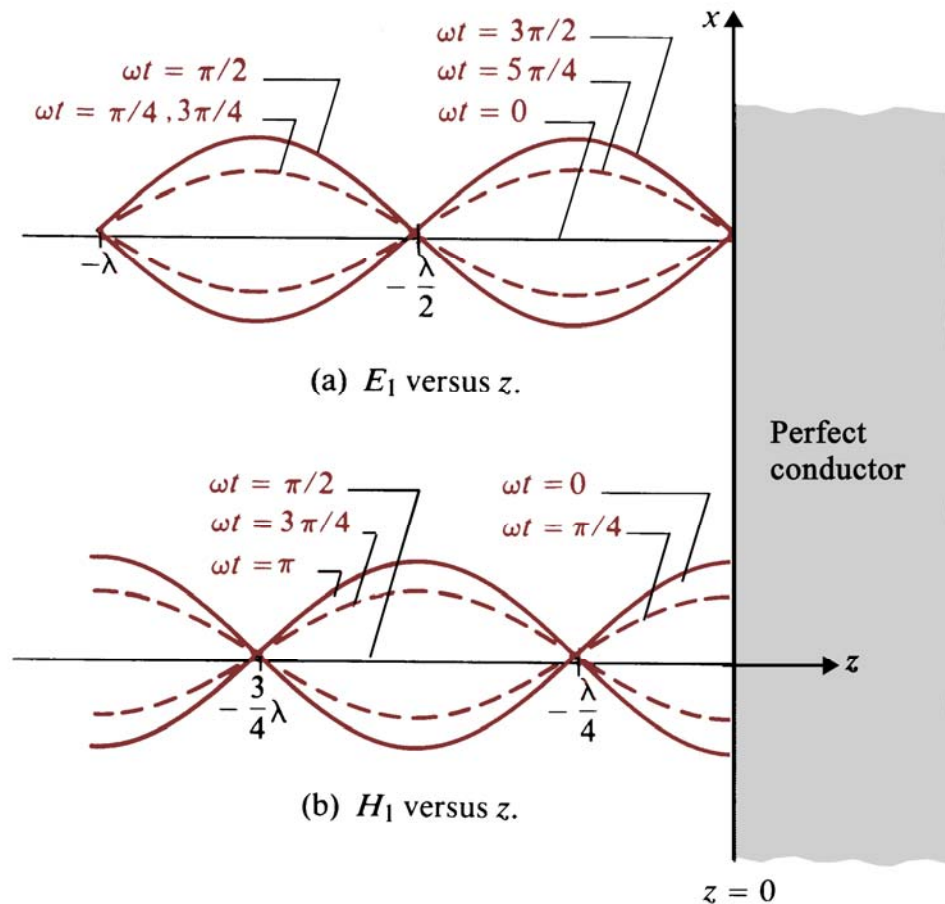


FIGURE 8-10

Standing waves of $E_1 = \mathbf{a}_x E_1$ and $H_1 = \mathbf{a}_y H_1$ for several values of ωt . 11

Points about Standing Wave

- E_1 vanishes on conducting boundary ($E_{r0} = -E_{i0}$) as well as points multiples of $\lambda/2$ from boundary
- H_1 max on conducting boundary ($H_{r0} = H_{i0} = E_{i0}/\eta_1$)
- standing waves of E_1 & H_1 in time quadrature (90° phase diff), and shifted in space by quarter wavelength

Example: Plane Wave Incidence on Conducting Plane (#1: prelims)

A y -polarized uniform plane wave ($\mathbf{E}_i, \mathbf{H}_i$), freq 100 MHz, propagates in air in $+x$ direction, impinges normally on perfect conducting plane at $x = 0$. Assume amplitude of $\mathbf{E}_i$ to be 6 mV/m. Write phasor & instantaneous expressions for (a) $\mathbf{E}_i$ & $\mathbf{H}_i$ of incident wave, (b) $\mathbf{E}_r$ & $\mathbf{H}_r$ of reflected wave, and (c) $\mathbf{E}_1$ & $\mathbf{H}_1$ of total wave in air. (d) Determine location nearest to conducting plane where $\mathbf{E}_1$ is zero.

Solution

Preliminaries:

$$\omega = 2\pi f = 2\pi \times 10^8 \text{ (rad/s)}$$

$$\beta_1 = k_0 = \omega/c = \frac{2\pi \times 10^8}{3 \times 10^8} = \frac{2\pi}{3} \text{ (rad/m)}$$

$$\eta_1 = \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi \text{ } (\Omega)$$

Example: Plane Wave Incidence on Conducting Plane (#2: Incident Wave)

(a) For *incident* wave (traveling wave):

(i) Phasor expressions:

$$\vec{E}_i(x) = \vec{a}_y \overbrace{6 \times 10^{-3}}^{\text{given amplitude}} e^{-j \overbrace{\left(\frac{2\pi}{3}\right)x}^{\beta_1}} \quad (\text{V/m})$$

$$\vec{H}_i(x) = \frac{1}{\underbrace{\eta_1}_{120\pi}} \vec{a}_x \times \underbrace{\vec{E}_i(x)}_{\substack{\text{from} \\ \text{above}}} = \overbrace{\vec{a}_x \times \vec{a}_y}^{\vec{a}_z} \frac{6 \times 10^{-3}}{\underbrace{120\pi}_{\frac{10^{-3}}{20\pi} = \frac{10^{-4}}{2\pi}}} e^{-j \frac{2\pi}{3} x} = \vec{a}_z \frac{10^{-4}}{2\pi} e^{-j \frac{2\pi}{3} x} \quad (\text{A/m})$$

(ii) Instantaneous expressions:

$$\vec{E}_i(x, t) = \text{Re} \left[\underbrace{\vec{E}_i(x)}_{\substack{\text{from} \\ \text{above}}} e^{j \overbrace{\frac{2\pi \times 10^8}{\omega}} t} \right] = \vec{a}_y 6 \times 10^{-3} \cos \left(\underbrace{2\pi \times 10^8}_{\omega} t - \frac{2\pi}{3} x \right) \quad (\text{V/m})$$

$$\vec{H}_i(x, t) = \text{Re} \left[\underbrace{\vec{H}_i(x)}_{\substack{\text{from} \\ \text{above}}} e^{j \overbrace{\frac{2\pi \times 10^8}{\omega}} t} \right] = \vec{a}_z \frac{10^{-4}}{2\pi} \cos \left(\underbrace{2\pi \times 10^8}_{\omega} t - \frac{2\pi}{3} x \right) \quad (\text{A/m})$$

Example: Plane Wave Incidence on Conducting Plane (#3: Reflected Wave)

(a) For *reflected* wave (*also* traveling wave):

(i) Phasor expressions:

$$\vec{E}_r(x) = -\vec{a}_y \overbrace{6 \times 10^{-3}}^{\text{given amplitude}} e^{+j \overbrace{\left(\frac{2\pi}{3}\right)x}^{\beta_1}} \quad (\text{V/m})$$

$$\vec{H}_r(x) = \frac{1}{\underbrace{\eta_1}_{120\pi}} (-\vec{a}_x) \times \underbrace{\vec{E}_r(x)}_{\substack{\text{from} \\ \text{above}}} = \underbrace{\vec{a}_x \times \vec{a}_y}_{\vec{a}_z} \frac{6 \times 10^{-3}}{\underbrace{120\pi}_{\frac{10^{-3}}{20\pi} = \frac{10^{-4}}{2\pi}}} e^{+j \frac{2\pi}{3} x} = \vec{a}_z \frac{10^{-4}}{2\pi} e^{+j \frac{2\pi}{3} x} \quad (\text{A/m})$$

(ii) Instantaneous expressions:

$$\vec{E}_r(x, t) = \text{Re} \left[\underbrace{\vec{E}_r(x)}_{\substack{\text{from} \\ \text{above}}} e^{j \underbrace{\frac{2\pi \times 10^8}{\omega} t}_{\omega t}} \right] = -\vec{a}_y 6 \times 10^{-3} \cos \left(\underbrace{2\pi \times 10^8 t}_{\omega t} + \frac{2\pi}{3} x \right) \quad (\text{V/m})$$

$$\vec{H}_r(x, t) = \text{Re} \left[\underbrace{\vec{H}_r(x)}_{\substack{\text{from} \\ \text{above}}} e^{j \underbrace{\frac{2\pi \times 10^8}{\omega} t}_{\omega t}} \right] = \vec{a}_z \frac{10^{-4}}{2\pi} \cos \left(\underbrace{2\pi \times 10^8 t}_{\omega t} + \frac{2\pi}{3} x \right) \quad (\text{A/m})$$

Example: Plane Wave Incidence on Conducting Plane (#4: Total Wave)

(a) For *total* wave (*standing* wave):

(i) Phasor expressions:

$$\vec{E}_1(x) = \underbrace{\vec{E}_i(x)}_{\text{found earlier}} + \underbrace{\vec{E}_r(x)}_{\text{found earlier}} = -\vec{a}_y 6 \times 10^{-3} \overbrace{\left(e^{+j\frac{2\pi}{3}x} - e^{-j\frac{2\pi}{3}x} \right)}^{2j \sin \frac{2\pi}{3}x} = -\vec{a}_y j 12 \times 10^{-3} \sin \frac{2\pi}{3}x \quad (\text{V/m})$$

$$\vec{H}_1(x) = \underbrace{\vec{H}_i(x)}_{\text{found earlier}} + \underbrace{\vec{H}_r(x)}_{\text{found earlier}} = \vec{a}_z \frac{10^{-4}}{2\pi} \overbrace{\left(e^{+j\frac{2\pi}{3}x} + e^{-j\frac{2\pi}{3}x} \right)}^{2 \cos \frac{2\pi}{3}x} = \vec{a}_z \frac{10^{-4}}{\pi} \cos \frac{2\pi}{3}x \quad (\text{A/m})$$

(ii) Instantaneous expressions:

$$\vec{E}_1(x, t) = \text{Re} \left[\underbrace{\vec{E}_1(x)}_{\text{from above}} \underbrace{e^{j \frac{2\pi \times 10^8}{\omega} t}}_{\cos \omega t + j \sin \omega t} \right] = +\vec{a}_y 12 \times 10^{-3} \sin \left(\frac{2\pi}{3}x \right) \sin \left(\underbrace{2\pi \times 10^8 t}_{\omega} \right) \quad (\text{V/m})$$

$$\vec{H}_1(x, t) = \text{Re} \left[\underbrace{\vec{H}_1(x)}_{\text{from above}} \underbrace{e^{j \frac{2\pi \times 10^8}{\omega} t}}_{\cos \omega t + j \sin \omega t} \right] = \vec{a}_z \frac{10^{-4}}{2\pi} \cos \left(\frac{2\pi}{3}x \right) \cos \left(\underbrace{2\pi \times 10^8 t}_{\omega} \right) \quad (\text{A/m})$$

Example: Plane Wave Incidence on Conducting Plane (#5: nearest null)

- Electric field vanishes at surface of conducting plane at $x = 0$
- In medium #1, first null occurs at 1st integer multiple of half-wavelength, i.e. at $x = -\lambda_1/2 = -\pi/\beta_1 = -3/2$ (m)

Part 6.2: Oblique Incidence on Plane Conducting Boundary

Opening Remarks

- When uniform plane wave incident on plane conducting surface obliquely, behavior of reflected wave depends on polarization of incident wave
- Define *plane of incidence* as: plane containing
 - vector along propagation direction of incidence, and
 - normal to boundary surface
- Incident E_i polarized in arbitrary direction can be decomposed into two components: (1) perpendicular $\perp$ & (2) parallel $\parallel$ to plane of incidence $\Rightarrow \therefore$ consider these 2 cases separately

Part 6.2-1: Perpendicular Polarization

Incident $\mathbf{E}_i \perp$ plane of incidence

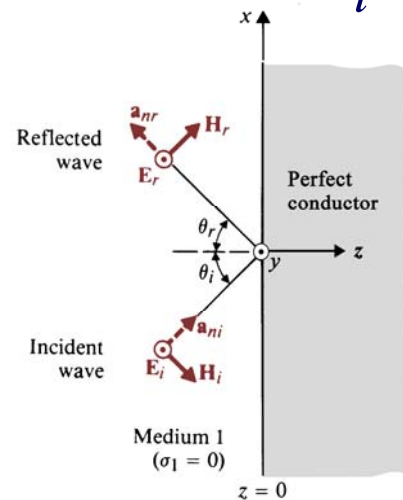


FIGURE 8-11
Plane wave incident obliquely on a plane conducting boundary (perpendicular polarization).

Incident E & H Fields

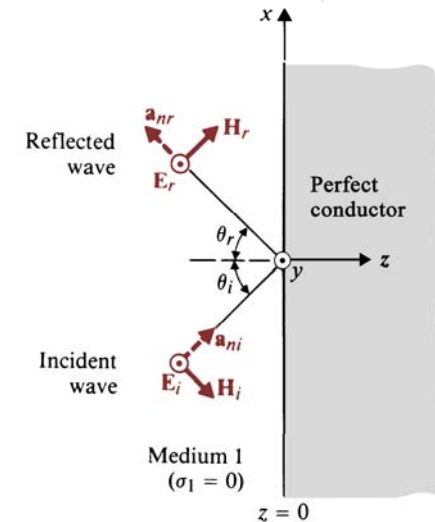
Unit vector along incidence direction: $\mathbf{a}_{ni} = \mathbf{a}_x \sin \theta_i + \mathbf{a}_z \cos \theta_i$. (6.2.1)

From (1.18):

$$\underbrace{\vec{E}(\vec{R}) = \overbrace{\vec{E}_0}^{\text{const vector}} e^{-j\vec{k} \cdot \vec{R}} = \vec{E}_0 e^{-j\overbrace{\vec{k} \cdot \vec{R}}^{\text{present case}}}}_{\text{Eq.(1.18)}} \Rightarrow \underbrace{\vec{E}_i(x, z)}_{\substack{\text{func of} \\ \text{these 2 only} \\ \text{2D case} \\ \text{here}}} = \underbrace{\hat{y}}_{\substack{\text{only} \\ \text{has} \\ \text{this} \\ \text{comp}}} E_{i0} e^{-j\beta_1 \underbrace{\vec{a}_{ni} \cdot \vec{R}}_{\substack{\begin{pmatrix} \sin \theta_i \\ 0 \\ \cos \theta_i \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ \text{incidence angle}}}} = \hat{y} E_{i0} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} \quad (6.2.2)$$

From (1.22):

$$\underbrace{\vec{H}(\vec{R}) = \frac{1}{\eta} \vec{a}_n \times \vec{E}(\vec{R})}_{\text{Eq.(1.22)}} \Rightarrow \underbrace{\vec{H}_i(x, z)}_{\substack{\text{func of} \\ \text{these 2 only} \\ \text{2D case} \\ \text{here}}} = \frac{1}{\underbrace{\eta_1}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}}} \underbrace{\vec{a}_{ni}}_{\substack{\text{above} \\ \begin{pmatrix} \sin \theta_i \\ 0 \\ \cos \theta_i \end{pmatrix}}} \times \underbrace{\vec{E}_i(x, z)}_{\substack{\text{of above} \\ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} E_{i0} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)}}}$$



$$= \underbrace{\frac{E_{i0}}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} \underbrace{\begin{pmatrix} -\cos \theta_i \\ 0 \\ \sin \theta_i \end{pmatrix}}_{\substack{\text{above} \\ \begin{pmatrix} -\hat{x} \cos \theta_i + \hat{z} \sin \theta_i \end{pmatrix}}} = \underbrace{\frac{E_{i0}}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} (-\hat{x} \cos \theta_i + \hat{z} \sin \theta_i) \quad (6.2.3)$$

Reflected E -Field & Snell's Reflection Law

Unit vector along reflection direction: $\mathbf{a}_{nr} = \mathbf{a}_x \sin \theta_r - \mathbf{a}_z \cos \theta_r$. (6.2.4)

Again from (1.18):

$$\underbrace{\vec{E}(\vec{R}) = \overbrace{\vec{E}_0}^{\text{const vector}} e^{-j\vec{k} \cdot \vec{R}} = \vec{E}_0 e^{-j\vec{k} \cdot \vec{R}}}_{\text{Eq.(1.18)}} \xRightarrow[\text{of reflec field}]{\text{present case}} \vec{E}_r(x, z) = \underbrace{\hat{y}}_{\text{only has this comp}} E_{r0} e^{-j\beta_1 \underbrace{\vec{a}_{nr} \cdot \vec{R}}_{\substack{\begin{pmatrix} \sin \theta_r \\ 0 \\ -\cos \theta_r \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ \text{reflection angle}}}} = \hat{y} E_{r0} e^{-j\beta_1 (x \sin \theta_r - z \cos \theta_r)} \quad (6.2.5)$$

At boundary surface $z = 0$, total E -field must vanish

$$\underbrace{\vec{E}_1^{tot}(x, z=0)}_{\substack{\text{boundary condition} \\ E_{\tan}=0 \\ \text{on conduc surface}}} = 0 \Rightarrow \underbrace{\hat{y} E_{i0} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)}}_{\vec{E}_i(x, z=0)} + \underbrace{\hat{y} E_{r0} e^{-j\beta_1 (x \sin \theta_r - z \cos \theta_r)}}_{\vec{E}_r(x, z=0)} = 0$$

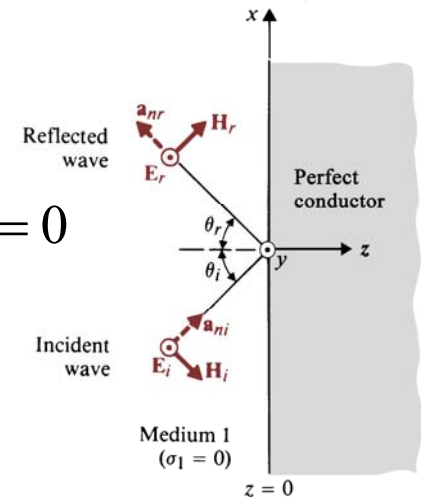
$$\Rightarrow \hat{y} (E_{i0} e^{-j\beta_1 x \sin \theta_i} + E_{r0} e^{-j\beta_1 x \sin \theta_r}) = 0 \Rightarrow E_{i0} e^{-j\beta_1 x \sin \theta_i} = -E_{r0} e^{-j\beta_1 x \sin \theta_r}$$

only way
is to

$\Rightarrow$
equate
magnitude
& phase
on both
sides

$$E_{i0} = -E_{r0} \quad \& \quad \theta_i = \theta_r \quad (6.2.6)$$

Snell's law of reflection: reflection angle = incidence angle



Reflected H -Field

Sub this (6.2.6) into (6.2.5) for reflected E -field in terms of parameters of incident field:

$$\vec{E}_r(x, z) = \underbrace{\hat{y}(-E_{i0})}_{E_{r0}} e^{-j\beta_1(x \sin \vec{\theta}_i - z \cos \vec{\theta}_i)} \quad (6.2.7)$$

From (1.22), reflected H -field:

$$\begin{aligned} \underbrace{\vec{H}(\vec{R}) = \frac{1}{\eta} \vec{a}_n \times \vec{E}(\vec{R})}_{\text{Eq. (1.22)}} &\overset{\substack{\text{present} \\ \text{case}}}{\Rightarrow} \underbrace{\vec{H}_r(x, z)}_{\substack{\text{func of} \\ \text{these 2 only} \\ \text{2D case} \\ \text{here}}} = \underbrace{\frac{1}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med1}}} \underbrace{\vec{a}_{nr}}_{\substack{\text{prev} \\ \text{slide}}} \times \underbrace{\vec{E}_r(x, z)}_{\substack{\text{of above}}} \\ &= \underbrace{\frac{E_{i0}}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med1}}} e^{-j\beta_1(x \sin \vec{\theta}_i - z \cos \vec{\theta}_i)} \underbrace{\begin{pmatrix} -\cos \theta_i \\ 0 \\ -\sin \theta_i \end{pmatrix}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med1}}} = -\underbrace{\frac{E_{i0}}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med1}}} e^{-j\beta_1(x \sin \vec{\theta}_i - z \cos \vec{\theta}_i)} (\hat{x} \cos \theta_i + \hat{z} \sin \theta_i) \quad (6.2.8) \end{aligned}$$

Total (Incident + Reflected) E -Field

$$\begin{aligned}
 \underbrace{\vec{E}_1^{tot}(x, z)}_{\vec{E}_i(x, z) + \vec{E}_r(x, z)} &= \underbrace{\hat{y} E_{i0} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}}_{\substack{\text{from } \updownarrow (6.2.2) \\ \vec{E}_i(x, z)}} + \underbrace{\hat{y} \left(-E_{i0} \right) e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)}}_{\substack{\text{use } (6.2.5) \updownarrow \& (6.2.6) \\ \vec{E}_r(x, z)}} \\
 &\quad \underbrace{-2j \sin(\beta_1 z \cos \theta_i)}_{\updownarrow} \\
 &= \hat{y} E_{i0} e^{-j\beta_1 x \sin \theta_i} \left(e^{-j\beta_1 z \cos \theta_i} - e^{+j\beta_1 z \cos \theta_i} \right) = -\hat{y} j 2 E_{i0} \sin(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}
 \end{aligned}$$

(6.2.9)

Total (Incident + Reflected) H -Field

$$\begin{aligned}
 \underbrace{\vec{H}_1^{tot}(x, z)}_{\vec{H}_i(x, z) + \vec{H}_r(x, z)} &= \underbrace{\frac{E_{i0}}{\eta_1} e^{-j\beta_1 x \sin \theta_i} e^{-j\beta_1 z \cos \theta_i}}_{\substack{\text{from } \Downarrow(6.2.3) \\ \vec{H}_i(x, z)}} (-\hat{x} \cos \theta_i + \hat{z} \sin \theta_i) - \underbrace{\frac{E_{i0}}{\eta_1} e^{-j\beta_1 (x \sin \theta_i - z \cos \theta_i)}}_{\substack{\text{from } \Downarrow(6.2.8) \\ \vec{H}_r(x, z)}} (\hat{x} \cos \theta_i + \hat{z} \sin \theta_i) \\
 &= \frac{E_{i0}}{\eta_1} e^{-j\beta_1 x \sin \theta_i} \left\{ -\hat{x} \cos \theta_i \overbrace{\left(e^{-j\beta_1 z \cos \theta_i} + e^{+j\beta_1 z \cos \theta_i} \right)}^{2 \cos(\beta_1 z \cos \theta_i)} + \hat{z} \sin \theta_i \overbrace{\left(e^{-j\beta_1 z \cos \theta_i} - e^{+j\beta_1 z \cos \theta_i} \right)}^{-2j \sin(\beta_1 z \cos \theta_i)} \right\} \\
 &= -\frac{2E_{i0}}{\eta_1} e^{-j\beta_1 x \sin \theta_i} \left[\hat{x} \cos \theta_i \cos(\beta_1 z \cos \theta_i) + \hat{z} j \sin \theta_i \sin(\beta_1 z \cos \theta_i) \right]
 \end{aligned}$$

(6.2.10)

Interpretive Comments/Notes (#1)

1. In z -direction normal to boundary:

- a. E_{1y}^{tot} & H_{1x}^{tot} maintain standing-wave patterns
 according to $\sin[(\beta_1 \cos \theta_i)z]$ & $\cos[(\beta_1 \cos \theta_i)z]$
- b. No average power propagated in this direction
 since E_{1y}^{tot} & H_{1x}^{tot} out of time phase by 90° .
- pure imag: see (6.2.9) pure real: see (6.2.10) pure imag too: see (6.2.10)

2. In x -direction parallel to boundary: E_{1y}^{tot} & H_{1z}^{tot}
 in time & space phase

- a. Propagates with phase velocity: $u_{1x} = \frac{\omega}{\beta_{1x}} = \frac{\omega/\beta_1}{\sin \theta_i} = \frac{u_1}{\sin \theta_i}$
- b. Wavelength: $\lambda_{1x} = \frac{2\pi}{\beta_{1x}} = \frac{2\pi/\beta_1}{\sin \theta_i} = \frac{\lambda_1}{\sin \theta_i}$
- both for x -direction
- (6.2.12) (6.2.11)

Interpretive Comments/Notes (#2)

3. Amplitude of propagating wave along x direction varies with $z \Rightarrow \therefore$ a *nonuniform plane wave*.
4. Functional term of E_{1y}^{tot} : $\sin(\beta_1 z \cos \theta_i) \rightarrow$ see (6.2.9), which vanishes for $\beta_1 z \cos \theta_i = -m\pi$ where $m = 0, 1, 2, \dots$ $\underbrace{\beta_1}_{2\pi/\lambda_1}$
 $\Rightarrow \therefore$ a conducting plate could be inserted at:

$$z = -\frac{m\lambda_1}{2 \cos \theta_i}, \quad m = 1, 2, 3, \dots \quad (6.2.13)$$

without changing field pattern between this conducting plate & conducting boundary at $z = 0$

A **transverse electric (TE) wave** with respect to x propagation direction ($E_{1x} = 0$) bounces back & forth between conducting planes, and propagates along $x \rightarrow$ in effect, a parallel-plate waveguide

Example: $\perp$ Polarized Plane-Wave Incidence on Conducting Wall (#1)

A uniform plane wave ($\mathbf{E}_i, \mathbf{H}_i$), of angular frequency ω incident from free-space on infinitely large perfectly electric conducting (PEC) wall at incidence angle θ_i with perpendicular polarization. Find (a) current induced on wall surface, (b) time-average Poynting vector in free-space (medium of incidence)

Solution

Conditions of this problem identical to those just discussed; $\therefore$ can use presented formulas directly. Define scenario as follow:

- Let $z = 0$ be plane representing PEC wall
- Assume y -polarized incidence: $\mathbf{E}_i$ has y comp only

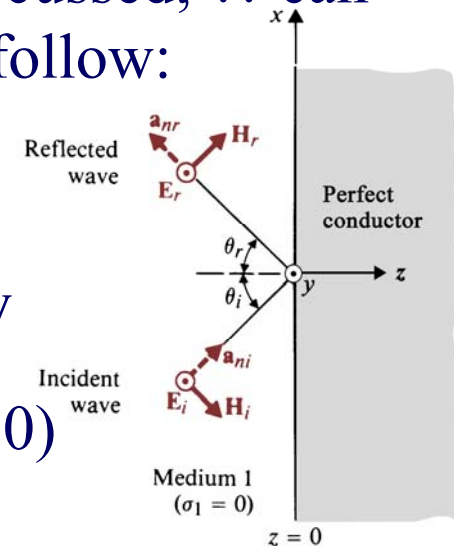
At $z = 0$ (PEC surface): $E_1^{tot} = 0$ & H_1^{tot} from (6.2.10)

$$\vec{H}_1^{tot}(x, z = 0)$$

medium1 is now freespace

$$= -\frac{2E_{i0}}{\underbrace{\eta_1}_{\eta_0}} e^{-j\beta_1 x \sin \theta_i} \left[\hat{x} \cos \theta_i \underbrace{\cos(\underbrace{\beta_1 z \cos \theta_i}_{\omega/c \beta_0 \cos \theta_i})}_1 + \hat{z} j \sin \theta_i \underbrace{\sin(\underbrace{\beta_1 z \cos \theta_i}_{\omega/c \beta_0 \cos \theta_i})}_0 \right] = -\hat{x} \frac{2E_{i0}}{\underbrace{\eta_1}_{\eta_0}} \cos \theta_i e^{-j\beta_1 x \sin \theta_i} \quad (6.2.14)$$

medium1 is now freespace



Example: $\perp$ Polarized Plane-Wave Incidence on Conducting Wall (#2)

Inside PEC wall (medium2), both $\mathbf{E}_2$ & $\mathbf{H}_2$ must vanish $\Rightarrow \therefore$

$\mathbf{H}_1^{tot}(z=0) \neq 0$ from previous eq., $\therefore$ discontinuity in tangential H -field across PEC boundary $\Rightarrow$ this amount of discontinuity equals surface current density $\rightarrow$ standard well-known boundary-condition!

From (5.12b): $\mathbf{a}_{n2} \times \mathbf{H}_1 = \mathbf{J}_s$ on dielectric side of PEC surface, and $H_{2t} = 0$ on other side, i.e., within PEC

$$\therefore \vec{J}_s(x) = \underbrace{\hat{\mathbf{a}}_{n2}}_{\substack{\text{present} \\ \text{case} \\ -\hat{z}}} \times \underbrace{\vec{H}_1^{tot}(x, z=0)}_{\substack{\text{prev. eqn} \\ -\hat{x} \frac{2E_{i0}}{\eta_0} \cos \theta_i e^{-j \frac{\omega/c}{\beta_0} x \sin \theta_i}}} = \overbrace{\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \times \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}}^{\hat{y}} \underbrace{\frac{2E_{i0}}{\eta_0}}_{\substack{120\pi}} \cos \theta_i e^{-j \frac{\omega/c}{\beta_0} x \sin \theta_i} = \hat{y} \frac{E_{i0}}{60\pi} \cos \theta_i e^{-j(\omega/c)x \sin \theta_i} \quad \textbf{(6.2.15a)}$$

Example: $\perp$ Polarized Plane-Wave Incidence on Conducting Wall (#3)

$$\underbrace{\left\{ \begin{array}{l} \cos \left[\omega \left(t - \frac{x}{c} \sin \theta_i \right) \right] \\ + j \sin \left[\omega \left(t - \frac{x}{c} \sin \theta_i \right) \right] \end{array} \right\}}_{\downarrow \uparrow} \underbrace{j \omega \left(t - \frac{x}{c} \sin \theta_i \right)}_{\downarrow \uparrow} \underbrace{\hat{y} \frac{E_{i0}}{60\pi} \cos \theta_i e^{-j(\omega/c)x \sin \theta_i + j\omega t}}_{\downarrow \uparrow}$$

Instantaneous form of this electric current density:

$$\vec{J}_s(x, t) = \text{Re} \left[\vec{J}_s(x) e^{j\omega t} \right] = \text{Re} \left[\underbrace{\hat{y} \frac{E_{i0}}{60\pi} \cos \theta_i e^{-j(\omega/c)x \sin \theta_i}}_{\vec{J}_s(x)} e^{j\omega t} \right]$$

$$= \hat{y} \frac{E_{i0}}{60\pi} \cos \theta_i \cos \left[\omega \left(t - \frac{x}{c} \sin \theta_i \right) \right] \quad (\text{A/m}) \quad \mathbf{(6.2.15b)}$$

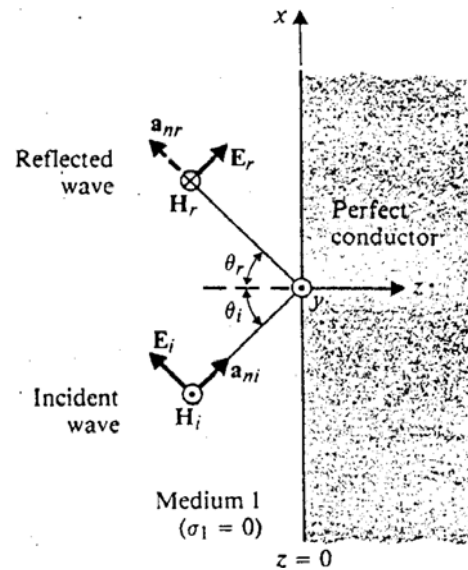
Example: $\perp$ Polarized Plane-Wave Incidence on Conducting Wall (#4)

Time-average Poynting vector in free-space (medium of incidence) found using (2.7) of earlier chapter:

$$\overbrace{\vec{W}_{av} = \frac{1}{2} \text{Re} [\vec{E}(x, y, z) \times \vec{H}^*(x, y, z)]}^{\text{Eq. (2.7)}} \xRightarrow[\text{case}]{\text{present}} \vec{W}_{av} = \frac{1}{2} \text{Re}$$

$$\left\{ \underbrace{4|E_{i0}|^2}_{\downarrow} \underbrace{e^{-j\beta_1 x \sin \theta_i} e^{+j\beta_1 x \sin \theta_i}}_1 \left[\sin(\beta_1 z \cos \theta_i) \begin{pmatrix} 0 \\ j \\ 0 \end{pmatrix} \times \frac{1}{\eta_1} \begin{pmatrix} \cos \theta_i \cos(\beta_1 z \cos \theta_i) \\ 0 \\ -j \sin \theta_i \sin(\beta_1 z \cos \theta_i) \end{pmatrix} \right] \right\} \times \left\{ \underbrace{\frac{2E_{i0}^*}{\eta_1} e^{+j\beta_1 x \sin \theta_i}}_{\text{of (6.2.9)}} \times \underbrace{\begin{bmatrix} \hat{x} \cos \theta_i \cos(\beta_1 z \cos \theta_i) \\ -\hat{z} j \sin \theta_i \sin(\beta_1 z \cos \theta_i) \end{bmatrix}}_{\text{of (6.2.10)}} \right\}^*$$

$$\xRightarrow[\text{case}]{\text{present}} \vec{W}_{av} = \hat{x} 2 |E_{i0}|^2 \frac{\overbrace{\sin(\beta_1 z \cos \theta_i)}^{(\beta_1 \cos \theta_i)z}}{\eta_1} \sin \theta_i \sin \underbrace{(\beta_1 z \cos \theta_i)}_{(\beta_1 \cos \theta_i)z} = \hat{x} \frac{2 |E_{i0}|^2}{\eta_1} \sin \theta_i \sin^2(\beta_{1z} z) \quad (6.2.16)$$



Part 6.2-2: Parallel Polarization

Incident $E_i \parallel$ plane of incidence
(lies in it)

Incident Electric Field

Same unit vectors $\mathbf{a}_{ni}$ & $\mathbf{a}_{nr}$ along incidence & reflection directions in (6.2.1) & (6.2.4) as were for perpendicular polarization reapply

here: $\mathbf{a}_{ni} = \mathbf{a}_x \sin \theta_i + \mathbf{a}_z \cos \theta_i$. (6.2.1)

repeat

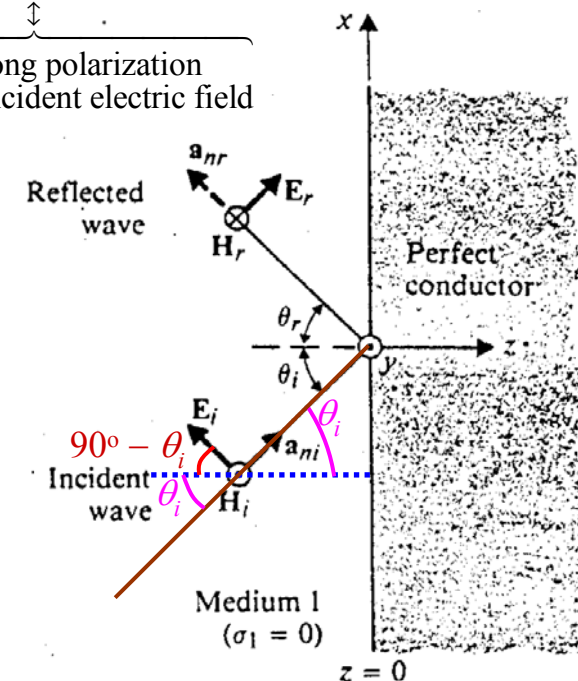
$\mathbf{a}_{nr} = \mathbf{a}_x \sin \theta_r - \mathbf{a}_z \cos \theta_r$. (6.2.4)

repeat

$$\underbrace{\vec{E}(\vec{R}) = \overbrace{\vec{E}_0}^{\text{const vector}} e^{-j\vec{k} \cdot \vec{R}} = \vec{E}_0 e^{-j\overbrace{\vec{k}\vec{a}_n}^{\vec{k}} \cdot \vec{R}}}_{\text{Eq.(1.18)}} \xRightarrow[\text{of incid field}]{\text{present case}} \vec{E}_i \underbrace{(x, z)}_{\substack{\text{func of} \\ \text{these 2 only} \\ \text{2D case} \\ \text{here}}}$$

$$= \underbrace{(\hat{x} \overbrace{\cos \theta_i}^{\sin(90^\circ - \theta_i)} - \hat{z} \overbrace{\sin \theta_i}^{\cos(90^\circ - \theta_i)})}_{\substack{\text{unit vector along polarization} \\ \text{direction of incident electric field}}} E_{i0} e^{-j\beta_1 \underbrace{\vec{a}_{ni} \cdot \vec{R}}_{\substack{\begin{pmatrix} \sin \theta_i \\ 0 \\ \cos \theta_i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ \updownarrow \\ \vec{a}_{ni} \cdot \vec{R}}}}$$

$$= (\hat{x} \cos \theta_i - \hat{z} \sin \theta_i) E_{i0} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} \quad (6.2.17)$$



Incident Magnetic Field

$$\begin{aligned}
 \underbrace{\vec{H}(\vec{R}) = \frac{1}{\eta} \vec{a}_n \times \vec{E}(\vec{R})}_{\text{Eq.(1.22)}} &\xRightarrow[\text{of incid field}]{\text{present case}} \underbrace{\vec{H}_i(x, z)}_{\substack{\text{func of} \\ \text{these 2 only} \\ \text{2D case} \\ \text{here}}} = \underbrace{\frac{1}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}} \underbrace{\vec{a}_{ni}}_{\substack{\text{of} \\ (6.2.1) \\ \updownarrow \\ \begin{pmatrix} \sin \theta_i \\ 0 \\ \cos \theta_i \end{pmatrix}}} \times \underbrace{\vec{E}_i(x, z)}_{\substack{\text{of previous slide} \\ \updownarrow \\ \begin{pmatrix} \cos \theta_i \\ 0 \\ -\sin \theta_i \end{pmatrix} E_{i0} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}}} \\
 = \underbrace{\frac{E_{i0}}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \underbrace{\begin{pmatrix} 0 \\ \cos^2 \theta_i + \sin^2 \theta_i = 1 \\ 0 \end{pmatrix}}_{\substack{\updownarrow \\ \text{just } \hat{y} \\ \text{indeed!}}} = \underbrace{\hat{y} \frac{E_{i0}}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \quad \textbf{(6.2.18)}
 \end{aligned}$$

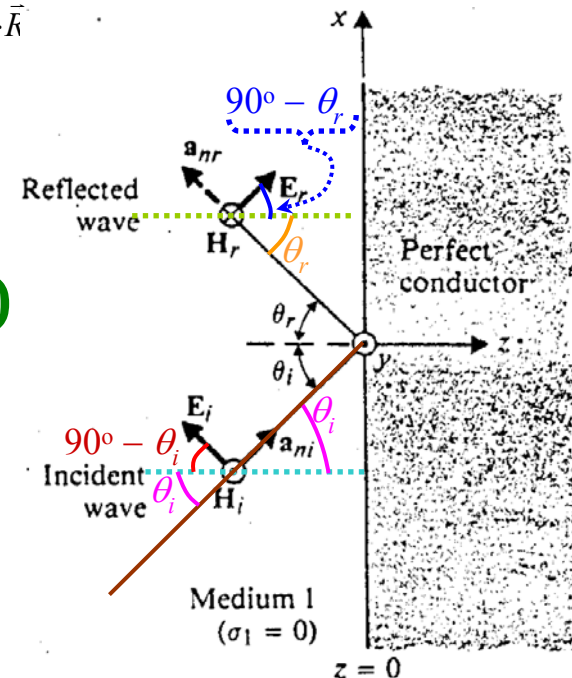
Reflected Electric Field

$$\underbrace{\vec{E}(\vec{R}) = \overbrace{\vec{E}_0}^{\text{const vector}} e^{-j\vec{k} \cdot \vec{R}} = \vec{E}_0 e^{-j\overbrace{\vec{k} \cdot \vec{a}_n}^{\vec{k}} \cdot \vec{R}}}_{\text{Eq.(1.18)}}$$

present
case

$$\Rightarrow \underbrace{\vec{E}_r}_{\substack{\text{of} \\ \text{reflec} \\ \text{field}}} \underbrace{(x, z)}_{\substack{\text{func of} \\ \text{these 2 only} \\ \text{2D case} \\ \text{here}}} = \underbrace{(\hat{x} \overbrace{\cos \theta_r}^{\sin(90^\circ - \theta_r)} + \hat{z} \overbrace{\sin \theta_r}^{\cos(90^\circ - \theta_r)})}_{\substack{\text{unit vector along polarization} \\ \text{direction of reflected electric field}}} E_{r0} e^{-j\beta_1 \underbrace{\begin{pmatrix} \sin \theta_r \\ 0 \\ -\cos \theta_r \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{\vec{a}_{nr} \cdot \vec{K}}}$$

$$= (\hat{x} \cos \theta_r + \hat{z} \sin \theta_r) E_{r0} e^{-j\beta_1 (x \sin \theta_r - z \cos \theta_r)} \quad (6.2.19)$$



Reflected Magnetic Field

$$\begin{aligned}
 \underbrace{\vec{H}(\vec{R}) = \frac{1}{\eta} \vec{a}_n \times \vec{E}(\vec{R})}_{\text{Eq.(1.22)}} &\xRightarrow[\text{of reflec field}]{\text{present case}} \underbrace{\vec{H}_r(x, z)}_{\substack{\text{func of} \\ \text{these 2 only} \\ \text{2D case} \\ \text{here}}} = \underbrace{\frac{1}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}} \underbrace{\vec{a}_{nr}}_{\substack{\text{from} \\ (6.2.4) \\ \updownarrow \\ \begin{pmatrix} \sin \theta_r \\ 0 \\ -\cos \theta_r \end{pmatrix}}} \times \underbrace{\vec{E}_r(x, z)}_{\substack{\text{of previous slide} \\ \updownarrow \\ \begin{pmatrix} \cos \theta_r \\ 0 \\ \sin \theta_r \end{pmatrix} E_{r0} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}}} \\
 = \underbrace{\frac{E_{r0}}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)} \underbrace{\begin{pmatrix} 0 \\ -\cos^2 \theta_i - \sin \theta_i = -1 \\ 0 \end{pmatrix}}_{\substack{\updownarrow \\ -\hat{y}}} = -\hat{y} \underbrace{\frac{E_{r0}}{\eta_1}}_{\substack{\text{wave} \\ \text{imped} \\ \text{of} \\ \text{med 1}}} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}
 \end{aligned}
 \tag{6.2.20}$$

Boundary Condition:

$E_{tan} = 0$ on PEC Surface

At boundary surface $z = 0$, total E -field must vanish

$$\begin{aligned}
 & \underbrace{\vec{E}_1^{tot}(x, z=0)}_{\substack{\vec{E}_i(x, z=0) + \vec{E}_r(x, z=0) \\ \text{conduc} \\ \text{surface}}} \stackrel{\substack{\text{boundary} \\ \text{condition}}}{=} \underset{\substack{E_{tan}=0 \\ \text{on}}}{0} \\
 & \Rightarrow \underbrace{(\hat{x} \cos \theta_i - \hat{z} \sin \theta_i) E_{i0} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}}_{\substack{\text{from } \Downarrow (6.2.17) \\ \vec{E}_i(x, z=0)}} + \underbrace{(\hat{x} \cos \theta_r + \hat{z} \sin \theta_r) E_{r0} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}}_{\substack{\text{from } \Downarrow (6.2.19) \\ \vec{E}_r(x, z=0)}} = 0
 \end{aligned}$$

$$\begin{aligned}
 & \stackrel{\substack{\text{equate} \\ x \\ \text{comp} \\ \text{to } 0}}{\Rightarrow} E_{i0} \cos \theta_i e^{-j\beta_1 x \sin \theta_i} + E_{r0} \cos \theta_r e^{-j\beta_1 x \sin \theta_r} = 0 \quad \stackrel{\substack{\text{only way} \\ \text{is to} \\ \text{equate} \\ \text{magnitude} \\ \& \text{ phase} \\ \text{on both} \\ \text{sides}}}{\Rightarrow} \boxed{E_{i0} = -E_{r0} \quad \& \quad \theta_i = \theta_r} \\
 & \quad \quad \quad \text{same as (6.2.6) earlier!}
 \end{aligned}$$

$\therefore$ as before:

Snell's law of reflection: reflection angle = incidence angle

Total (Incident + Reflected) E -Field

$$\begin{aligned}
 \underbrace{\vec{E}_1^{tot}(x, z)}_{\vec{E}_i(x, z) + \vec{E}_r(x, z)} &= \underbrace{(\hat{x} \cos \theta_i - \hat{z} \sin \theta_i) E_{i0}}_{\substack{\text{from } \Downarrow (6.2.17) \\ \vec{E}_i(x, z)}} \underbrace{e^{-j\beta_1 x \sin \theta_i} e^{-j\beta_1 z \cos \theta_i}}_{\Downarrow} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)} + \underbrace{(\hat{x} \cos \theta_r + \hat{z} \sin \theta_r) (-E_{i0})}_{\substack{\text{use } (6.2.19) \Downarrow \text{ and } (6.2.6) \\ \vec{E}_r(x, z)}} e^{-j\beta_1 (x \sin \theta_r - z \cos \theta_r)} \underbrace{e^{-j\beta_1 x \sin \theta_i} e^{+j\beta_1 z \cos \theta_i}}_{\Downarrow} \\
 &= \hat{x} E_{i0} \cos \theta_i e^{-j\beta_1 x \sin \theta_i} \underbrace{\left(e^{-j\beta_1 z \cos \theta_i} - e^{+j\beta_1 z \cos \theta_i} \right)}_{\substack{-2j \sin(\beta_1 z \cos \theta_i) \\ \Downarrow}} - \hat{z} E_{i0} \sin \theta_i e^{-j\beta_1 x \sin \theta_i} \underbrace{\left(e^{-j\beta_1 z \cos \theta_i} + e^{+j\beta_1 z \cos \theta_i} \right)}_{\substack{2 \cos(\beta_1 z \cos \theta_i) \\ \Downarrow}} \\
 &= -2 E_{i0} e^{-j\beta_1 x \sin \theta_i} \left[\hat{x} j \cos \theta_i \sin(\beta_1 z \cos \theta_i) + \hat{z} \sin \theta_i \cos(\beta_1 z \cos \theta_i) \right] \quad \textbf{(6.2.21)}
 \end{aligned}$$

Total (Incident + Reflected) H -Field

$$\begin{aligned}
 \underbrace{\vec{H}_1^{tot}(x, z)}_{\vec{H}_i(x, z) + \vec{H}_r(x, z)} &= \underbrace{\hat{y} \frac{E_{i0}}{\eta_1} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}}_{\substack{\text{from (6.2.18)} \\ \vec{H}_i(x, z)}} \underbrace{- \hat{y} \frac{E_{r0}}{\eta_1} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}}_{\substack{\text{from (6.2.6) and (6.2.20)} \\ \vec{H}_r(x, z)}} \\
 &= \hat{y} \frac{E_{i0}}{\eta_1} e^{-j\beta_1 x \sin \theta_i} \underbrace{\left(e^{-j\beta_1 z \cos \theta_i} + e^{+j\beta_1 z \cos \theta_i} \right)}_{2 \cos(\beta_1 z \cos \theta_i)} = \hat{y} \frac{2E_{i0}}{\eta_1} \cos(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}
 \end{aligned}$$

(6.2.22)

Interpretive Comments/Notes (#1)

1. In z -direction normal to boundary:

- a. E_{1x}^{tot} & H_{1y}^{tot} maintain standing-wave patterns according to $\sin[(\beta_1 \cos \theta_i)z]$ & $\cos[(\beta_1 \cos \theta_i)z]$ → transverse ($\perp$) to z under consideration
- b. No average power propagated in this direction → β_{1z} since E_{1x}^{tot} & H_{1y}^{tot} out of time phase by 90° .
- pure imag: see (6.2.21) pure real: see (6.2.22) pure real too: see (6.2.21)

2. In x -direction parallel to boundary: E_{1z}^{tot} & H_{1y}^{tot} in time & space phase

- a. Propagates with phase velocity: $u_{1x} = \frac{\omega}{\beta_{1x}} = \frac{\overbrace{\omega/\beta_1}^{u_1}}{\sin \theta_i} = \frac{u_1}{\sin \theta_i}$ both for x -direction
- b. Wavelength: $\lambda_{1x} = \frac{2\pi}{\beta_{1x}} = \frac{\overbrace{2\pi/\beta_1}^{\lambda_1}}{\sin \theta_i} = \frac{\lambda_1}{\sin \theta_i}$ same as (6.2.12) for $\perp$ pol
- same as (6.2.11) for $\perp$ pol

Interpretive Comments/Notes (#2)

3. Amplitude of propagating wave along x direction varies with $z \Rightarrow \therefore$ a *nonuniform plane wave*.
4. Functional term of E_{1x}^{tot} : $\sin(\beta_1 z \cos \theta_i) \rightarrow$ see (6.2.21), which vanishes for $\beta_1 z \cos \theta_i = -m\pi$ where $m = 0, 1, 2, \dots$
 $\Rightarrow \therefore$ a conducting plate could be inserted at:

$$z = -\frac{m\lambda_1}{2 \cos \theta_i}, \quad m = 1, 2, 3, \dots \quad (6.2.13)$$

without changing field pattern between this conducting plate & conducting boundary at $z = 0$

A **transverse magnetic (TM) wave** with respect to x propagation direction ($H_{1x} = 0$) bounces back & forth between conducting planes, and propagates along $x \rightarrow$ in effect, a parallel-plate waveguide

Part 7.1: Normal Incidence on Plane Dielectric Boundary

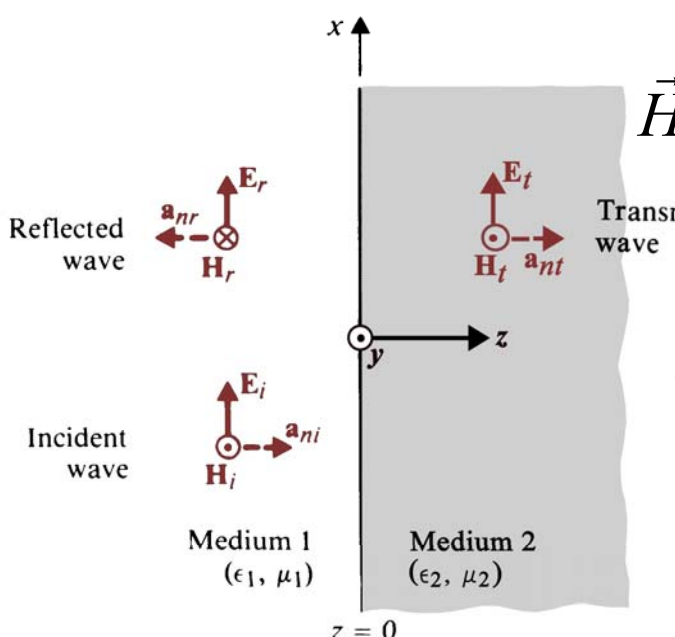
Opening Remarks

- When EM wave incident on surface of dielectric medium with intrinsic impedance (material parameters) different from that of medium in which wave originated, part of incident power reflected, part transmitted

Incident Fields in Medium#1

Consider below, incident wave travels +z direction, boundary surface is $z = 0$ plane, with incident field phasors:

$$\vec{E}_i(z) = \vec{a}_x \overbrace{E_{i0}}^{\text{field amplit}} e^{-j \overbrace{\beta_1}^{\text{phase const of medium \#1}} z} \quad (7.1.1)$$



$$\vec{H}_i(z) = \vec{a}_y \left(\underbrace{\frac{E_{i0}}{\eta_1}}_{\substack{\text{field amplitude} \\ \text{intrinsic impedance of medium \#1}}} \right) e^{-j \overbrace{\beta_1}^{\text{phase const of medium \#1}} z} \quad (7.1.2)$$

being same as (6.1.1) & (6.1.2) earlier

FIGURE 8-14
Plane wave incident normally on a plane dielectric boundary.

Reflected Fields Back into Medium #1

- $\therefore$ medium discontinuity at $z = 0$, incident wave partly reflected back into medium #1, partly transmitted into medium #2

- Reflected E -field:

$$\vec{E}_r(z) = \vec{a}_x \underbrace{E_{r0}}_{\substack{\text{field} \\ \text{amplit}}} e^{+j \underbrace{\beta_1}_{\substack{\text{phase} \\ \text{const} \\ \text{of} \\ \text{medium} \\ \text{\#1}}} z} \quad \text{being same as (6.1.3) earlier} \quad (7.1.3)$$

plus sign $\rightarrow$ reflected waves travels in $-z$ direction

- Reflected magnetic field $\vec{H}_r$ related to reflected electric field, $\vec{E}_r$, via (1.22) earlier this chapter:

$$\vec{H}_r(z) = \frac{1}{\eta_1} \underbrace{\vec{a}_{nr}}_{\substack{-\vec{a}_z \\ \therefore \text{reflected} \\ \text{wave in} \\ z \text{ direction}}} \times \vec{E}_r(z) = \frac{1}{\eta_1} (-\vec{a}_z) \times \underbrace{\vec{E}_r(z)}_{\vec{a}_x E_{r0} e^{+j\beta_1 z}} = -\vec{a}_y \frac{E_{r0}}{\eta_1} e^{+j\beta_1 z} \quad (7.1.4)$$

Transmitted Fields into Medium #2

$$\vec{E}_t(z) = \vec{a}_x E_{t0} e^{-j\beta_2 z} \quad (7.1.5a)$$

$$\vec{H}_t(z) = \vec{a}_z \times \frac{1}{\eta_2} \underbrace{\vec{E}_t(z)}_{\substack{\vec{a}_x E_{t0} e^{-j\beta_2 z} \\ \text{above}}} = \vec{a}_y \frac{E_{t0}}{\eta_2} e^{-j\beta_2 z} \quad (7.1.5b)$$

where E_{t0} = magnitude of $\vec{E}_t$ at $z = 0$, β_2 & η_2 are phase constant & intrinsic impedance of medium #2, respectively

Boundary Conditions

- Two equations needed to determine both unknown amplitudes E_{r0} & E_{t0} .
- These equations obtained from satisfying boundary conditions on E & H fields
- At dielectric interface $z = 0$, tangential components (x -directed) of E & H fields continuous:

$$\checkmark E_i(0) + E_r(0) = E_t(0) \Rightarrow E_{i0} + E_{r0} = E_{t0}. \quad (7.1.6)$$

$$\checkmark H_i(0) + H_r(0) = H_t(0) \Rightarrow (E_{i0} - E_{r0})/\eta_1 = E_{t0}/\eta_2. \quad (7.1.7)$$

-

Reflection & Transmission Coefficients

- Solving these latter 2 equations:

$$E_{r0} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} E_{i0} \quad (7.1.8)$$

$$E_{t0} = \frac{2\eta_2}{\eta_2 + \eta_1} E_{i0} \quad (7.1.9)$$

- Ratios E_{r0}/E_{i0} & E_{t0}/E_{i0} : ***reflection & transmission coefficients***, respectively
- In terms of intrinsic impedances:

$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (\text{dimensionless}) \quad (7.1.10)$$

$$\tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad (\text{dimensionless}) \quad (7.1.11)$$

Remarks about Γ & τ

- Reflection coefficient Γ can be positive or negative, depending whether η_2 greater or less than η_1 .
- Transmission coefficient τ however always > 0
- Formulas (7.1.10) & (7.1.11) apply even when media dissipative, i.e. η_1 &/or η_2 complex
 - ✓ $\therefore \Gamma$ & τ may be complex (in general)
 - ✓ Complex Γ (or τ) $\rightarrow$ phase shift introduced at interface upon reflection (or transmission).
- Γ & τ related via: $1 + \Gamma = \tau$ (dimensionless)
(7.1.12)

Medium #2 Perfect Conductor

- If medium #2 perfect conductor, $\eta_2 = 0$
- Eqs. (7.1.10) & (7.1.11) yield: $\Gamma = -1$ & $\tau = 0$
- $E_{r0} = -E_{i0}$ & $E_{t0} = 0$
- Incident wave totally reflected
- Standing wave in medium #1, with zeros & maxima points

Medium #2 non conducting

- If medium #2 non-conducting, partial reflection
- Total E -field in medium #1:

$$\vec{E}_1(z) = \underbrace{\vec{E}_i(z)}_{\substack{\text{from (7.1.1)} \\ \text{field amplit} \\ \text{phase const of medium \#1}}} + \underbrace{\vec{E}_r(z)}_{\substack{\text{from (7.1.3)} \\ \text{field amplit} \\ \text{phase const of medium \#1}}} = \vec{a}_x \underbrace{E_{i0}}_{\text{field amplit}} e^{-j \underbrace{\beta_1}_{\text{phase const of medium \#1}} z} + \vec{a}_x \underbrace{E_{r0}}_{\text{field amplit}} e^{+j \underbrace{\beta_1}_{\text{phase const of medium \#1}} z} = \vec{a}_x E_{i0} (e^{-j\beta_1 z} + \Gamma e^{+j\beta_1 z}) \quad (7.1.13a)$$

$$\underbrace{E_{r0}}_{\substack{\Gamma \\ \text{from (7.1.8) \& (7.1.10)}}} = \underbrace{\left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right)}_{\Gamma} E_{i0}$$

$$= \vec{a}_x E_{i0} \left[\underbrace{(1 + \Gamma)}_{\substack{\tau \\ \text{from (7.1.12)}}} e^{-j\beta_1 z} + \Gamma \underbrace{(e^{+j\beta_1 z} - e^{-j\beta_1 z})}_{2j \sin \beta_1 z} \right] = \vec{a}_x \underbrace{E_{i0} \tau}_{\text{traveling wave}} e^{-j\beta_1 z} + j \vec{a}_x \underbrace{2E_{i0} \Gamma \sin \beta_1 z}_{\text{standing wave}}$$

(7.1.13b) $\therefore E_1$ doesn't vanish at fixed distances from interface

Rewrite E_1 :

- From (7.1.13a), rewrite E_1 as:

$$\vec{E}_1(z) = \vec{a}_x E_{i0} e^{-j\beta_1 z} (1 + \Gamma e^{j2\beta_1 z}) \quad (7.1.14)$$

- For dissipationless media, η_1 & η_2 real $\rightarrow$ both Γ & τ also real
- But Γ can be positive or negative
- Consider following two cases (next slides ...)

Case 1: $\Gamma > 0$ ($\eta_2 > \eta_1$)

- From (7.1.14),
 - ✓ max value of $|E_1(z)| = E_{i0}(1 + \Gamma)$, which occurs when $2\beta_1 z_{max} = -2n\pi$ ($n = 0, 1, 2, \dots$), or at: $z_{max} = -n\pi/\beta_1 = -n\lambda_1/2$; $n = 0, 1, 2, \dots$ (7.1.15)
 - ✓ min value of $|E_1(z)| = E_{i0}(1 - \Gamma)$, which occurs when $2\beta_1 z_{min} = -(2n + 1)\pi$ ($n = 0, 1, 2, \dots$), or: $z_{min} = -(2n + 1)\pi/(2\beta_1) = -(2n + 1)\lambda_1/4$; $n = 0, 1, 2, \dots$ (7.1.16)
- Note that $\Gamma > 0$ for present case
- Inverse case of Case 2 next slide ... i.e. locations for $|E_1(z)|_{max}$ & $|E_1(z)|_{min}$ when $\Gamma > 0$ & $\Gamma < 0$ interchanged

Case 2: $\Gamma < 0$ ($\eta_2 < \eta_1$)

- From (7.1.14),
 - ✓ min value of $|E_1(z)| = E_{i0}(1 + \Gamma)$, which occurs when $2\beta_1 z_{max} = -2n\pi$ ($n = 0, 1, 2, \dots$), or at:
 $z_{max} = -n\pi/\beta_1 = -n\lambda_1/2$; $n = 0, 1, 2, \dots$ (7.1.17)
 - ✓ max value of $|E_1(z)| = E_{i0}(1 - \Gamma)$, which occurs when $2\beta_1 z_{min} = -(2n + 1)\pi$ ($n = 0, 1, 2, \dots$), or: $z_{min} = -(2n + 1)\pi/(2\beta_1) = -(2n + 1)\lambda_1/4$; (7.1.18)
 $n = 0, 1, 2, \dots$
- Note that $\Gamma < 0$ for present case
- Inverse case of Case 1 previous slide, i.e. locations for $|E_1(z)|_{max}$ & $|E_1(z)|_{min}$ when $\Gamma > 0$ & $\Gamma < 0$ interchanged

Standing Wave Ratio (SWR)

- Ratio of max to min value of E -field of standing wave: *standing wave ratio (SWR)*, S .

$$S = \frac{\overbrace{|E|_{\max}}^{\text{use (7.1.15) \& (7.1.18)}}}{\underbrace{|E|_{\min}}_{\text{use (7.1.16) \& (7.1.17)}}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad (\text{dimensionless}) \quad (7.1.19)$$

- Inverse relation of this (7.1.19):

$$|\Gamma| = \frac{S - 1}{S + 1} \quad (\text{dimensionless}) \quad (7.1.20)$$

Ranges of Γ & S

- From (7.1.10): $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$, Γ ranges from -1 to $+1$
- $\therefore$ from (7.1.19): $S = (1 + |\Gamma|)/(1 - |\Gamma|)$, S ranges from 1 to ∞
- Customary to express S on logarithmic scale:
 $S_{\text{dB}} = 20\log_{10}S.$

Total Magnetic Field in Medium #1

- Combining (7.1.2) & (7.1.4):

$$\begin{aligned}\vec{H}_1(z) &= \underbrace{\vec{H}_i(z)}_{\substack{\vec{a}_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} \\ \text{from (7.1.2)}}} + \underbrace{\vec{H}_r(z)}_{\substack{-\vec{a}_y \frac{E_{r0}}{\eta_1} e^{+j\beta_1 z} \\ \text{from (7.1.4)}}} = \vec{a}_y \frac{1}{\eta_1} (E_{i0} e^{-j\beta_1 z} - \underbrace{E_{r0}}_{\substack{\Gamma E_{i0} \\ \text{from (7.1.8) \& (7.1.10)}}} e^{+j\beta_1 z}) \\ &= \vec{a}_y \frac{E_{i0}}{\eta_1} (e^{-j\beta_1 z} - \Gamma e^{+j\beta_1 z}) = \vec{a}_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} (1 - \Gamma e^{+j2\beta_1 z}) \quad (7.1.21)\end{aligned}$$

- Comparing this with (7.1.14), in dissipationless media, Γ real $\rightarrow |\mathbf{H}_1(z)|$ min at locations where $|\mathbf{E}_1(z)|$ max, & vice-versa

$$\vec{E}_1(z) = \vec{a}_x E_{i0} e^{-j\beta_1 z} (1 + \Gamma e^{j2\beta_1 z}) \quad (7.1.14)\text{repeated}$$

Transmitted Fields in Medium #2

- Using (7.1.11): $\tau = E_{t0}/E_{i0}$,

✓ From (7.1.5a):
$$\vec{E}_t(z) = \vec{a}_x \overbrace{\tau E_{i0}}^{\substack{\text{from} \\ (7.1.11) \\ E_{t0}}} e^{-j\beta_2 z} \quad (7.1.22)$$

✓ From (7.1.5b):
$$\vec{H}_t(z) = \vec{a}_y \frac{\overbrace{\tau E_{i0}}^{\substack{\text{from} \\ (7.1.11) \\ E_{t0}}}}{\eta_2} e^{-j\beta_2 z} \quad (7.1.23)$$

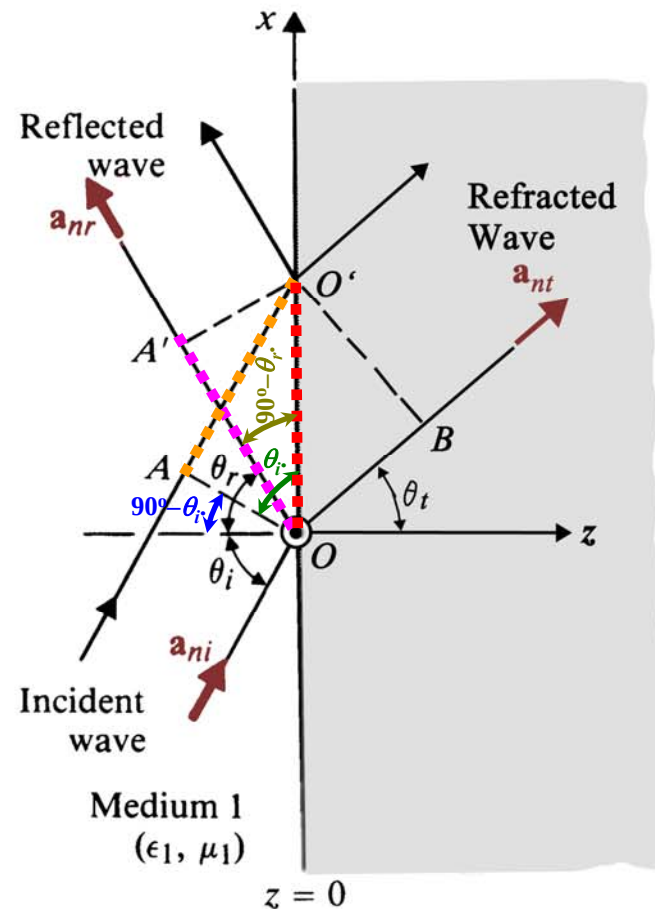
Part 7.2: Oblique Incidence at Plane Dielectric Boundary

Opening Comments

- Now consider case of plane wave incident obliquely at arbitrary angle of incidence θ_i on plane interface between two dielectric media
- Both media lossless, with (μ_1, ε_1) & (μ_2, ε_2)
- $\therefore$ discontinuity at interface, part of incident wave reflected, part transmitted

Snell's Law of Reflection

See below schematic: $\because$ both incident & reflected waves propagate in medium 1 with same phase velocity u_{p1} , $\therefore$ the following two distances must be equal: $\overline{OA'} = \overline{AO'}$



$$\therefore \overbrace{\overline{OO'}}^{\overline{OA'}} \underbrace{\sin \theta_r}_{\cos(90^\circ - \theta_r)} = \overbrace{\overline{OO'}}^{\overline{AO'}} \sin \theta_i$$

$$\Rightarrow \boxed{\theta_r = \theta_i} \quad (7.2-1)$$

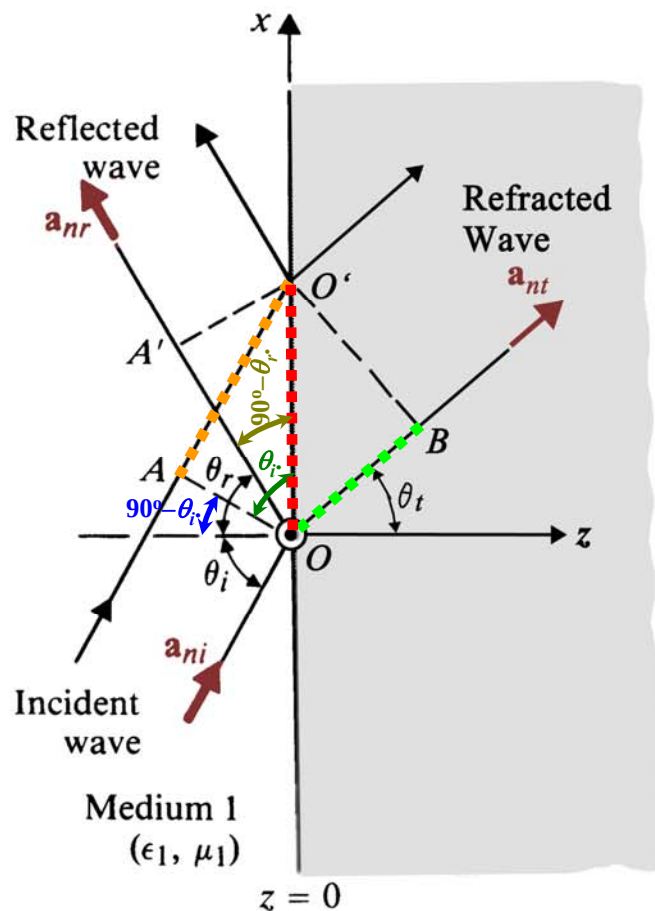
Angle of reflection = angle of incidence

SNELL'S LAW OF REFLECTION

Uniform plane wave incident obliquely on a plane dielectric boundary.

Snell's Law of Refraction

In medium 2, **time** taken for transmitted wave to travel from O to B equals **time** for incident wave to travel from A to O .



$$\therefore \frac{\overline{OB}}{u_{p2}} = \frac{\overline{AO'}}{u_{p1}} \Rightarrow \frac{\overbrace{\overline{OO'}}^{\overline{OB}} \cos(90^\circ - \theta_t)}{\underbrace{\overline{OO'}}_{\overline{AO'}} \sin \theta_i} = \frac{u_{p2}}{u_{p1}} \quad \text{refractive index of \#1}$$

$$\Rightarrow \frac{\sin \theta_t}{\sin \theta_i} = \frac{\overbrace{u_{p2}}^{\omega/\beta_2}}{\underbrace{u_{p1}}_{\omega/\beta_1}} = \frac{\overbrace{\beta_1}^{\omega\sqrt{\mu_1\epsilon_1}}}{\underbrace{\beta_2}_{\omega\sqrt{\mu_2\epsilon_2}}} = \sqrt{\frac{\mu_1\epsilon_1}{\mu_2\epsilon_2}} = \frac{\overbrace{\sqrt{\mu_{r1}\epsilon_{r1}}}^{n_1}}{\underbrace{\sqrt{\mu_{r2}\epsilon_{r2}}}_{n_2}} \quad \text{refrac. index of \#2}$$

(7.2-2)

SNELL'S LAW OF REFRACTION

Uniform plane wave incident obliquely on a plane dielectric boundary.

Refractive Index

- $n_{\#}$: refractive index of medium#
- represents ratio of speed of light (EM wave) in free space to that in that medium#, i.e., $n = c/u_p = [1/\sqrt{(\mu_0\epsilon_0)}]/[1/\sqrt{(\mu_{r\#}\mu_0\epsilon_{r\#}\epsilon_0)}] = \sqrt{(\mu_{r\#}\epsilon_{r\#})}$
- Eq. (7.2-2) previous slide: *Snell's law of refraction* $\rightarrow$
- Statement of *Continuity of Tangential Phase Constant Across Interface Boundary*.

$$\beta_1 \sin \theta_i = \beta_2 \sin \theta_t. \quad (7.2-3)$$

Total Reflection

*Read up this part in textbook
yourselves ...*

Part 7.2-1: Perpendicular Polarization

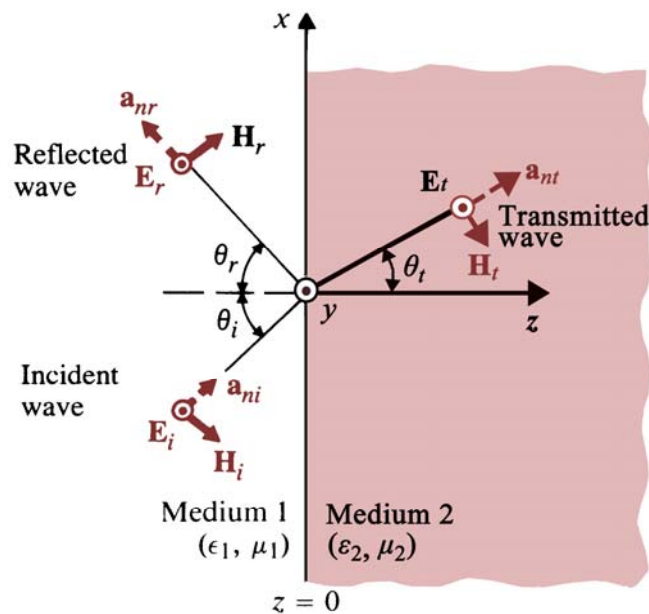


FIGURE 8-20
Plane wave incident on a plane dielectric boundary 165
(perpendicular polarization).

Incident Fields

Incident fields from (6.2.2) & (6.2.3) earlier (when dealt with oblique incidence on plane conducting boundary):

$$\overbrace{\vec{E}_i(x, z) = \hat{y}E_{i0}e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)}}^{\text{of (6.2.2) earlier}} \quad \begin{matrix} \text{(7.2-4)} \\ \text{from (6.2.2)} \end{matrix}$$

$$\underbrace{\vec{H}_i(x, z) = \frac{E_{i0}}{\eta_1}e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)}(-\hat{x}\cos\theta_i + \hat{z}\sin\theta_i)}_{\text{of (6.2.3) earlier}} \quad \begin{matrix} \text{(7.2-5)} \\ \text{from (6.2.3)} \end{matrix}$$

Reflected Fields

Reflected fields from (6.2.5) & (6.2.8) earlier (when dealt with oblique incidence on plane conducting boundary), but with exception that E_{r0} no longer equals to $-E_{i0}$:

$$\overbrace{\vec{E}_r(x, z) = \hat{y} E_{r0} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}}^{\text{of (6.2.5) earlier}} \quad \begin{matrix} \text{(7.2-6)} \\ \text{from (6.2.5)} \end{matrix}$$

$$\underbrace{\vec{H}_r(x, z) = \frac{\overbrace{E_{r0}}^{\substack{\text{no longer} \\ \text{equal to} \\ -E_{i0}}} }{\eta_1} e^{-j\beta_1(x \sin \overset{\theta_r}{\theta_i} - z \cos \overset{\theta_r}{\theta_i})} (\hat{x} \cos \theta_r + \hat{z} \sin \theta_r)}_{\text{of (6.2.8) earlier}} \quad \begin{matrix} \text{(7.2-7)} \\ \text{from (6.2.8)} \end{matrix}$$

Transmitted Fields

Transmitted fields written directly by making following script changes to incident fields of (7.2-4) & (7.2-5):

- $i \rightarrow t$ (former *incident* now *transmitted*)
- $1 \rightarrow 2$ (former medium#1 now medium#2)

$$\vec{E}_t(x, z) = \hat{y}E_{t0}e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)} \quad (7.2-8)$$

$$\vec{H}_t(x, z) = \frac{E_{t0}}{\eta_2}(-\hat{x}\cos\theta_t + \hat{z}\sin\theta_t)e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)} \quad (7.2-9)$$

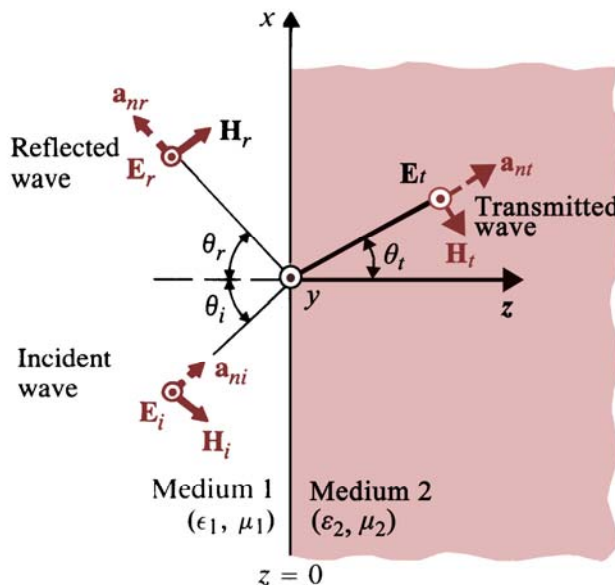


FIGURE 8-20
Plane wave incident on a plane dielectric boundary
(perpendicular polarization).

Unknowns Solved by Boundary Conditions

- Four unknowns in (7.2-4) through (7.2-9):
 - E_{r0} ,
 - E_{t0} ,
 - θ_r ,
 - θ_t .
- Solved for by enforcing boundary conditions:
continuity of tangential E & H components
across interface boundary at $z = 0$

Continuity of Tangential E -Field

$$E_{iy}(x, z = 0) + E_{ry}(x, z = 0) = E_{ty}(x, z = 0)$$

$$\Rightarrow \overbrace{\underbrace{E_{i0} e^{-j\beta_1(x \sin \theta_i + \cancel{z \cos \theta_i})}}_{\substack{E_{iy} \\ \text{of (7.2.4)}}} + \underbrace{E_{r0} e^{-j\beta_1(x \sin \theta_r - \cancel{z \cos \theta_r})}}_{\substack{E_{ty} \\ \text{of (7.2.6)}}}}^{\text{all backward slashes to zero } \because z=0} = \underbrace{E_{t0} e^{-j\beta_2(x \sin \theta_t + \cancel{z \cos \theta_t})}}_{\substack{E_{ty} \\ \text{of (7.2.8)}}$$

(7.2-10)

Continuity of Tangential H -Field

$$H_{ix}(x, z = 0) + H_{rx}(x, z = 0) = H_{tx}(x, z = 0)$$

$$\begin{aligned}
 & \overbrace{\frac{E_{i0}}{\eta_1} e^{-j\beta_1(x \sin \theta_i + \cancel{z \cos \theta_i})} \left(-\cancel{\hat{x}} \cos \theta_i + \hat{z} \sin \theta_i \right)}^{H_{ix} \text{ from (7.2-5)}} + \overbrace{\frac{E_{r0}}{\eta_1} e^{-j\beta_1(x \sin \hat{\theta}_r - \cancel{z \cos \hat{\theta}_r})} \left(-\cancel{\hat{x}} \cos \theta_r + \hat{z} \sin \theta_i \right)}^{H_{rx} \text{ from (7.2-7)}} \\
 &= \underbrace{\frac{E_{t0}}{\eta_2} \left(-\cancel{\hat{x}} \cos \theta_t + \hat{z} \sin \theta_t \right) e^{-j\beta_2(x \sin \theta_t + \cancel{z \cos \theta_t})}}_{H_{tx} \text{ from (7.2-9)}} \\
 &\Rightarrow \frac{1}{\eta_1} \left(-E_{i0} e^{-j\beta_1 x \sin \theta_i} \cos \theta_i + E_{r0} e^{-j\beta_1 x \sin \hat{\theta}_r} \cos \theta_r \right) = -\frac{E_{t0}}{\eta_2} e^{-j\beta_2 x \sin \theta_t} \cos \theta_t \quad \textbf{(7.2-11)}
 \end{aligned}$$

whereby

- forward slashes (/) to 0 ∴ only tangential x comp considered now,
- backward slashes (\) to 0 ∴ $z = 0$ (at boundary interface)
- cross-out ($\times$) to 1 ∴ common unit vector along x cancelled throughout

Phase Matching

$\therefore$ both (7.2-10) & (7.2-11) must be satisfied for all x , all three exponential terms that are functions of x must be equal: *phase matching*: $\therefore$

$$\beta_1 x \sin \theta_i = \beta_1 x \sin \theta_r = \beta_2 x \sin \theta_t, \quad (7.2-12)$$

leading to Snell's laws of:

- reflection: $\theta_r = \theta_i$, (tie up 1st two)
- refraction: $\beta_1 x \sin \theta_i = \beta_2 x \sin \theta_t$, (tie up 1st & 3rd)

Reflection & Transmission

With this (7.2.12),

- Eq. (7.2-10) simplified to: $E_{i0} + E_{r0} = E_{t0}$. **(7.2-13)**

- Eq. (7.2-11) simplifies to: $\frac{1}{\eta_1}(E_{i0} - E_{r0})\cos\theta_i = \frac{E_{t0}}{\eta_2}\cos\theta_t$ **(7.2-14)**

From these two relations, E_{r0} & E_{t0} can each be expressed in terms of E_{i0} :

$$\Gamma_{\perp} = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t} = \frac{(\eta_2/\cos\theta_t) - (\eta_1/\cos\theta_i)}{(\eta_2/\cos\theta_t) + (\eta_1/\cos\theta_i)} \quad \textbf{(7.2-15)}$$

$$\tau_{\perp} = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t} = \frac{2(\eta_2/\cos\theta_t)}{(\eta_2/\cos\theta_t) + (\eta_1/\cos\theta_i)} \quad \textbf{(7.2-16)}$$

derive
these
yourselves;
simple!

Γ & τ Relatable to Normal Incidence

Comparing these (7.2-15) & (7.2-16) with formulas (7.1-10) & (7.1-11) for reflection & transmission coefficients at normal incidence, repeated below:

$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (\text{dimensionless}) \quad (7.1-10)\text{repeated}$$

$$\tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad (\text{dimensionless}) \quad (7.1-11)\text{repeated}$$

can see that same formulas apply if η_1 & η_2 changed to $\eta_1/\cos\theta_i$ & $\eta_2/\cos\theta_t$, respectively.

When $\theta_i = 0$, making $\theta_r = \theta_t = 0$, expressions of (7.2-15) & (7.2-16) reduce to those for normal incidence, as required.

Also, readily shown that $\Gamma_{\perp}$ & $\tau_{\perp}$ related by following way:

$$1 + \Gamma_{\perp} = \tau_{\perp}, \quad (7.2-17)$$

being same as (7.1-12) for normal incidence

If Medium 2 Perfect Conductor

- If medium 2 is perfect conductor, $\eta_2 = 0$
- (7.2-16) says: $\tau_{\perp} = 0 \Rightarrow E_{t0} = 0$ as well (as required by boundary condition!)

Angle θ_i that Vanishes $\Gamma_{\perp}$.

Let $\theta_{B\perp}$ denote the θ_i that makes $\Gamma_{\perp}$ of (7.2-15) vanish, i.e., its numerator = 0 $\Rightarrow \eta_2 \cos \theta_{B\perp} = \eta_1 \cos \theta_t$. **(7.2-18)**

By Snell's law of refraction,

$$\cos \theta_t = \sqrt{1 - \sin^2 \theta_t} \overset{\substack{\text{Snell's} \\ \text{Refraction Law} \\ n_1 \sin \theta_i = n_2 \sin \theta_t \\ \text{from (7.2-2)}}}{=} \sqrt{1 - \left(\frac{n_1}{n_2}\right)^2 \sin^2 \theta_i} = \sqrt{1 - \frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2} \sin^2 \theta_i} \quad \mathbf{(7.2-19)}$$

Brewster Angle

Then using (7.2-18) in this (7.2-19):

$$\begin{aligned}
 & \underbrace{\eta_1 \sqrt{1 - \frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2} \sin^2 \theta_{B\perp}}}_{\text{from (7.2-18)}} = \eta_2 \cos \theta_{B\perp} \xRightarrow[\text{both sides}]{\text{square}} \underbrace{\eta_1^2 \left[1 - \frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2} \sin^2 \theta_{B\perp} \right]}_{\frac{\mu_1}{\epsilon_1} \overbrace{\frac{\mu_1 - \mu_1^2}{\mu_2 \epsilon_2} \sin^2 \theta_{B\perp}}} = \underbrace{\eta_2^2 \cos^2 \theta_{B\perp}}_{\frac{\mu_2}{\epsilon_2} \overbrace{\frac{\mu_2 - \mu_2 \sin^2 \theta_{B\perp}}{1 - \sin^2 \theta_{B\perp}}}} \\
 & \Rightarrow \frac{\mu_1^2}{\mu_2 \epsilon_2} \sin^2 \theta_{B\perp} - \frac{\mu_2}{\epsilon_2} \sin^2 \theta_{B\perp} = \frac{\mu_1}{\epsilon_1} - \frac{\mu_2}{\epsilon_2} \xRightarrow[\mu_2 \epsilon_2 \text{ throughout}]{\text{multiply}} (\mu_1^2 - \mu_2^2) \sin^2 \theta_{B\perp} = \underbrace{\frac{\mu_1 \mu_2 \epsilon_2}{\epsilon_1} - \mu_2^2}_{\frac{\mu_1 \mu_2 \epsilon_2 - \mu_2^2 \epsilon_1}{\epsilon_1}}
 \end{aligned}$$

$$\Rightarrow \sin^2 \theta_{B\perp} = \frac{\mu_1 \mu_2 \epsilon_2 - \mu_2^2 \epsilon_1}{\epsilon_1 (\mu_1^2 - \mu_2^2)} \xrightarrow[\text{numer \& denom}]{\text{negate}} \frac{\mu_2^2 \epsilon_1 - \mu_1 \mu_2 \epsilon_2}{\mu_2^2 \epsilon_1 - \mu_1^2 \epsilon_1} = \frac{1 - \frac{\mu_1 \mu_2 \epsilon_2}{\mu_2^2 \epsilon_1}}{1 - \frac{\mu_1^2 \cancel{\epsilon_1}}{\mu_2^2 \cancel{\epsilon_1}}} = \frac{1 - \frac{\mu_1 \epsilon_2}{\mu_2 \epsilon_1}}{1 - \frac{\mu_1^2}{\mu_2^2}}$$

**Brewster angle of no reflection,
for perpendicular polarization**

End of Chapter

*Read up the rest in the textbook yourselves
... all quite trivial & simple materials
No time to cover everything, need to quickly
move on to next chapter!*